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OECD ECONOMIC SURVEYS

1999

SPECIAL FEATURES

Structural policies Sustainable economic growth: natural resources and the environment

NORWAY

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NORWAY

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF NORWAY (1997)

THE LAND

Area (1 000 km^2):	
Total	385.4
Mainland	323.8
Agricultural	10.4
Productive forests	73.6

Major cities (thousand inhabitants, 1.1.98):	
Oslo	499.7
Bergen	225.4
Trondheim	145.8

THE PEOPLE

Population (thousands, 1.1.98)	4 417.6
Number of inhabitants per km^2	11.5
Net natural increase (thousands)	15.2
Net migration (thousands, 1996)	5.8

Total labour force (thousands)	2 279.0
Civilian employment (thousands)	2 180.0
Civilian employment (per cent of total):	
Agriculture, forestry and fishing	4.5
Industry and construction	23.8
Services	71.7

PRODUCTION

Gross domestic product:	
NKr billion	1 084.8
Per head (US$)	34 918

Gross fixed capital investment:	
Per cent of GDP	23.0
Per head (US$)	8 045

THE GOVERNMENT

Public consumption (% of GDP)	20.2
General government (% of GDP):	
Current and capital expenditure	44.3
Current revenue	51.7

Last general elections: 15.9.97
Next general elections: 2001

Composition of Parliament (number of seats):	
Labour	65
Progress	25
Christian Democrats	25
Conservatives	23
Centre	11
Socialist Left	9
Other	7
Total	165

FOREIGN TRADE

Exports of goods and services (% of GDP)	41.3
of which: Gross freight and oil drilling	15.3
Main commodity exports (% of total):	
Forestry products	3.2
Fish and fish products	6.8
Base metals and products	10.7
Machinery and transport equipment (excluding ships)	8.9

Imports of goods and services (% of GDP)	34.2
Main commodity imports (% of total):	
Ships	5.3
Raw materials (including fuel and chemicals)	10.7
Base metals and products	9.4
Machinery and transport equipment (excluding ships)	34.5

THE CURRENCY

Monetary unit: Krone

Currency units per US$, average of daily figures:	
Year 1998	7.55
December 1998	7.60

Note: An international comparison of certain basic statistics is given in an Annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Norway by the Economic and Development Review Committee on 11 January 1999.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 26 January 1999.

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The previous Survey of Norway was issued in February 1998.

Assessment and recommendations

After a five-year boom the mainland economy has hit capacity limits...

The strong performance of the Norwegian economy since 1993 has lifted mainland GDP by 20 per cent in only five years, but a significant slowdown now appears to be underway. The impetus for the upswing came from robust foreign and domestic demand. Soaring oil and gas exports together with the increasing sale of non-oil "traditional" commodities contributed to a sharp widening of the current account surplus. The resulting upward pressure on the exchange rate contributed to an easy monetary policy stance, pushing short-term interest rates slightly below European levels. Fixed capital formation was boosted by high profits and rising capacity utilisation, while substantial gains in real earnings buoyed private consumption. Strong job creation raised the employment rate to almost 80 per cent of the working-age population, among the highest in the OECD area, facilitated by a flexible response of the labour force. At the same time, inflationary pressures remained subdued due to low imported inflation and moderate wage growth until the 1998 pay settlement. As the upturn developed, capacity constraints became increasingly evident, especially in labour-intensive sheltered sectors such as construction, health care and education. The potential for expanding the labour supply was almost exhausted in 1998, with vacancies soaring and the unemployment rate dropping below 3 per cent by the end of the year. Wages strongly accelerated in the wake of the 1998 central wage negotiations, whereas consumer price inflation remained low as profits were squeezed amid weakening external conditions.

... which together with an oil price shock, weakening external demand and financial turbulence...

In the course of 1998, growth in non-oil exports slowed down while the economy was also hit by a sharp drop in oil prices. Oil production, after having surged by 10 per cent per year from 1993 to 1996, has since levelled off, in part reflecting a government decision to limit production on the Norwegian continental shelf from May 1998. As a result, the current account surplus, having reached a record 6 per cent of GDP in the previous two years, vanished in 1998. Weakening external conditions, coupled with the sharp acceleration in wages and concerns over the stance of fiscal policy, triggered substantial capital outflows after the summer. The exchange rate plunged and interest rates soared, contributing to sharp losses in business and consumer confidence. While the growth of overall and mainland GDP (excluding oil and gas) may have reached 2¼ and 3 per cent, respectively, in 1998, the economy has already begun to show signs of weakening following the events after the summer: the asset boom started to unwind, the housing market has become less buoyant and investment activity has been floundering.

... will force the economy into a lower gear

With supply constraints binding, economic growth in the coming years is expected to be considerably weaker than in recent years. The OECD Secretariat's central projection suggests a "soft landing" for the Norwegian economy, with mainland GDP growth of 1 and 1½ per cent, in 1999 and 2000, respectively. The unemployment rate should increase somewhat to around 4 per cent, but unless a very moderate wage agreement is struck in 1999, wage inflation will be sustained at around 6 per cent, while consumer price inflation could rise to close to 3 per cent over the projection period. Based on the assumptions that the interest rate hike will prove short-lived and the exchange rate remain unchanged over the next two years, non-oil exports and private consumption should provide some growth impetus, but fixed investment is expected to decline, reflecting *inter alia* lower profits, weak commodity prices and falling capacity utilisation rates. With oil production expected to show a temporary rebound in 1999, overall real GDP growth is projected to reach 2¼ per cent in 1999 and 1¾ per cent in 2000. Another high wage settlement or a new bout of financial market turbulence, however, would probably lead to

considerable further policy tightening and could result in an outright recession.

Stated macroeconomic policy objectives should be pursued rigorously:

A return to robust but sustainable, non-inflationary growth calls for a better mix of macroeconomic policies in the years ahead. Against a backdrop of historically high unemployment in the early 1990s, the government, in co-operation with the social partners, formulated a macroeconomic policy framework drawing on the experiences of tripartite co-operation since the late 1980s. It aimed at improving the international competitiveness of mainland industry by a combination of moderate wage growth and a stable exchange rate *vis-à-vis* trading partners, while fiscal policy was geared to smoothing domestic demand, with budget surpluses associated with the government's oil and gas revenues reinvested in financial assets abroad. This framework, the so-called *Solidarity Alternative*, helped to restore financial market confidence and to pull the economy out of the protracted slump that followed the oil price collapse of 1986. However, while the upswing gathered momentum, fiscal policy was not sufficiently tight to offset strong growth in private demand associated *inter alia* with the decline in interest rates below European levels due to upward pressure on the exchange rate. By prompting the sharp rise in interest rates in 1998, which may have nipped a vicious wage-price spiral in the bud, the timing of the oil price collapse and the associated weakening of the exchange rate may even appear to be fortuitous, but it masks the failure of fiscal policy to cool off a booming economy in a timely fashion.

– fiscal policy must reassert its role of stabilising demand

The potential demand impetus stemming from oil and gas activity is very large and fiscal policy is, in the context of the Solidarity Alternative, the designated instrument to ensure balanced economic growth over the medium run. Fiscal policy has successfully played this role for nearly a decade but has, unfortunately, been too loose amid increasing signs of economic overheating in recent years. In 1998, the cyclically-adjusted general government surplus fell and public expenditure rose as a share of GDP for the first time since 1993. As public finances in Norway, unlike in most other OECD countries, are fundamentally sound, it proved difficult to sustain the budgetary rigour implied by the

Solidarity Alternative. Indeed, the public balance sheet has remained very solid with net financial assets exceeding 35 per cent of GDP, and is staying on a rising trend owing to the central government's annual allocations of its surpluses to the Government Petroleum Fund – the fund established in 1990 to accumulate public financial wealth abroad. The Budget envisages the resumption of sizeable fiscal restraint in 1999; it is important that the government, as a minimum, sticks to the budget and ensures an appropriate fiscal stance also in later years, in particular if, as expected, oil and gas exports and associated fiscal proceeds pick up.

– monetary policy should be tight in the present juncture to underpin exchange rate stability

Since the krone was floated in the wake of the turmoil in European exchange markets in late 1992, monetary policy has aimed at maintaining a stable exchange rate against European currencies to anchor inflation expectations and ensure stable business conditions. However, soaring oil revenues prompted upward pressure on the krone in the winter of 1996/97 and interest rates were cut below the EU average, even though economic activity was much more buoyant than elsewhere in Europe. In contrast, as the fundamentals weakened over the summer of 1998, capital outflows sparked downward pressure on the krone and short-term interest rates were raised to 8 per cent. The monetary authorities thereby exploited the room for manoeuvre inherent in the existing monetary policy framework. Continuation of this flexible approach would allow controlled exchange rate movements around a medium-term target range. While the interest rate hike has, so far, not sufficed to steer the exchange rate back to its previous trading range, in the present circumstances the authorities should persist in gearing monetary policy to restoring economic fundamentals in order to obtain exchange rate stability over time.

– incomes policy needs to regain credibility

A new tripartite incomes policy framework was adopted in 1992 in line with Norway's long-standing tradition in this regard. Wage agreements were moderate until the mid-1990s, but this has been harder to achieve subsequently. The main incomes policy concession offered by the government in 1997 – an extension of the early retirement programme – did not succeed in preventing stronger wage pressure, partly because it also resulted in workers with-

drawing from the labour market. The social partners have consistently emphasised that continued co-operation with the government is based on a stable exchange rate and the use of fiscal policy to maintain sustainable economic growth – the other two legs of the Solidarity Alternative. In turn, they should deliver a moderate wage deal in the upcoming negotiation round, in order to preserve competitiveness, contain inflationary pressure and support the efforts of monetary policy to maintain a stable exchange rate.

Labour market reform should aim to reduce inflow into the generous welfare system

In order to facilitate the task of macroeconomic policies to sustain solid, non-inflationary growth in the longer run, labour market policies should aim to durably relieve labour supply constraints. A key feature of the remarkable flexibility of labour supply during the past economic upswing has been the entry of "outsiders" into the labour force, lifting the employment rate to unprecedented heights. On the other hand, despite the tighter eligibility conditions introduced in 1992, the rise in the number of persons on disability has continued, while a stepwise reduction of the minimum entry age in the early retirement scheme to 62 years – without actuarial adjustment of pension benefits – has induced a sharp increase in early retirement. As a result, the effective retirement age of Norwegian workers, while still substantially higher than the EU average, is rapidly falling. The recent strengthening of incentives for earlier retirement runs counter to the adjustments needed in view of the population ageing and the expected decline in oil revenues in the next century. At the same time the Family Cash Benefit scheme introduced in 1998, rewarding parents who decide not to use the public day-care scheme, discourages job search – though in part it may merely lead to a shift towards private day-care. Policies that reduce labour supply limit the room for pursuing new initiatives, such as the competence reform, a measure that aims to allow active workers and other adults to raise their skill level through participation in public education programmes.

Product and financial market policies could be more cost-effective

While Norway has been a forerunner in terms of applying market principles for the important hydro power supply and distribution sector, state ownership and support has remained high in most sectors of the economy. Regional support, which re-distributes part of the economic rent deriving from the exploitation of oil and gas wealth to sustain dispersed economic activities, remains substantial. In particular, despite some decline recently, support to agriculture remains high: at 2 per cent of GDP, it is significantly higher than in the European Union. While this reflects the predominance of regional policy goals, there is scope for achieving them at a lower cost by reducing border protection and better targeting income support not linked to production, as this would widen the scope for Norwegian consumers to benefit from lower prices of imported goods. Regional considerations also motivate substantial credit subsidies for companies in remote areas. To achieve regional development objectives in ways that distort the provision of financial services less, policy options should be explored which level the playing field between public and private financial institutions. More generally, regional policies should be reassessed with a view to raising efficiency without jeopardising its objectives. This could include a greater role of firm-level wage bargaining to preserve equilibrium in local labour markets. Other structural initiatives should focus on, *inter alia*, further privatisation of state-owned banks and enterprises as well as strengthening the framework for competition policy.

Sustainable development figures high on the political agenda...

The principal viewpoints laid down in the 1987 UN Report *Our Common Future*, the "Brundtland Report", have inspired the overall orientation of economic policy in Norway. The Report defines sustainable development as one that meets the needs of the present without compromising the ability of future generations to meet their own needs, by maintaining an appropriate stock of natural resources, environmental and man-made capital. In order to maximise this combined wealth of present and future generations, it is important to pursue sustainability in a cost-effective way. Norway has traditionally held a leading position in the field of sustainable development. With the establishment of a Ministry of Environment in 1972, it set up a comprehensive framework

ensuring compliance with elaborate environmental rules. Efforts have been made to co-ordinate environmental policies across the public administration, and a framework has been established for examining the "environmental profile" of the Budget. Since the early 1990s the government has set up several high-level committees to evaluate the cost-effectiveness of its environmental and resource management policies. Such action has helped to build a consensus about the need to integrate environmental and other structural policies, and should pave the way for a better trade-off between vested interests and the requirements of sustainable development. While direct regulation has remained an important policy instrument since the late 1980s, tax instruments, including a carbon dioxide (CO_2) tax, have played an increasingly important role in providing incentives for cleaner production and consumption patterns. Moreover, tax instruments have served to extract the "rents" generated by Norway's important oil and gas resources from the outset. As Norway is a large net importer of pollution and shares its second-most important natural resource, the fish stock, with other nations, policy has also focused on negotiating, implementing and complying with international agreements.

... including the preservation of oil and gas wealth for future generations

Norway is endowed with major crude oil and natural gas stocks on the continental shelf. In the area of oil and gas extraction the concept of sustainability allows for substitution of man-made for natural capital as a viable way to preserve the total national wealth. Indeed, rather than keeping the stock in the ground, the extraction rate is high and the oil and gas wealth has been transformed into various forms of man-made capital. The latter also comprises "social capital"; the government's oil and gas proceeds have to a considerable extent been used to sustain traditional economic and social structures in remote areas, which would otherwise have been a victim of crowding out due to the increase in the real exchange rate associated with the oil and gas wealth. Moreover, general government net financial assets have expanded since the mid-1990s through the Government Petroleum Fund, and are projected to reach 60 per cent of GDP by the year 2002. This suggests that the government's strategy to achieve sound public finances in

the long run could succeed. Vigilance is needed, however, as political pressure to unduly liquidate assets in the Fund to finance current public expenditure could re-emerge, thereby compromising the interests of future generations and contributing to greater economic volatility.

Compliance with international agreements on pollution abatement should receive top priority...

Norway stands out for its valuable efforts to stimulate international co-operation on environmental and resource management policies. This is deemed to be in the country's best interest, given its vulnerability to air and marine pollution originating in neighbouring countries. Moreover, the country – being susceptible to a relatively high marginal cost of abatement of climate gases due to its reliance on hydro electricity – has a distinct interest in international agreements that induce stronger abatement efforts in countries where marginal abatement costs are lower. Norway has succeeded in fulfilling its commitments arising from all international agreements to date – including those on acidification, ozone layer depletion and marine eutrophication – with the exception of the targets concerning emissions that contribute to tropospheric ozone formation. Climate change policies in the years ahead will be geared towards meeting the Kyoto Protocol targets established in 1997. A new set of instruments to achieve these targets will be designed in the course of 1999, and is likely to include the creation of a domestic market for tradeable emission quotas. If no global trading system is implemented, Norway should still aim to adopt an economy-wide national trading system or extend the CO_2 tax to the currently exempted industries so as to minimise the costs of meeting the targets.

... together with improving the local environment and ensuring biodiversity

Improvements in Norway's local environment result from long established national policies, as well as policies geared towards complying with international agreements, or stem from the adoption of EU regulations following Norway's membership in the European Economic Area (EEA). In addition, hazardous substances have been banned, the use of leaded petrol significantly reduced and waste reduction and treatment policies are well advanced. Policy also aims at protecting endangered and vulnerable species and to preserve biodiversity, for instance by expanding national parks. However, even in a sparsely

populated country like Norway, loss of biological diversity due to, for example, the expansion of towns and other changes in land use, is significant. Moreover, a few urban areas in Norway occasionally face serious air and noise pollution associated with car traffic and heating, which is only in part explained by the geographical characteristics of these areas. Hence, there remains significant scope for enhancing the environmental sustainability of local economic activities. Increased use of long-term planning guided by extensive cost-benefit analysis should be instrumental in achieving this.

There is also scope for greater cost-effectiveness in environmental and natural resource policies

There is scope for greater cost-effectiveness in a number of areas even though this may compromise other policy objectives, in particular, the preservation of traditional economic activities on Norway's mainland. Such potential resides in a greater use of instruments that aim to equalise marginal abatement costs and marginal benefits from abatement across all economic activities. A number of industries, such as heavy metals, fisheries and shipping, have been exempted from several "green taxes", including the CO_2 and sulphur tax, in order to support economic activity often located in remote areas. The removal of such exemptions would contribute to lowering the overall cost of abatement, the more so if tax rate differentiation better reflected the content of polluting substances in products and activities. Cost-effectiveness could be enhanced also in a number of domains by moving from "command-and-control" to economic instruments, including green taxes for all mobile sources of pollution, deposit-refund schemes and tradeable emission permits. In the same spirit, to the extent regional considerations allow this, tradeable catch quotas could be introduced in the fishing industry in order to facilitate a reallocation of quotas to the more efficient parts of the fleet. Given the sheer size of the oil and gas wealth, raising cost-effectiveness of oil and gas extraction could be very beneficial; this has, therefore, been the guiding principle for the licensing of oil and gas fields. While the allocation of acreage on the Norwegian continental shelf between oil companies was improved in 1998 by abandoning the system of "arranged marriages" also in the North Sea, this reform could be extended to the whole drilling area.

Summary

In sum, the boom that has been ongoing since 1993 began to unwind in 1998, as capacity limits were reached, oil prices plummeted and international financial turbulence hit the economy. A return to sustainable, non-inflationary growth requires a bolstering of the macroeconomic policy framework. In particular: fiscal policy needs to reassert its role of stabilising demand; monetary policy needs to remain tight in the present juncture to underpin exchange rate stability; and incomes policy must deliver moderate wage growth. Over the longer haul, labour market reform should aim to reduce the flow of labour resources into the generous welfare system, while other structural policies, including environmental, resource management and regional policies, could be more cost-effective. Indeed, while the huge resource wealth has contributed to raising living standards in Norway to one of the highest in the OECD area, staying ahead and developing the full potential of the economy will require further efforts to enhance economic efficiency.

I. Macroeconomic performance

After over five years of vigorous growth, the Norwegian economy is headed for a slowdown. Investment appears to have peaked in the residential and mainland business sectors as well as the offshore oil and gas sector. While private consumption has remained buoyant due to robust employment and earnings growth to date, it is expected to weaken in the wake of sharp increases in interest rates in the late summer of 1998. Monetary policy has been considerably tightened to offset downward pressure on the currency associated with important capital outflows, which were induced by concerns over the impact of the oil price collapse on the economy, mounting wage inflation amid tight labour market conditions and concerns over the stance of fiscal policy.

The Norwegian business cycle, once again, is shaped to a large extent by developments in global energy markets, in particular the crude oil market. The 1974 and 1979 oil shocks, which practically coincided with major oil discoveries on the Norwegian continental shelf, propelled the mainland economy into a pronounced upswing which lasted roughly until the 1986 oil price collapse. Norway subsequently entered into a protracted slump, from which it only began to recover in 1993, spurred by easier monetary conditions after the financial turmoil in Europe in late 1992. While the recovery gathered momentum after the mid-1990s, soaring oil prices, associated, *inter alia*, with the Iraq boycott and the upswing in the emerging economies, lifted economic growth in Norway to unsustainable heights. This happened through three main channels: *i)* ballooning government revenues from the oil and gas activities eventually contributed to an insufficiently tight fiscal policy; *ii)* the huge current account surplus prompted an easy stance of monetary policy to avoid an appreciation of the currency that could have led to a squeeze of "traditional" mainland industries exposed to international competition; and *iii)* the oil and gas sector itself provided a strong demand impetus through investment and other spending. As in 1986, the recent oil price collapse occurred at a critical time, when the labour market was overheated. As a result, economic developments over the last twelve months or so have been marked by a combination of several adverse shocks: deteriorating terms of trade

coupled with accelerating wages squeezed profits, interest rates soared and the boom in asset markets unwound. As a result, overall sentiment weakened.

While there are significant risks that the Norwegian economy will move into a recession, especially if external conditions continue to weaken, there is also ground for optimism. As compared with the mid-1980s, the balance sheets of households, businesses and financial institutions are in much better shape after a decade of wage moderation, rationalisation and net financial asset formation, which should provide a strong buffer against a cyclical downswing. The government also contributed in this regard, by establishing the Petroleum Fund, which has been used as a vehicle to transform the fiscal surpluses of recent years into foreign financial assets – an arrangement which now proves its usefulness as a "hedge" against a decrease in the value of domestic financial assets. In addition, the huge official reserves built up while oil exports were booming, have facilitated the efforts to stabilise the currency in a credible manner. A prerequisite for an easing in the macroeconomic policy stance and an associated "soft landing" of the economy is that the social partners internalise the recession risk in the wage negotiations, due in spring 1999, and agree on ways to arrest the current upward momentum of wages. The government, in turn, needs to continue to keep a tight rein on public expenditure, following the adoption of the relatively tight 1999 Budget in order to avoid further pressure on the over-stretched labour market. An easing of wage pressure, which is currently most pronounced in the sheltered sectors of the economy, would contribute to restoring the confidence of financial markets and allow a decline in short-term interest rates from their present level of around 8 per cent. The sections that follow discuss the recent macroeconomic developments and policy responses in more detail. The chapter concludes with a review of the economic prospects for 1999 and 2000 and an assessment of the risks surrounding these projections, and considers options for the conduct of macroeconomic policies in the years ahead.

The economy has peaked

The economic expansion underway since 1993 showed first signs of weakening in the course of 1998. Although mainland GDP growth has remained robust at about 3 per cent in 1998, edging down from 3¾ per cent the year before, this reflected in part the significant carry-over from the very strong second half of 1997 (Table 1 and Figure 1, Panel A). Buoyant private consumption and surging investment in the oil and gas sector continued to boost mainland activity in 1998, but a downturn in residential and other construction activity began to unfold while "traditional" (non-oil) commodity exports were hit by the crisis in emerging market economies. As a result, the mainland economy has become increasingly consumption driven. Moreover, with the upward trend in oil production stalling

Table 1. **Demand and output**

Percentage change from previous period, volume

	1995 Current prices NKr billion	1996	1997	1998[1]	Seasonally adjusted, annual rate		
					1997 2nd half	1998 1st half	1998[1] 2nd half
Private consumption	458.5	4.7	3.4	3.7	4.2	3.9	2.9
Government consumption	194.5	3.2	3.0	2.4	2.6	3.0	1.0
Gross fixed capital formation	192.5	9.6	12.6	5.0	8.8	5.2	1.1
Public sector	29.9	3.7	12.1	0.3	–7.4	20.6	–25.6
Petroleum production and pipeline transport	47.9	1.5	15.5	12.5	5.7	31.1	–12.0
Residential	26.5	–1.2	9.0	2.0	7.7	–0.4	1.3
Other private[2]	88.2	19.3	12.4	4.3	16.0	–8.6	21.8
Stockbuilding[3,4]	27.4	–0.6	0.2	0.5	1.2	0.5	–0.6
Total domestic demand	873.0	4.7	5.6	4.2	6.2	4.5	1.7
Exports of goods and services	353.4	9.8	5.8	2.4	3.2	2.0	2.4
Traditional goods	143.4	10.0	8.0	4.3	7.7	2.6	4.4
Crude oil and natural gas	113.2	15.6	2.3	–1.4	–0.9	0.2	–5.0
Ships and platforms	10.6	–16.2	11.7	–7.1	–40.7	55.6	–49.5
Services	86.2	5.2	6.3	5.3	6.9	–0.7	16.4
Imports of goods and services	297.7	8.3	12.3	7.2	8.1	9.9	1.2
Foreign balance[3]	55.8	1.1	–1.8	–1.5	–1.5	–2.7	0.5
GDP	928.7	5.5	3.4	2.3	4.3	1.6	1.8
of which: Mainland	790.1	4.1	3.7	2.9	5.0	2.2	2.3

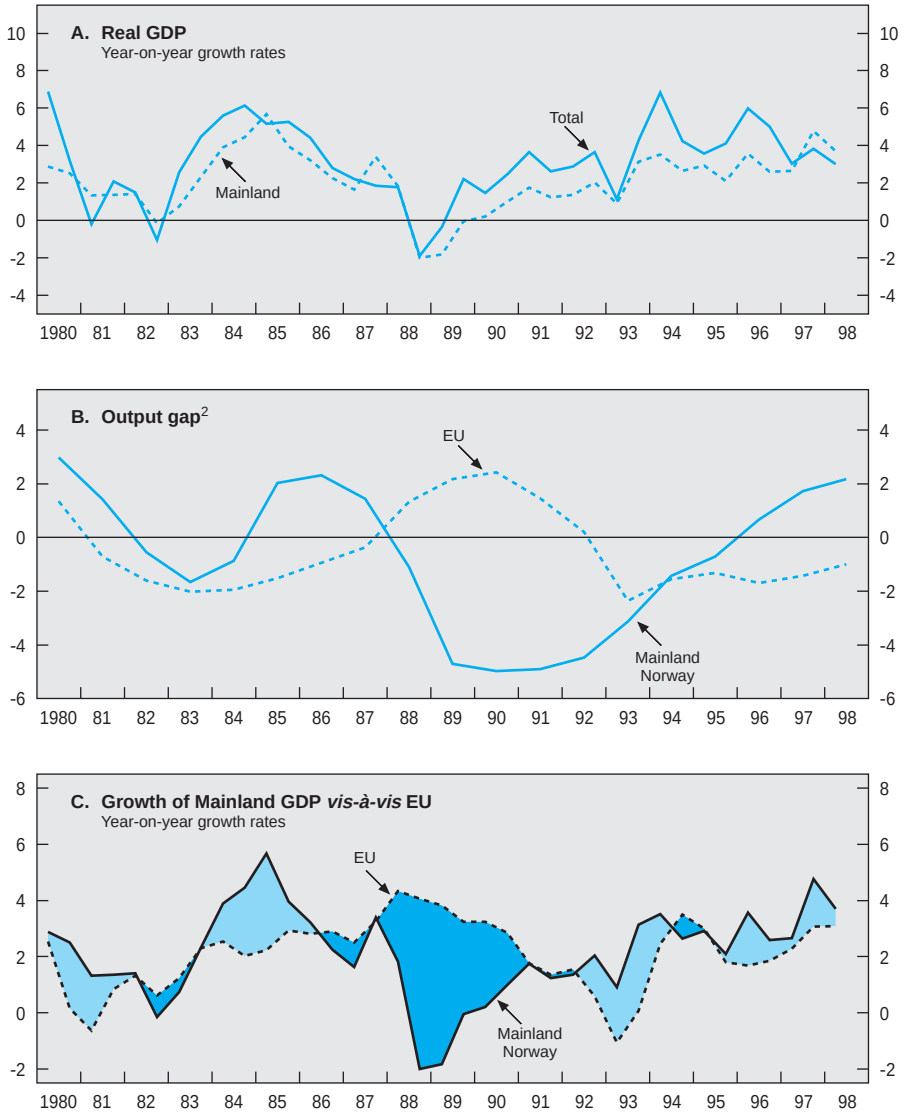
1. OECD Secretariat estimates.
2. Including shipping.
3. Contribution to GDP growth.
4. Includes statistical discrepancy.

Source: Statistics Norway and OECD Secretariat.

and oil prices plummeting, the current account surplus virtually disappeared in 1998 while growth of overall GDP (including oil and gas) eased to an estimated 2¼ per cent from 3½ per cent in 1997. The extent to which the Norwegian economic cycle has remained out of tune with the EU area is striking. Indeed, after Norway had posted much more rapid economic growth than the EU in the 1993-97 period, the growth differential has now virtually disappeared and may be set to turn negative in the near future (Figure 1, Panels B and C).

With the mainland economy growing strongly, the excess of actual over potential output has continued to widen to around 2 per cent in 1998, suggesting that the economy has stretched its capacity limits (Figure 1, Panel B). The constraints are primarily related to very tight labour market conditions, amid fairly

Figure 1. **Growth performance**[1]
Per cent



1. Mainland GDP excludes exports and imports of petroleum, oil rigs and ships.
2. The difference between actual and trend GDP.

Source: OECD, *Quarterly National Accounts* and OECD Secretariat.

high capacity utilisation rates overall. Indeed, the unemployment rate has averaged less than 3½ per cent in 1998, below the OECD Secretariat's estimate of the non-accelerating wage rate of unemployment (NAWRU) of around 4 per cent. The emerging labour scarcity is reflected in an increasing number of employers reporting difficulties in recruiting qualified labour, soaring unfilled vacancies and a rapid acceleration of wage growth to around 6 per cent in 1998. Wage developments in 1998 have begun to undermine the profitability of mainland industry, even though the more than 10 per cent depreciation of the currency since the beginning of the year has more than offset the wage increases. At the same time, narrowing profit margins, coupled with virtually stable import prices, have helped to keep consumer price inflation within the 2 to 2½ per cent range.

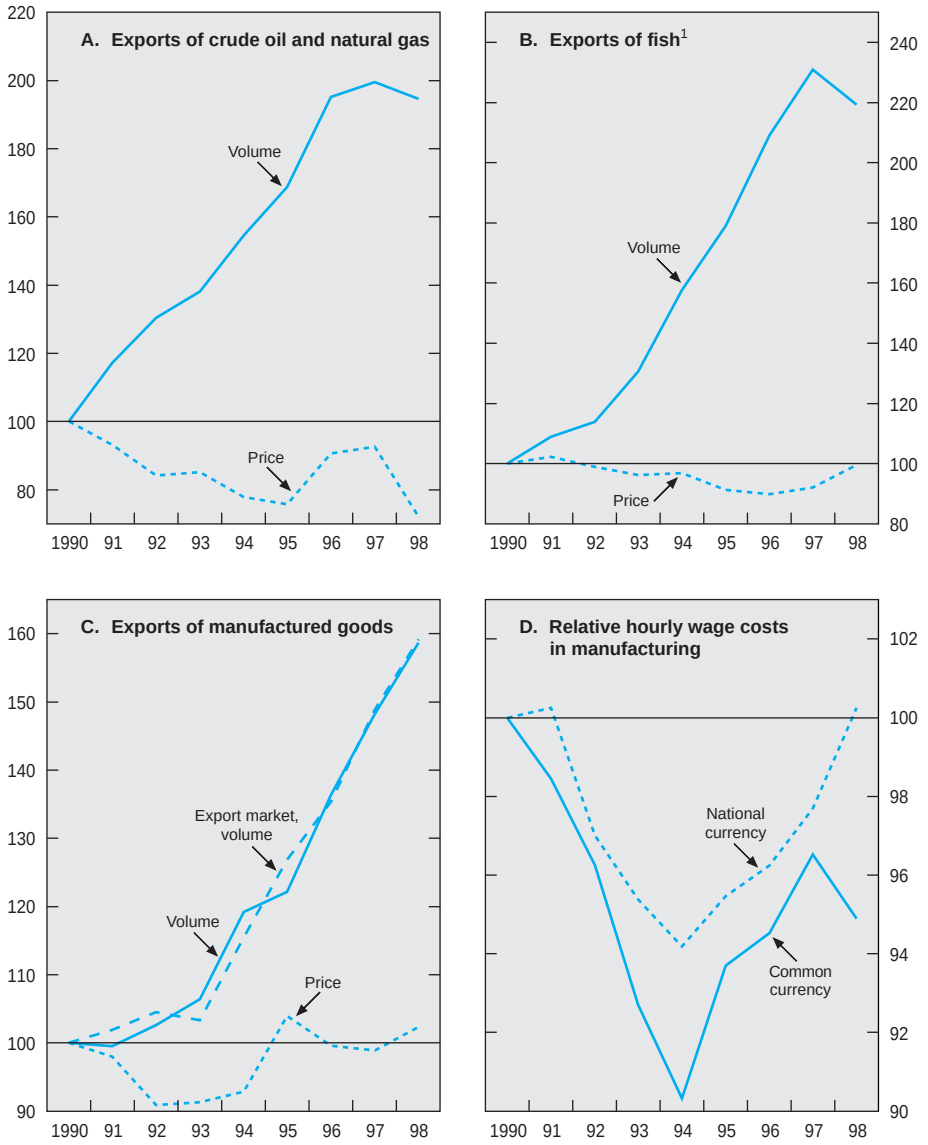
External conditions have weakened

Oil and gas production, which represents roughly one-third of Norwegian exports, showed practically no change between 1996 and 1998 after expanding at an annual rate of around 10 per cent over the 1993-96 period (Figure 2, Panel A). The hike in oil and gas activity experienced in recent years was underpinned by high oil prices, with the Brent Blend averaging slightly over NKr 120 (US$18) per barrel between 1993 and 1997, and strong world demand, fuelled by developments in the Asian and other emerging economies. However, the oil bonanza is now over. The oil price fell by almost 30 per cent in 1998, leaving the Brent Blend at around NKr 100 (US$12) per barrel in November 1998. While oil extraction has been roughly stable in the first half of 1998, low crude oil prices and a government decision to reduce production by 100 000 barrels per day effective from 1 May until the end of 1998 – following production cuts by other oil exporting countries – contributed to a decline in the value of oil exports in the second half of the year.

Growth in "traditional" non-oil exports almost halved from 8 per cent in 1997 to 4¼ per cent in 1998 as several categories of commodity exports were hit by the crisis in emerging economies. Traditional merchandise exports of goods grew at an annual rate of 7½ per cent between 1993-97, with the impetus provided by a pronounced expansion in world markets and despite a decline in cost competitiveness (Table 1; Figure 2, Panels C and D). In recent years, export growth has been particularly strong in refined oil, basic metals, machinery and chemicals, while exports of fish – the second largest export category in Norway after oil and gas – also posted a considerable expansion as catch quotas were raised and the farmed salmon industry boomed (Figure 2, Panel B). The crisis in Asia and other emerging economies has been reflected also in weak spot prices for many Norwegian export products, including aluminium and other metals where spot prices have come down some 30 per cent from the mid-1997 peak. Nevertheless, non-oil export prices remained stable or even increased slightly during 1998 in krone-terms due to the depreciation of the currency and time lags

Figure 2. **Export performance and cost competitiveness**

1990 = 100



1. Fresh, cooled or frozen fish.

Source: Statistics Norway, Ministry of Finance and OECD Secretariat.

between spot and contractual prices. Evidence provided by business surveys suggests that labour supply has become a serious constraining factor limiting manufactured exports and, as a result, export industries have not been able so far to reap the benefits of improved cost competitiveness associated with the recent depreciation of the currency, which more than offset the impact of accelerating wages (Figure 2, Panel D).

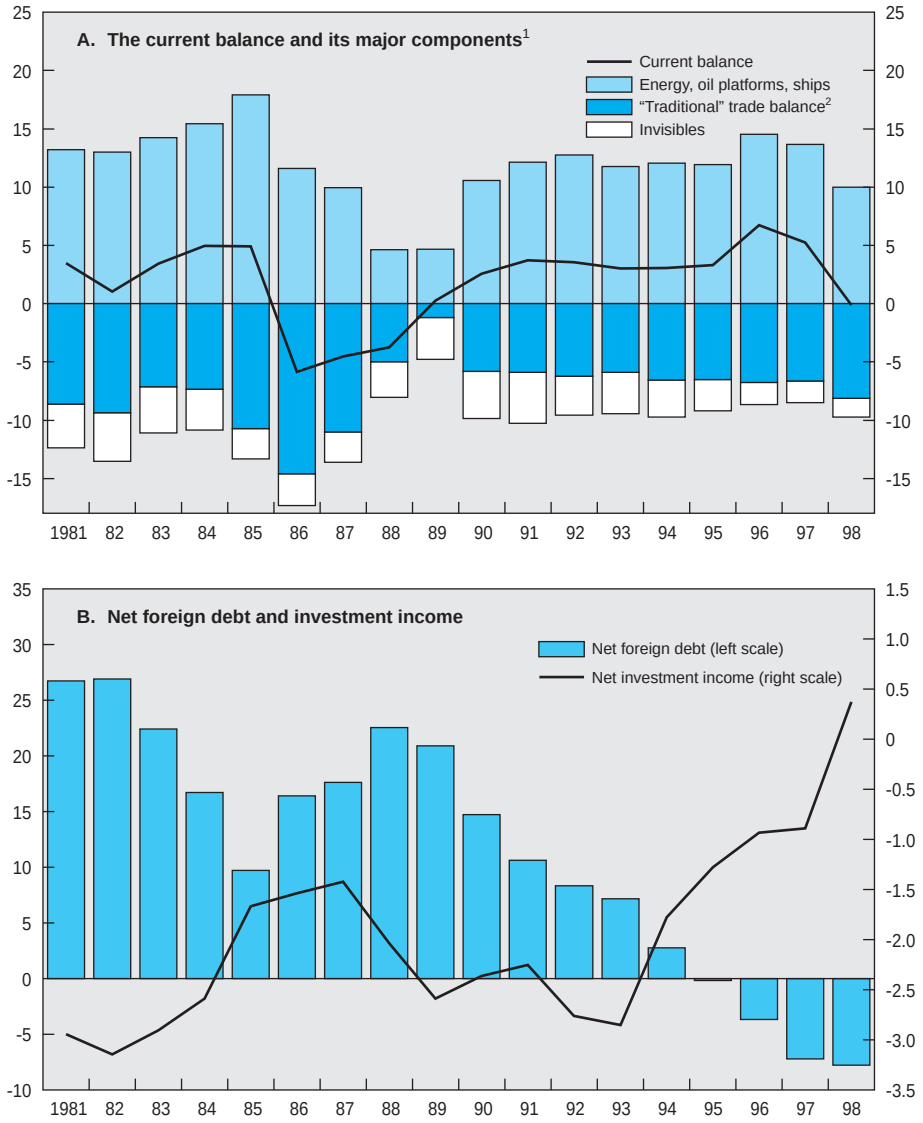
After ballooning to 6 per cent of GDP on average in 1996 and 1997, the current account surplus has almost disappeared due mostly to the decline in the "offshore" trade surplus (petroleum, oil platforms and ships), which narrowed from an average of almost 15 per cent of GDP in 1996 and 1997 to 10 per cent in 1998. The weak "traditional" trade balance, which posted an estimated 8 per cent of GDP deficit in 1998, reflected not only slack exports, but also surging imports as strong household demand spilled over into purchases abroad due to capacity constraints until mid-1998 (Figure 3, Panel A). Owing to the huge current account surplus in past years, Norway's net foreign asset position has swung to an estimated 7½ per cent of GDP in 1998 after hovering around a 20 per cent net debtor position in the 1980s and early 1990s (Figure 3, Panel B). Until 1997 the current account surpluses were accompanied by substantial capital outflows, including both direct and portfolio investment. In 1997, however, this trend ended abruptly because speculation on an appreciation of the krone prompted substantial capital inflows, and official reserves soared due to interventions by the Bank of Norway. In 1998, this trend was again reversed as the decline of the current account surplus coincided with strong net capital outflows, notably in the second half of the year, triggering a fall in official reserves due to offsetting interventions. Although Norway has been a net foreign creditor since 1995, its net foreign interest payments turned positive only in 1998, as high interest-rate loans were redeemed.[1]

The asset boom is unwinding

Norway, as other OECD countries, has experienced a sharp turnaround in asset price developments in the second half of 1998. The stock market has been particularly hard hit. After the drop in oil prices and the acceleration in wage inflation had weakened the profit prospects of companies operating both on the mainland and the continental shelf, domestic investors began to liquidate their domestic-currency denominated financial assets which, as noted, sparked substantial capital outflows. Due to continued capital flight later in 1998, including by foreign investors, the Oslo stock index fell 37 per cent from its peak in early May to December 1998 (Figure 4), outstripping the drop in most other stock markets in the OECD area. Since then, the Norwegian stock market index hardly recovered unlike developments elsewhere. Bond prices have also fallen, with the long-term

Figure 3. **External position**

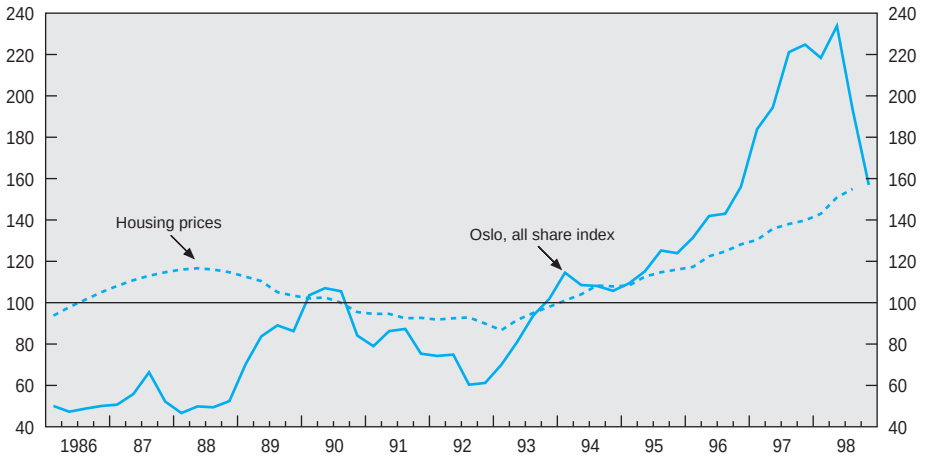
Per cent of GDP



1. Break in series in 1990. Figures for 1998 are estimates based on data up to October.
2. Excluding exports and imports of petroleum, oil rigs and ships.

Source: Statistics Norway and OECD Secretariat.

Figure 4. **Asset prices**
1990 = 100



Source: Ministry of Finance and OECD, *Main Economic Indicators.*

interest rate moving up by a ¼ percentage point since the beginning of the year to around 5½ per cent by December 1998. As a result, the differentials *vis-à-vis* equivalent interest rates in Germany and the United States have moved from around 20 and –50 basis points to 220 and 110 basis points, respectively, over the same period. House prices, after a brisk rise during the past five years, have not declined so far, but commercial activity in the secondary housing market has been muted, suggesting a turning point. Moreover, Norway's National Association of Real Estate Agents expects a sharp drop in house prices during the winter of 1998/99.

Capital formation has slowed down

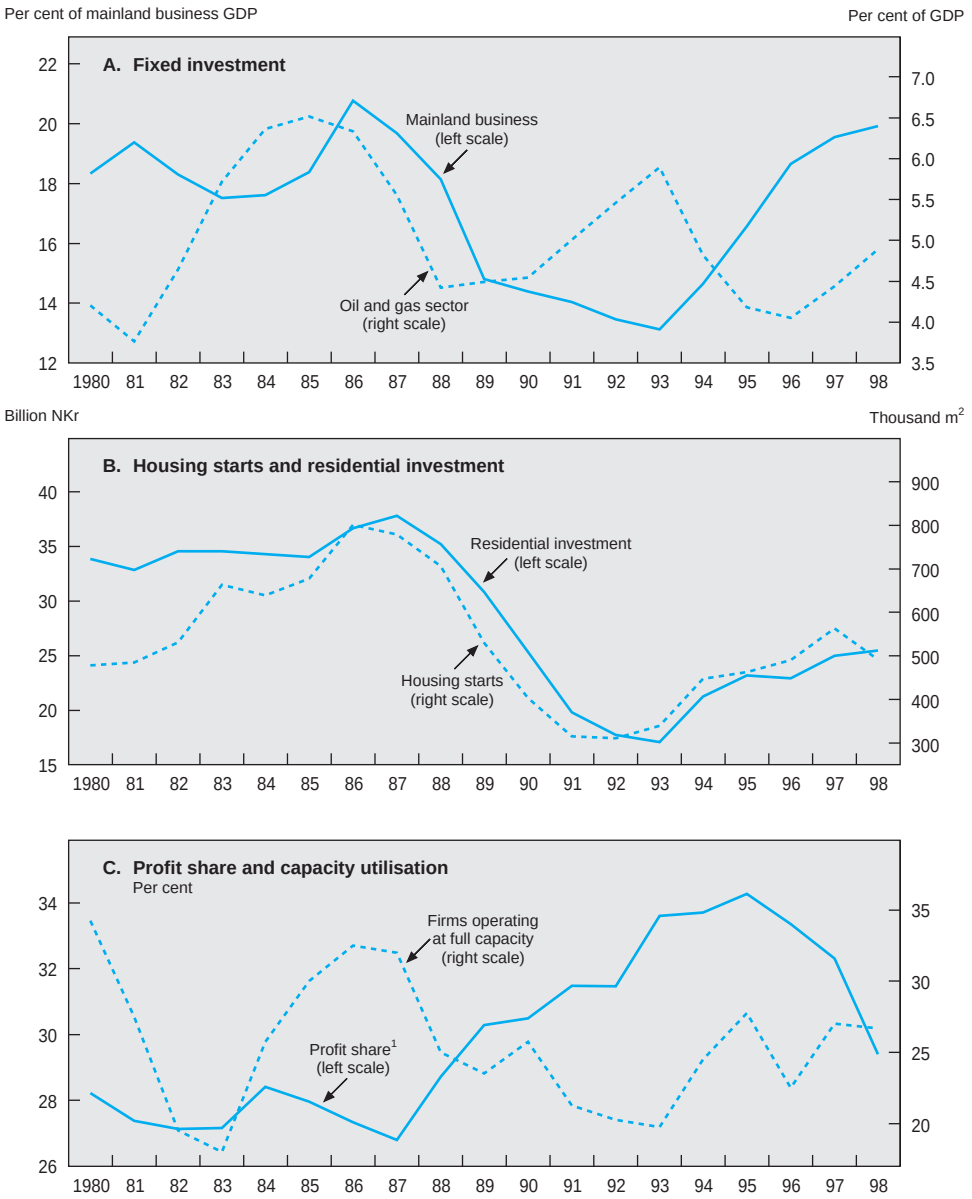
Gross capital formation has lost momentum in 1998, growing by an estimated 5 per cent for the whole year compared to 12½ per cent the year before. In the second half of 1998 investment growth slowed down to an estimated 1 per cent, as a sharp reduction in public sector investment, due to the completion of major government investment projects – such as the Oslo airport at Gardermoen and the National Hospital – was coupled with weaker private investment. In particular:

– growth of mainland business investment slowed down from 12½ per cent in 1997 to 4½ per cent in 1998. This has been associated with the completion of major investment projects in tandem with the drop in public investment, as well as a sharp reversal of earlier increases in profitability due to accelerating wage growth (Figure 5, Panel A). Recent investment surveys, moreover, show downward revisions of planned manufacturing investment;
– while house prices continued to rise in the first half of 1998, construction starts fell sharply, contributing to an estimated slowdown in the annual growth rate of residential investment from 9 per cent in 1997 to 2 per cent in 1998 (Figure 5, Panel B). The downward trend began well before the recent interest rate hike and has been associated with labour market constraints, administrative delays in the issuing of construction permits and developers anticipating a slowdown in economic activity;
– investment in petroleum activities – which accounts for nearly a quarter of total fixed investment – posted an estimated 12½ per cent increase for 1998 as a whole, but investment intentions fell sharply in the second half of the year. While the government took measures to postpone petroleum investment by one year in order to cool off mainland economic activity, investment cuts will be mostly felt in 1999. Indeed, weaker oil market conditions prompted many companies to cut back investment plans, especially on exploration activities.

Household consumption has stayed buoyant amid weaker sentiment

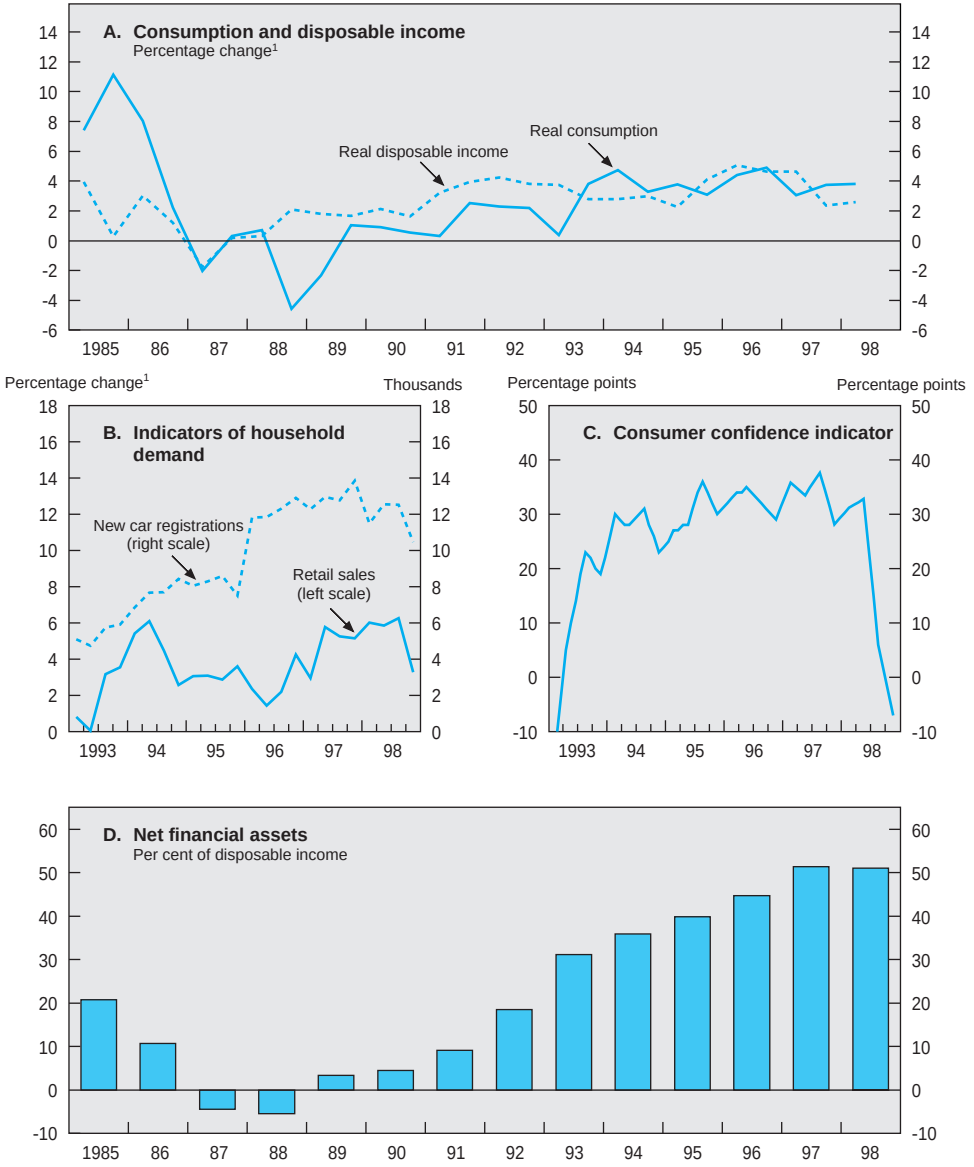
Private consumption grew by an estimated 3¾ per cent in 1998, broadly matching the 4 per cent annual growth over the 1993-97 period (Table 2; Figure 6, Panels A and B). Spurred by strong real wage and employment growth, households' real disposable income expanded by around 5 per cent in 1998, about 1½ percentage points above the average over the last five years. Households' disposable income and consumption have grown broadly in concert for several years, with the household saving ratio remaining broadly stable at around 6 per cent (Table 2). As a result, households' balance sheets have continued to strengthen. Net financial assets relative to disposable income reached 51 per cent in 1998, considerably exceeding the 20 per cent peak achieved in the mid-1980s and sharply contrasting with the 1988 low of –5½ per cent (Figure 6, Panel D). Despite the comfortable net financial asset position, recent indicators suggest that the growth in consumption may be about to edge down. Indeed, household confidence, after soaring unabated since the upswing started in 1993, declined markedly following the turbulence in financial markets over the summer of 1998 (Figure 6, Panel C).

Figure 5. **Residential and business investment**



1. Mainland business GDP less compensation as a percentage of mainland business GDP.
Source: Statistics Norway; OECD, *Main Economic Indicators* and OECD Secretariat.

Figure 6. **Household consumption**



1. Percentage change over same period of previous year.

Source: Statistics Norway; Ministry of Finance; OECD, *National Accounts* and *Main Economic Indicators.*

Table 2. **Household appropriation account**

Percentage changes

	1992 NKr billion	1993	1994	1995	1996	1997	1998[1]
Disposable income	417.1	5.4	3.6	5.6	6.4	6.6	5.7
Contributions from:							
Wages	319.9	2.7	3.6	4.5	5.7	6.2	..
Property income	87.0	1.8	0.9	1.1	0.7	1.2	..
Current transfers received (net)	150.7	2.0	0.2	1.2	2.3	1.8	..
Interest on consumer debt (net)	–19.4	0.6	1.0	0.9	0.3	0.2	..
Direct taxes	–121.0	–1.7	–2.0	–2.2	–2.6	–2.7	..
Real disposable income	393.6	3.2	2.4	3.1	4.9	4.1	3.2
Real consumption expenditure	370.4	2.0	3.9	3.4	4.7	3.4	3.7
Saving ratio[2]	5.9	6.9	5.9	5.8	6.1	6.6	6.3
Net financial assets[2]	..	31.2	35.9	39.8	44.7	51.4	51.0

1. Estimates.
2. Per cent of disposable household income.

Source: Statistics Norway and OECD Secretariat.

The labour market is still very tight

Employment growth has tapered off in the course of 1998, but is still set to reach 2 per cent for the year as a whole (Table 3; Figure 7, Panel A). Employment has increased by around 11 per cent over the 1993-98 period, boosting the employment rate (the ratio of the number of employed persons to the working age population) to 78 per cent. This is 6½ percentage points above the low reached in 1993 and it is now among the highest in the OECD area. During the upswing, job creation was most pronounced in the private sector, although employment in this sector has remained below its 1987 peak. With the ongoing expansion of the health care and education systems, leading to sharp increases in expenditure by local governments, public employment remained on a strong upward trend.

Employment growth has been largely matched by a flexible response of labour supply. The labour force increased by over 8 per cent between 1993 and 1998 and the labour force participation rate reached a record high of 80½ per cent, 4 percentage points above the 76½ per cent trough in 1993. The labour force growth has, however, shown a distinct slowdown in 1998 and, as a result, labour shortages are increasingly felt. Over one-third of all companies reported recruitment problems in 1998 and more than half of the enterprises in the machinery and construction industries. Bottlenecks have also been a major problem in

Table 3. **Labour market developments**

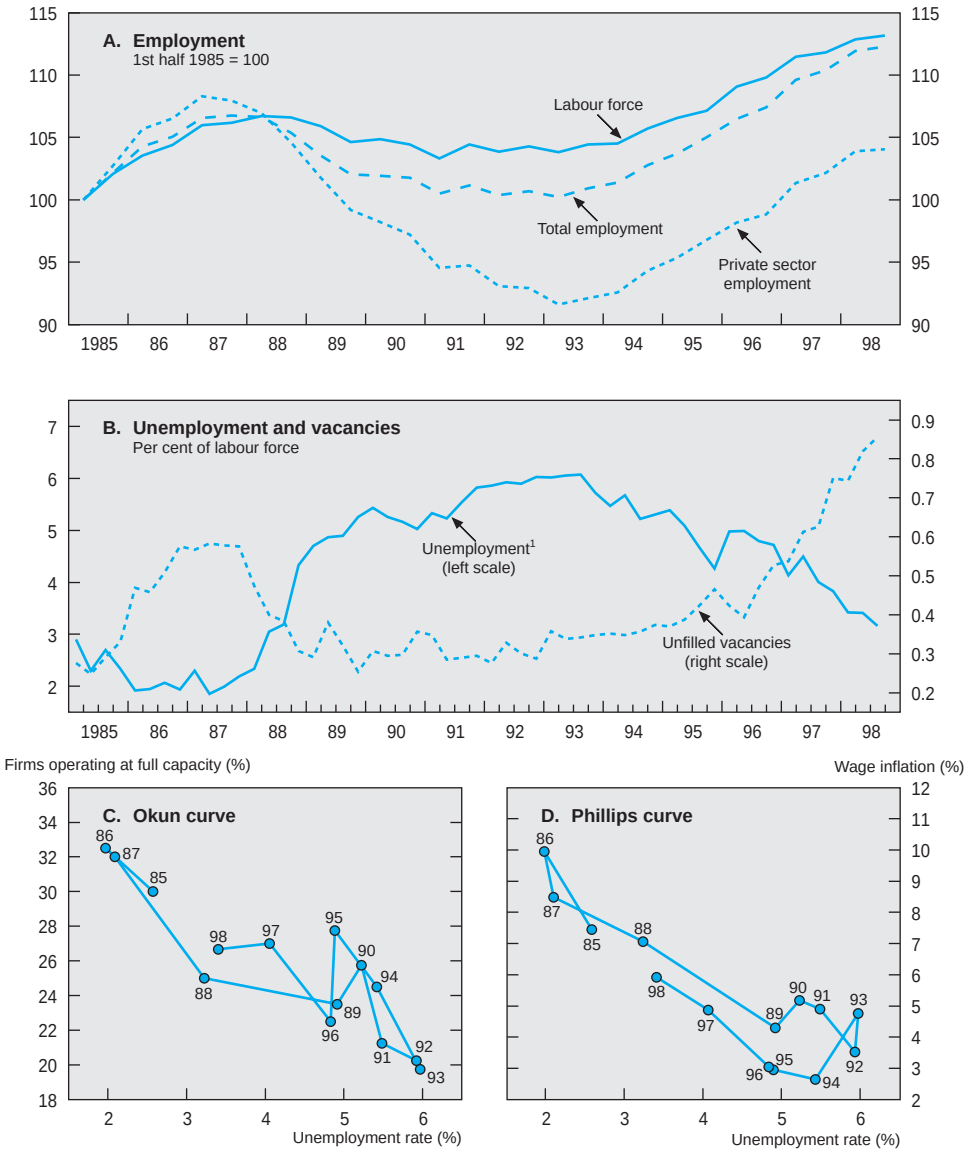
	1992 Thousand persons	1993	1994	1995	1996[1]	1997	1998[2]
		Percentage change					
Labour force	2 130	0.1	0.9	1.7	2.4	2.0	1.2
Employment	2 004	0.0	1.5	2.2	2.5	2.8	1.9
GDP mainland	..	2.8	4.1	3.1	4.1	3.7	2.9
Labour productivity	..	2.7	3.9	1.6	2.9	0.6	0.3
		Per cent of labour force in same category					
Unemployment	126	6.0	5.4	4.9	4.8	4.1	3.4
By sex							
Male	76	6.6	6.0	5.2	4.8	4.0	3.4
Female	50	5.2	4.7	4.6	4.9	4.2	3.6
By age group							
16-24	44	13.7	12.6	11.8	12.4	10.9	10.3
25-54	74	5.0	4.5	4.1	3.8	3.2	2.4
55-74	8	2.5	2.5	2.6	2.4	1.8	2.0
By duration							
Persons unemployed for more than 6 months (as a per cent of total)	45	39	38	37	29	25	20
		Per cent of working age population[3]					
Active labour market programmes							
Participation	..	1.8	1.7	1.3	1.1	0.7	0.5
Including special measures	..	3.4	3.4	2.9	2.5	2.0	1.8

1. Break in series for Labour Force Survey data.
2. Estimate based on average of first three quarters.
3. Population aged 16-74.

Source: Statistics Norway, Ministry of Labour and Government Administration, and OECD Secretariat.

health and social care as well as in the forestry industry. These shortages are reflected in the sharp rise in the number of unfilled vacancies during 1998 (Figure 7, Panel B). The unemployment rate, already very low relative to the average in Europe, continued to decline to just below 3½ per cent on average in 1998 – 3 percentage points down from the 1993 peak level (Figure 7, Panel B). The scope for further increases in employment is, thus, almost exhausted. Indeed, unemployment spells have continued to shorten, as the incidence of long-term unemployment declined further to 20 per cent in 1998, half the level of the 1993 trough, with most of the decline among the young and prime-age workers.

Figure 7. **Labour market**



1. Break in series in January 1996 due to a broadening of the definition of unemployment in the Labour Force Survey which is estimated to have raised the unemployment rate by 0.5 percentage point.

Source: Statistics Norway; OECD, *Main Economic Indicators, Quarterly Labour Force Statistics* and OECD Secretariat.

With the remaining long-term unemployment concentrated among older workers, reflecting skill-mismatches and low incentives for job search, there appears to be only a limited scope for a further reduction in long-term unemployment. Other sources for increased labour inputs have also become very limited. For example, the incidence of part-time and temporary employment has sharply declined in 1998 as many of these workers moved to steady, full-time jobs, after the number of part-time workers had rapidly increased until 1997 due to the entry of large cohorts of young people and women into the labour market. Moreover, the scope for further increases in cross-border labour supply from the other Nordic countries, which has made a substantial contribution over the last two years, may be exhausted – the more so since the labour markets in these countries are also becoming tighter.

At present, the unemployment rate has dropped below most estimates of the structural rate. The consensus estimate of the structural unemployment rate by the social partners in Norway is around 3½ to 4 per cent, close to the OECD Secretariat's estimate of a NAWRU of 4 per cent. This indicates that a further reduction in unemployment would be likely to contribute to wage inflation. In fact, the "Phillips" and "Okun" curves depicted in Figure 7, Panels C and D, suggest that further significant reductions in unemployment associated with higher rates of capacity utilisation, to levels prevailing in the mid-1980s, would risk boosting wage inflation to two-digit levels.

Incomes policy fails to contain wage demands

Norway has a long-standing tradition of centralised wage bargaining (Thomas, 1998). Since the early 1960s, the contract period has been for two years, with intermediate bargaining after twelve months to adjust to, for example, changes in the outlook for inflation. Since 1973 the government has often played an active role in wage negotiations. To help induce the unions to accept moderate wage agreements, the government offered price subsidies and lower taxes in the period 1973-77 and a stepwise reduction in the early retirement age (the AFP scheme, see Chapter II) towards the end of the 1980s. Moreover, whereas the government has not been involved in wage negotiations in the period 1980-87, it legislated wage freezes in the late 1970s and late 1980s. During the recession of the early 1990s, a broad agreement emerged among the unions, the employers and the government on the need for even more co-operation in incomes policies. This led to the publication of a policy document in August 1992 that paved the way for a tripartite incomes policy framework, known as the *Solidarity Alternative*, aiming at bolstering international competitiveness through lower wage growth in Norway than in its major trading partners. Both the unions and the employers' organisation have consistently emphasised the contingency of their co-operation

with the government on the continuation of the long-standing tradition of a stable exchange rate policy *vis-à-vis* Norway's major trading partners. The government, therefore, agreed to gear monetary policy to keep the exchange rate stable and to use fiscal policy to smooth domestic demand. The stable exchange rate policy, in turn, was deemed to prompt unions in the exposed sector to internalise profit considerations into their wage demands.

While substantial slack remained in the labour market in the first half of the 1990s, moderate wage agreements were reached with minimal government concessions on taxes or expenditures. Since labour shortages emerged, however, moderate wage agreements have become harder to reach. During the 1997 intermediate wage round, a lowering of the entry age to the early retirement pension was necessary to strike a moderate wage accord. During the spring 1998 main wage round, the Labour Union organisation (LO) and the Confederation of Norwegian Business and Industry (NHO) reached a central agreement for an action plan on a so-called *Competence Reform*, aimed at providing education leave for workers. With the labour market very tight and economic prospects still very favourable at the time, the LO was unable to hold back calls for higher wage increases by independent unions and its own "rank and file", and finally agreed to allow branch level wage negotiations. Hourly wages, as a result, accelerated from the 4½ to 5 per cent range in 1997 to 6 per cent in 1998 (Table 4 and Figure 8, Panel A). Moreover, as pay increases agreed during the spring partly took effect later in the year, there is a significant carry-over effect on 1999 wage growth, on average of the order of 3 to 3¼ per cent.

Table 4. **Prices, wages and costs**

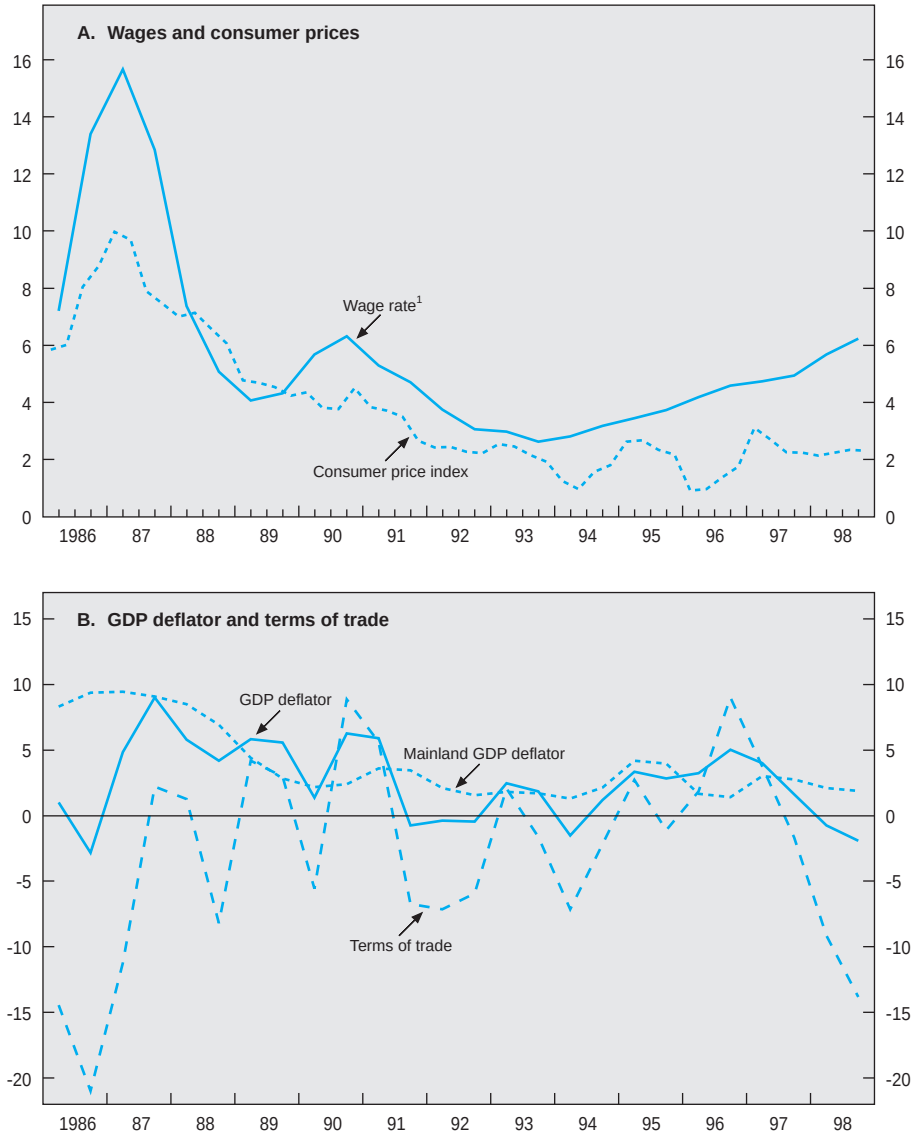
Percentage change from previous year

	1992	1993	1994	1995	1996	1997	1998[1]
Consumer price index	2.3	2.3	1.4	2.5	1.3	2.6	2.3
Hourly earnings per employee	2.7	3.5	3.0	4.6	4.7	5.0	6
of which:							
Mainland	2.7	3.3	3.0	4.6	4.7	4.9	6
Manufacturing	2.8	2.8	2.9	4.6	4.7	4.5	6
Personal services	2.9	3.0	2.8	4.7	4.4	4.5	6½
Public sector	2.8	3.4	3.2	4.6	4.6	4.9	6
Relative compensation per worker[2]							
Local currency	–3.2	–1.7	–1.3	1.3	0.8	1.6	2¾
Common currency	–2.2	–3.7	–2.6	3.7	0.8	2.2	–1¾

1. Estimate.
2. Manufacturing.

Source: Bank of Norway, Ministry of Finance and OECD Secretariat.

Figure 8. **Wage and price developments**
Per cent change over corresponding period of previous year



1. Manufacturing.
Source: OECD, *National Accounts* and *Main Economic Indicators.*

The ongoing hike in wage inflation has not led to higher consumer price inflation to date, which has remained moderate throughout 1998 at around 2¼ per cent, helped by a reversal of drought-related increases in hydro-electricity prices and subdued import price developments. As a result, the increase in real wages is estimated to have reached 3¾ per cent in 1998, approximately twice the average of the 1990s, the strongest since 1976, but almost matching the hike in real wages in the years 1984-86. Since the adverse terms-of-trade shock associated with the recent oil price collapse has also been of similar magnitude to that in 1986, history seems to be repeating itself in some respects, with the economy peaking under conditions of high real wage growth, falling terms of trade and eroding profits (Figure 8, Panel B).

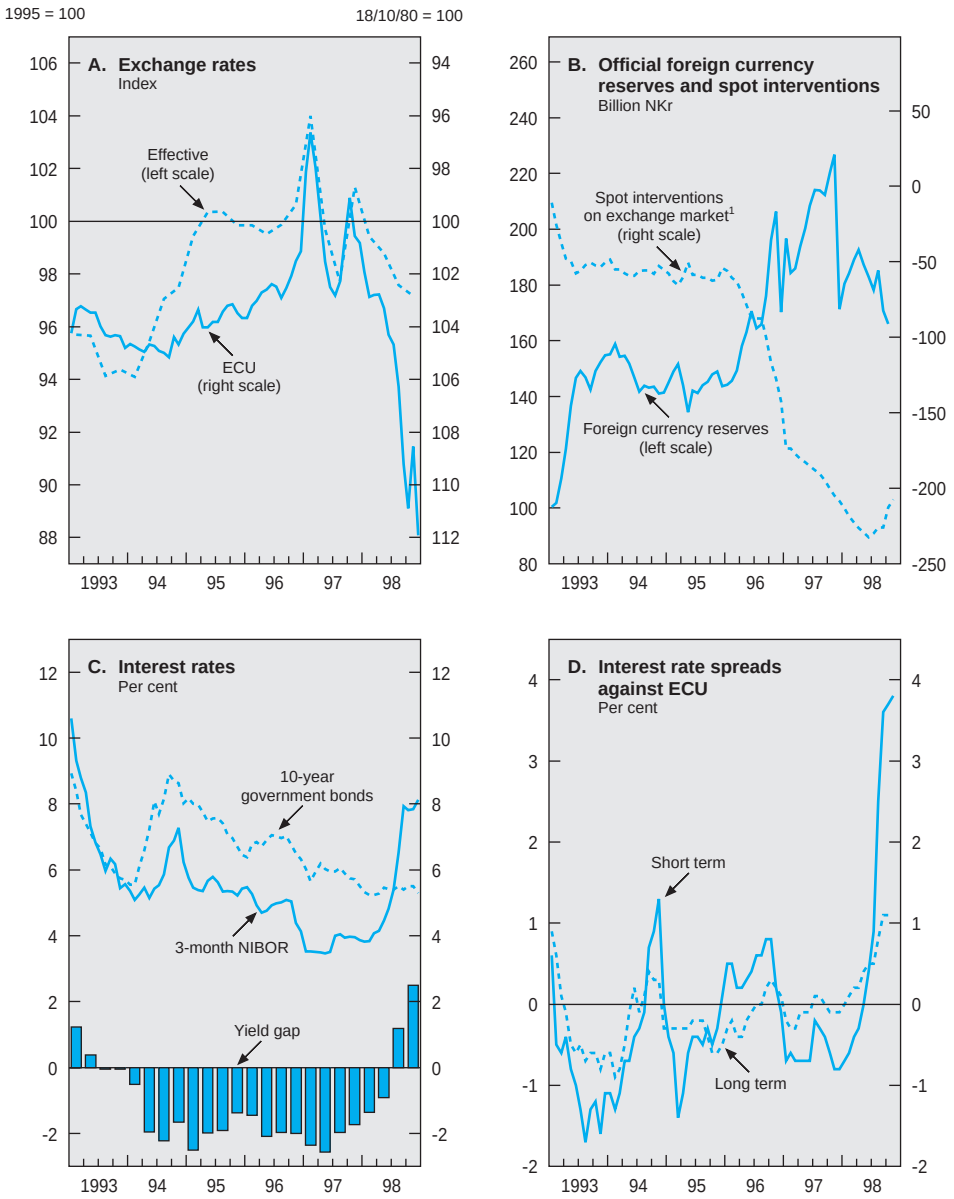
Both monetary and fiscal policy move to a restrictive stance

Monetary policy is geared to stabilise the currency

Since the turmoil in European exchange markets in late 1992 forced the Norwegian authorities to let the krone float, monetary policy has aimed at maintaining a stable value against European currencies. The government has formalised this orientation in official guidelines contained in a Royal Decree of 6 May 1994, stipulating that "the monetary policy to be conducted by the Bank of Norway shall be aimed at maintaining a stable krone exchange rate against European currencies, based on the range of the exchange rate maintained since the krone was floated on 10 December 1992. In the event of significant changes in the exchange rate, monetary policy instruments will be oriented with a view to returning the exchange rate over time into its initial range. No fluctuation margins are established, nor is there an obligation on the Norges Bank to intervene in the foreign exchange market".

Based on these guidelines, the Bank of Norway has aimed to maintain the ECU/krone exchange rate within the trading range that had been observed in the period from 10 December 1992 to 6 May 1994 (Figure 9, Panel A).[2] Initially the Bank of Norway adhered to this rule without having recourse to major interventions in the exchange market or official interest rate changes. In the winter of 1996/97, however, soaring petroleum exports prompted speculation on an appreciation of the krone, which required the Bank of Norway to purchase significant amounts of foreign currency. Official reserves (excluding Petroleum Fund assets) rose to a record high (Figure 9, Panel B).[3] In addition, official deposit and overnight lending rates were cut by 125 basis points to 3¼ and 5¼ per cent by January 1997, respectively. With the appreciation pressure gradually subsiding, official interest rates were raised in several steps to reach 4¼ and 6¼ per cent, respectively, by the early summer of 1998. The stance of monetary policy

Figure 9. **Exchange and interest rate developments**



1. Net accumulated sales of currency to banks from January 1993 onwards.
Source: Bank of Norway and OECD, *Main Economic Indicators.*

remained relatively easy, however, with short-term interest rates mostly hovering around 20 to 50 basis points above the official deposit rate – below the EU average over this period – and the yield curve portraying a pronounced upward slope (Figure 9, Panels C and D).

Given the weakening fundamentals through 1998, events over the summer led to renewed turbulence in the foreign exchange market, this time involving downward pressure on the krone. Monetary policy was tightened sharply to curb capital outflows and counter a major currency depreciation, through a significant rise in official interest rates. The Bank of Norway raised its deposit and overnight lending rates to 8 and 10 per cent, respectively, thereby lifting short-term interest rates by almost 3½ percentage points to over 8 per cent in August 1998.[4] The krone has recovered somewhat since, but major interventions in the exchange market were needed in mid-October to lean against a further slide. By the end of the year, the ECU/krone rate was approximately 3 per cent above its late August low and short-term interest rates were around 8¼ per cent. The monetary policy tightening is reflected in the yield curve which has been inverted since August 1998 (Figure 9, Panel C).

As a result, credit conditions have become tight. The rate of expansion of the money stock decelerated in the course of 1998, due mainly to slower growth in domestic credit (Table 5 and Figure 10, Panel A). Indeed, lending by commercial banks, savings banks and commercial mortgage institutions decelerated markedly

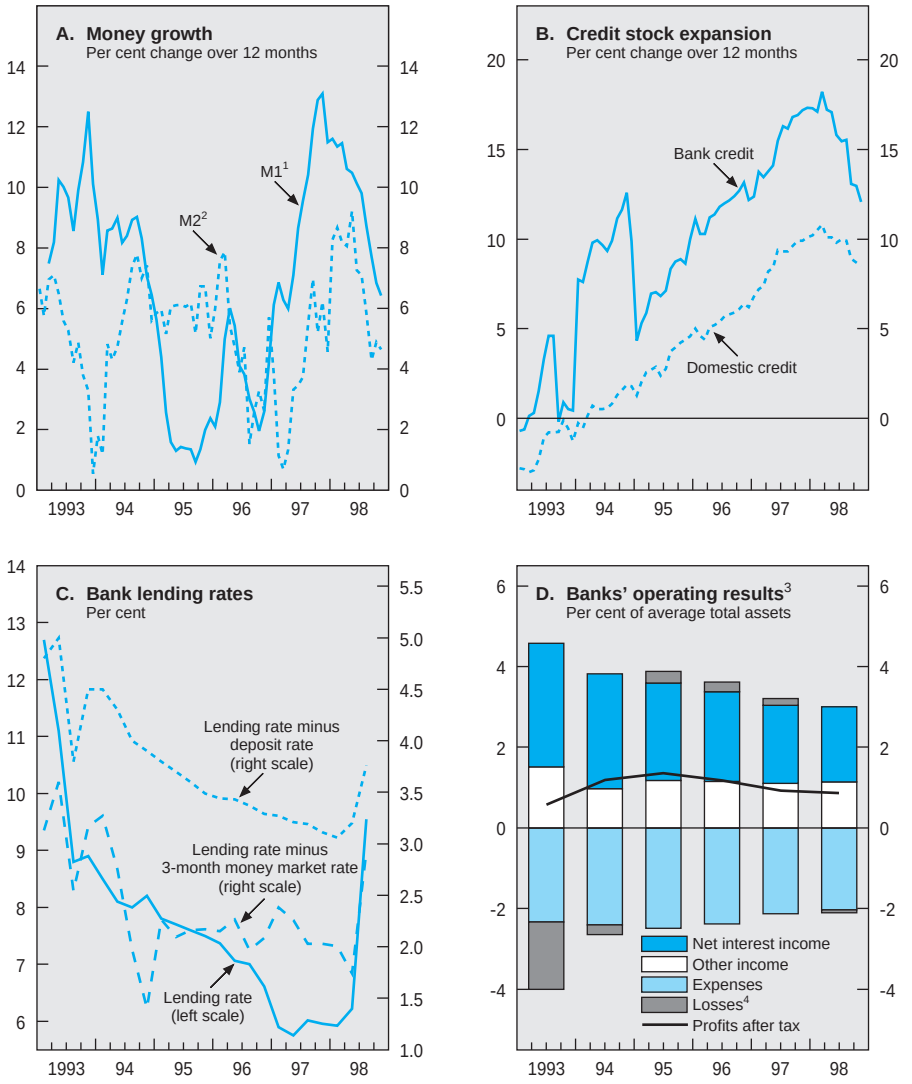
Table 5. **Money and credit**

Percentage change from previous year

	End-1992 NKr billion	1993	1994	1995	1996	1997	1998[1]
Total money stock (M2)	**550.1**	**0.5**	**6.6**	**5.1**	**5.7**	**4.6**	**4.6**
Percentage point contribution from:							
Central authorities	..	6.8	7.2	3.3	–7.1	–8.1	..
Commercial and savings banks	..	–4.8	3.8	8.0	10.0	14.2	..
Other sectors	..	–1.5	–4.4	–6.2	2.8	–1.5	..
Total domestic credit (C2)	**888.8**	**–1.8**	**2.3**	**4.9**	**6.2**	**10.1**	**8.5**
Commercial and savings banks	445.0	0.4	6.9	9.6	11.9	16.8	11.8
State lending insititutions	185.6	7.0	–2.2	–0.5	–1.9	0.2	3.8
Mortgage companies	83.1	–13.5	–11.8	–1.0	–4.4	17.6	8.7
Life insurance companies and pension funds	81.0	–7.7	–7.3	–16.2	–8.0	–29.5	–13.2
Bond and certificate market	62.3	13.4	4.3	6.4	10.5	1.5	–4.3
Other credit	31.8	10.8	16.2	18.4	–3.7	17.3	21.4

1. November. Growth over same period of previous year.
Source: Bank of Norway.

Figure 10. **Money and credit**



1. Cash, demand deposits and unused credit held by the public; each observation represents a 3-month right adjusted moving average.
2. M1 plus time deposits.
3. Commercial banks.
4. Losses on loans and guarantees; write-downs and gains/losses on securities held long-term.

Source: Bank of Norway and OECD, *Main Economic Indicators.*

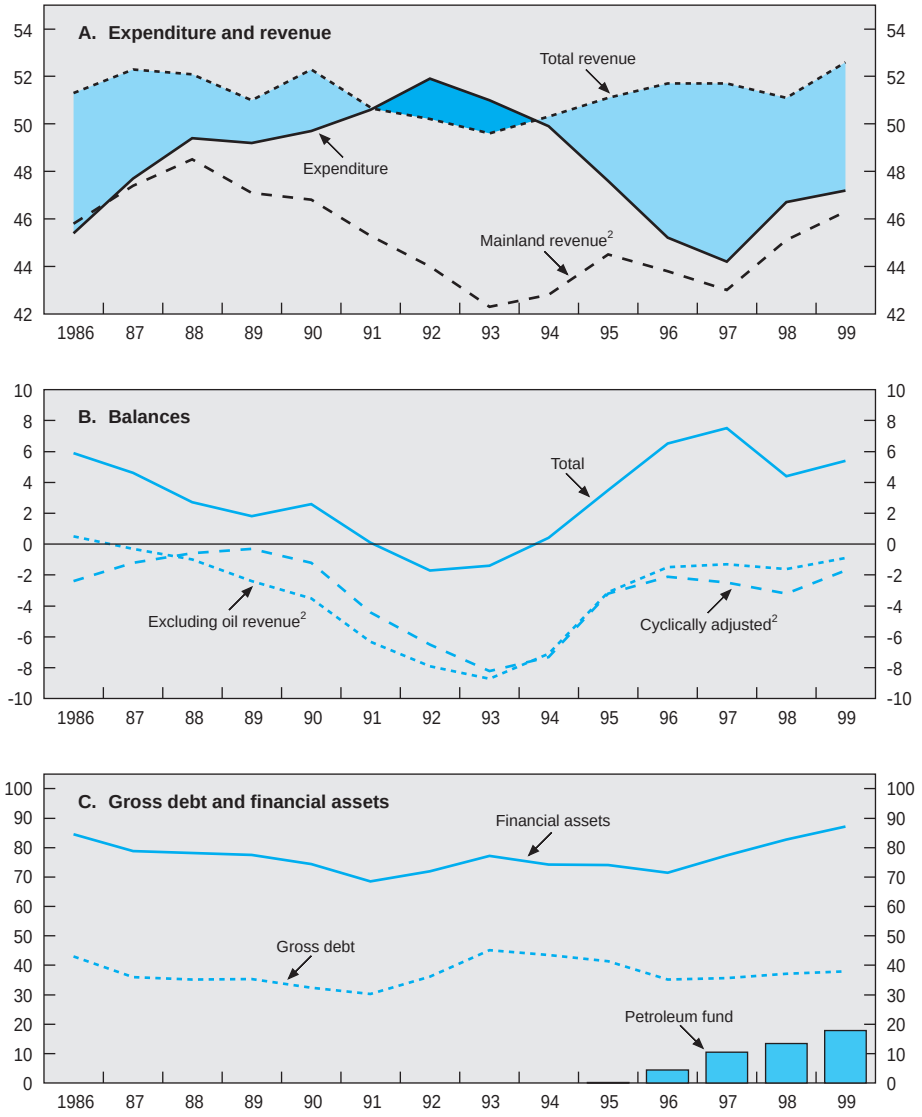
towards the end of the year, only partly offset by increased mortgage lending by the State Housing Bank – the largest of the "traditional" state-funded banks – that has allowed its clients to take advantage of the inversion of the yield curve by offering fixed interest-rate loans (Table 5 and Figure 10, Panel B).[5] Slower overall lending activity has been largely demand-driven and is not associated with credit restrictions on the part of the banking sector, which has continued to report satisfactory balance sheets and broadly stable interest margins (Figure 10, Panels C and D). A further decline in credit growth may be in the pipeline, however, and could intensify if the value of collateral declines due to the unwinding of the real estate market boom, but lending risks should remain manageable given the still good shape of household and business balance sheets. The exposure of Norwegian banks to setbacks in emerging markets is, moreover, very small.

Fiscal slippage in 1998 prompts a tighter budget for 1999

Among the factors that have contributed to financial turbulence in the second half of 1998 have been concerns over the relatively easy stance of fiscal policy amid increasing signs of economic overheating. In the period from 1994 to 1997 the general government financial surplus had widened substantially from 0.4 to 7.5 per cent of GDP, with the magnitude of the surpluses consistently exceeding initial budget estimates. As shown in Figure 11 (Panels A and B), this has been associated with soaring petroleum revenues coinciding with a significant reduction in the "non-oil" general government deficit – the latter underpinned by expenditure restraint, with the official measure of "underlying" government expenditure growth averaging ½ per cent per year in the 1994-97 period. The OECD Secretariat estimates the cyclically-adjusted deficit of the general government (excluding oil proceeds) to have narrowed from 7.3 per cent of GDP in 1994 to 2.5 in 1997, which points to a major fiscal tightening also by international standards. This assessment is in line with the official estimate of the change in the cyclically-adjusted non-oil budget deficit for the central government, where most of the expenditure restraint has occurred (Table 6).

In 1998, in contrast, the general government surplus dropped by as much as 3¼ percentage points to an estimated 4½ per cent of GDP. The bulk of this decline stemmed from plummeting oil revenues in the wake of the oil price collapse and subdued oil and gas activity in 1998. At the same time, whereas the official fiscal stance indicator for the central government showed a slight budgetary contraction in 1998, the fiscal stance of the general government was roughly neutral due to an increase in the local governments' deficit (Box 1 and Table 6). It is striking that the trend decline in the share of government expenditure in GDP observed since 1993 was reversed in 1998 (Figure 11). This reflects developments in local governments, where expenditure has been spurred by mounting income

Figure 11. **General government finances**[1]
Per cent of GDP



1. Data for 1998 and 1999 is estimated.
2. Per cent of mainland GDP.

Source: Ministry of Finance and OECD Secretariat.

Table 6. **Fiscal balance: projections and outcomes**

	Outcome			1998		1999	
	1995	1996	1997	Proposed[1]	Estimate[2]	Proposed[1]	Approved[2]
Central government (NKr billion)	**4.1**	**47.2**	**66.7**	**73.4**	**28.8**	**52.2**	**49.6**
Net proceeds from petroleum activities	38.5	69.9	86.8	85.6	45.7	57.7	56.1
Balance excluding petroleum activities	–34.4	–22.7	–20.1	–12.2	–17.0	–5.5	–6.5
Petroleum Fund (NKr billion)	**2.0**	**44.3**	**67.3**	**78.5**	**34.4**	**58.3**	**55.7**
Transfer from Central Government	2.0	44.2	64.0	73.4	28.8	52.2	49.6
Return on financial assets	0.0	0.1	3.3	5.1	5.6	6.1	6.1
General government[3] (per cent of GDP)	**3.3**	**5.9**	**7.5**	**8.3**	**4.4**	**6.0**	**5.6**
Central government	3.2	6.1	7.6	8.7	5.3	6.3	. .
Local government	0.1	–0.2	–0.1	–0.4	–0.9[4]	–0.3	. .
Memorandum items:							
Mainland GDP volume growth, per cent	2.9	4.1	3.7	3.0	3.0	1.3	1.3
Real "underlying" central government expenditure[5]	–1	1	2	1	2	1	3/4
Change in adjusted central government surplus[6]	2	1 1/2	3/4	3/4	1/2	1	3/4
Oil price (kroner per barrel)	108	134	136	125	100	110	110
Oil and gas production[7]	193	222	229	267	227	251	251

1. October of the preceding year.
2. December 1998.
3. Net lending.
4. Roughly half the defecit reflects a timing change in accrued revenues.
5. Deflated by the mainland GDP price index; per cent change.
6. Corrected for cyclical developments, net interest payments, petroleum revenues and technical adjustments; as a per cent of mainland GDP.
7. Million standard cubic meters of oil equivalents.

Source: Ministry of Finance.

from local taxes, supported by the favourable economic development. In 1998, moreover, substantial extra resources were provided by the central government, earmarked for local governments' health care expenditure.

The sharp deterioration in petroleum revenues and the exchange market turbulence in the second half of 1998, however, prompted a reassessment of the appropriateness of the fiscal stance. Accordingly, the 1999 draft Budget, presented to Parliament in October 1998, projected the general government surplus to rebound to 6 per cent of GDP in 1999, based on tax increases and renewed expenditure restraint, but also helped by an assumed recovery in oil prices and production (Box 2). Based on these official estimates, the OECD Secretariat projects the cyclically-adjusted non-oil deficit of the general government to narrow from 3 1/4 per cent of mainland GDP in 1998 to 1 3/4 per cent

Box 1. Fiscal developments in 1998

The budget presented to the Parliament by the outgoing Government on 3 October 1997 projected a central government surplus of NKr 73½ billion in 1998, reflecting net oil revenues of NKr 85½ billion and a non-oil deficit of NKr 12 billion (Table 6). The authorities estimated that this would correspond to a fiscal tightening of ¾ per cent of GDP as gauged by the change in the central government cyclically-adjusted non-oil budget deficit net of interest payments, based on a projected 1 per cent increase in real underlying expenditure (adjusted for expenditure on petroleum activities, unemployment benefits, interest payments and changes in accounting methods) and increasing tax revenues associated with the planned changes in the special tax regimes for liberal professions and shipping activities, and a modification in wealth taxation.

The new minority centre Coalition Government, which took office only a few days after the release of the 1998 Budget, presented an amended Budget in November 1997 showing a non-oil budget deficit of NKr 17 billion, NKr 5 billion more than the previous government, due to a rise in minimum pensions, the introduction of a special family allowance for small children not enrolled in subsidised public day care, and an increase in transfers to local governments to cover growing health and elderly care expenses. In addition, while a proposed increase in taxes and excises was rejected by parliament, social security contributions and some excise duties were raised. As a result, the estimated fiscal tightening was revised down to around ¼ per cent of mainland GDP in the approved Budget, with real underlying expenditure growing by 2¼ per cent.

Changes in the fiscal estimates since the approval of the 1998 Budget have resulted from a sharp decline in estimated net oil and gas revenues, which almost halved from NKr 87¼ billion to NKr 45¾ billion (estimate of December 1998). This reflected a 20 per cent drop in oil prices (local currency terms) and 15 per cent lower oil and gas production as compared to the approved budget. At the same time, non-oil expenditure slippage of more than NKr 6½ billion resulted from the spring income settlement, cost overruns on the railway line to the new national airport and unexpected expenditure increases in various other areas (social security, refugees and support to shipyards), but was more than offset by a tax windfall associated with sharp wage increases. As a result, the estimated central government fiscal surplus for 1998 amounts to NKr 28¾ billion or 3 per cent of GDP.

The surplus will be allocated to the Government Petroleum Fund in line with the approach adopted several years ago. The Fund will therefore have accumulated NKr 158½ billion (14½ per cent of GDP) by the end of 1998. With the budget proposal in October 1997 the Ministry of Finance issued new guidelines for the management of the Petroleum Fund by the Central Bank, which entered into force on 1 January 1998. According to the guidelines 30 to 50 per cent of the total fund will be invested in foreign equity markets, and the remainder primarily in foreign bonds. The Fund's investments in equities are limited to portfolio investments, as investments in individual companies have been capped at 1 per cent.*

* The Fund is investing in accordance with the following regional distribution: The Americas (United States and Canada) 20-40 per cent; Europe (Austria, Belgium, Denmark, Finland, France, Germany, Ireland, Italy, Netherlands, Portugal, Spain, Sweden, Switzerland and United Kingdom) 40-60 per cent; Asia and Oceania (Australia, Hong Kong, Japan, New Zealand and Singapore) 10-30 per cent.

Box 2. The 1999 Budget

The Budget presented to the Parliament in October 1998 projected the central government surplus to rebound by NKr 23½ billion to over NKr 52 billion (Table 6, last column). Almost half of the improvement, or NKr 11½ billion, is due to a narrowing of the non-oil budget deficit, as:

- non-oil revenue was expected to rise by NKr 33¼ billion (or around 8 per cent), of which NKr 4 billion result from increases in the tax rates on income, electricity, tobacco, cars and heating oil and the remainder from growth in the tax base and other proceeds. The NKr 4 billion tax increase was, however, not endorsed by parliament and hence not incorporated in the approved budget;
- non-oil expenditure was projected to increase by NKr 19¾ billion (or 4¾ per cent), despite a NKr 3½ billion expenditure cut related to *e.g.* a reduction in the government's contribution for sick pay, the abolition of the family allowance for children that already receive grants under the child-cash benefit scheme introduced in 1998, and a nominal freeze of the latter. The bulk of the spending increases reflect growing social security entitlements and other government transfer schemes, including health care, social services, regional support, child-cash benefit, disability and public pension programmes.

The government estimated the implied fiscal tightening to be NKr 9 billion, or 1 per cent of mainland GDP, as measured by the change in the central government cyclically adjusted non-oil budget deficit net of interest payments. The increase in the central government real underlying expenditure, was officially estimated at 1 per cent, down from 2¼ per cent in 1998.

The expected rebound in net oil and gas revenue by NKr 12 billion, or 26 per cent, is underpinned by a projected 10½ per cent growth in oil and gas production as new extraction capacity comes on stream, coupled with a recovery in oil prices from NKr 100 to NKr 110 per barrel. The latter reflects a projected US$2 per barrel price increase on the world market and a 5 per cent appreciation of the krone exchange rate against the dollar. As in 1998, the entire budget surplus will be allocated to the Government Petroleum Fund.

in 1999, with a fiscal tightening of the central government deficit by around 1 per cent of mainland GDP and an improvement of the local governments' deficit by over ½ per cent of GDP (Figure 11, Panel B). Thus, the tightening of fiscal policy envisaged in the draft Budget is marked and, if realised, should restrain domestic demand significantly in 1999. It should be noted, however, that planned tax increases were not endorsed by the parliament, resulting in a somewhat less tight stance of fiscal policy in the approved as compared to the draft Budget (Table 6).

Despite the less favourable fiscal outcome in 1998, the net financial asset position of the government has remained very comfortable. Financial assets held by the general government exceed 70 per cent of GDP, and have been on an upward trend since 1996 owing to the annual allocation of the state budget

surplus to the Government Petroleum Fund (Figure 11, Panel C). The purpose of the Fund, which was formally established in 1990 but initially remained empty due to the recession, is to build up foreign financial assets as a means to preserve the public wealth that is no longer embodied in the oil and gas reserves (see Chapter III). The Fund, which is managed by the Central Bank but separate from the official currency reserves, is required to invest exclusively in foreign financial assets in order to shield the mainland economy from variations in petroleum revenues while ensuring that these revenues contribute to diversifying the accumulation of financial wealth (see Box 1). Hence, the government has given priority to expanding the Petroleum Fund over reducing public debt, currently around 35 per cent of GDP, also to keep bonds in circulation so as to maintain a liquid bond market in Norway.

The policy framework faces new challenges as the economy cools off

The projection for 1999 and 2000

The OECD Secretariat's projections (Table 7), which are the same as those published in the OECD *Economic Outlook No. 64* (December 1998), assume that market growth for Norwegian manufacturing exports will decline from 7 per cent in 1998 to 5¼ and 6 per cent in 1999 and 2000, respectively. The oil price, which dropped from an average of US$19 per barrel in 1997 to around US$12 in November 1998, is projected to remain at that level in real terms (corrected for world inflation) during the projection period. Short-term interest rates are assumed to converge towards the average in the EU, reaching around 4¾ per cent by the end of the projection period, with the exchange rate *vis-à-vis* European currrencies remaining unchanged at its November 1998 level. Long-term interest rates are also projected to show a decline, with the interest rate differential against Germany narrowing to less than 1 percentage point by 2000 as inflationary pressures in Norway ease. The fiscal policy assumptions are based on the 1999 draft Budget which, as discussed above, points to a tightening of the policy stance by 1 per cent of mainland GDP in 1999. As noted, this translates into a 1½ per cent of mainland GDP fiscal tightening for the general government (including local governments). For 2000, the projections show a broadly neutral fiscal policy stance.

As new extraction capacity on the continental shelf comes on stream, a rebound in oil and gas production should underpin real GDP growth of 2¼ and 1¾ per cent in 1999 and 2000, respectively, following an estimated 2¼ per cent in 1998. Meanwhile, mainland GDP growth is projected to fall sharply from 3 per cent in 1998 to 1 per cent in 1999 before recovering to about 1½ per cent in 2000, reflecting a downturn in mainland business, residential and oil-related investment as well as weaker private consumption growth. Subdued investment activity

Table 7. **Short-term projections**[1]

Percentage change, constant prices

	1995 Current prices NKr billion	1996	1997	1998	1999	2000
Private consumption	458.5	4.7	3.4	3.7	1.9	2.3
Government consumption	194.5	3.2	3.0	2.4	1.1	2.6
Gross fixed investment[2]	192.5	9.6	12.6	5.0	–7.3	–4.3
of which:						
Oil sector[2]	47.9	1.5	15.5	12.5	–12.5	–3.3
Mainland business sector	84.5	17.3	9.1	5.3	–5.3	–6.9
Residential construction	26.5	–1.2	9.0	2.0	–6.4	1.3
Public sector	29.9	3.7	12.1	0.3	–3.8	1.6
Stockbuilding[3]	27.4	–0.6	0.2	0.5	0.2	0.0
Total domestic demand	873.0	4.7	5.6	4.2	–0.4	0.8
Exports of goods and services	353.4	9.8	5.8	2.4	6.1	4.0
of which:						
Non-manufactured goods (including energy)	157.0	14.0	2.6	0.3	6.7	4.0
Imports of goods and services	297.7	8.3	12.3	7.2	0.6	2.2
Foreign balance[4]	55.8	1.1	–1.8	–1.5	2.7	1.1
GDP	928.7	5.5	3.4	2.3	2.3	1.8
of which: Mainland	790.1	4.1	3.7	2.9	1.0	1.4
Memorandum items:						
GDP deflator	..	4.1	2.8	–1.3	2.0	2.8
Private consumption deflator	..	1.4	2.5	2.4	3.5	2.8
Employment	2 105.7[5]	2.5	2.9	1.9	–0.2	0.4
Unemployment rate	4.9	4.8	4.1	3.4	3.7	3.9
Private sector compensation per employee	..	3.1	4.8	6.1	6.6	5.6

1. Projections published in the OECD *Economic Outlook* No. 64, (December 1998).
2. Includes platforms under construction, crude oil production, oil drilling and pipeline transport.
3. Contribution to GDP growth, excluding platforms under construction.
4. Contribution to GDP growth.
5. In thousand persons.

Source: OECD Secretariat.

is associated, *inter alia*, with the completion of major projects on both the mainland and the continental shelf, a profit squeeze due to the rapid wage increases and weak commodity prices, lower capacity utilisation rates and the stock market correction. As a consequence, employment is projected to level off, but, unless a very moderate pay deal is struck next spring, wage inflation will be sustained at around 6 per cent, raising consumer price inflation to about 3 per cent.[6] The unemployment rate is expected to increase modestly, from 3¼ per cent in 1998 to just below 4 per cent in both 1999 and 2000. The current account surplus should recover, helped by a pick-up of oil production and growing net exports associated with weaker domestic demand and easing labour market constraints.

Main risks and challenges

The baseline projection is thus for a "soft landing" of the mainland economy, based on expectations of a moderate wage agreement, a tight stance of fiscal policy and broadly stable external conditions in 1999 – including flat oil prices and calm international financial markets. If one or several of these assumptions prove to be too optimistic, however, interest rates in Norway could stay high, asset prices drop further, wealth positions deteriorate and debt servicing expenditure of households and business rise during 1999 and beyond. In such a scenario, domestic demand would lose further momentum and the mainland economy could slide into an outright recession. The authorities need to take out insurance against such an eventuality, by keeping a tight rein on public expenditure and, especially, by persuading the social partners to keep wage growth moderate. But they must also avoid incomes policy concessions – such as a further extension of the early retirement scheme or a too rapid implementation of the "Competence Reform" – that would constrain labour supply and intensify labour market shortages. This would help avoid a further erosion of financial market confidence and safeguard a return to lower interest rates.

In order to pave the way for a return to solid, sustainable economic growth with low inflation in the medium run, the authorities need to embark on a careful assessment of the macroeconomic policy framework in place. The three-pronged policy approach adopted in 1993 – with: *i)* monetary policy geared to a stable exchange rate *vis-à-vis* the European currencies; *ii)* fiscal policy maintaining domestic demand at sustainable levels; and *iii)* incomes policy aimed at keeping wages in check to safeguard international competitiveness, low inflation and full employment – succeeded in pulling the mainland economy out of recession in the first half of the 1990s. However, the recent course of events underscores the fact that the framework has not succeeded in preventing a pro-cyclical stance of economic policies when the upswing matured. As full employment was reached from 1997 onwards, market forces overtook the incomes policy arrangement and wages started to accelerate. Meanwhile fiscal policy eased as the government took out a "growth dividend" and allowed expenditure discipline to slip, whereas the central bank kept short-term interest rates low to prevent an appreciation of the currency. In the event, the adverse terms-of-trade shock in 1998 prompted the sorely needed move towards tighter fiscal and monetary policies. In the absence of such an external shock, the Norwegian economy would have risked entering an episode of high wage and price inflation akin to the mid-1980s, and become even more susceptible to sustained high risk premia on interest rates, reflecting financial markets' concerns over a possible inflation-devaluation cycle.

The macroeconomic policy framework still receives broad support from the government and the social partners, and abandoning it without consensus over a possible alternative would probably not work well. The framework could

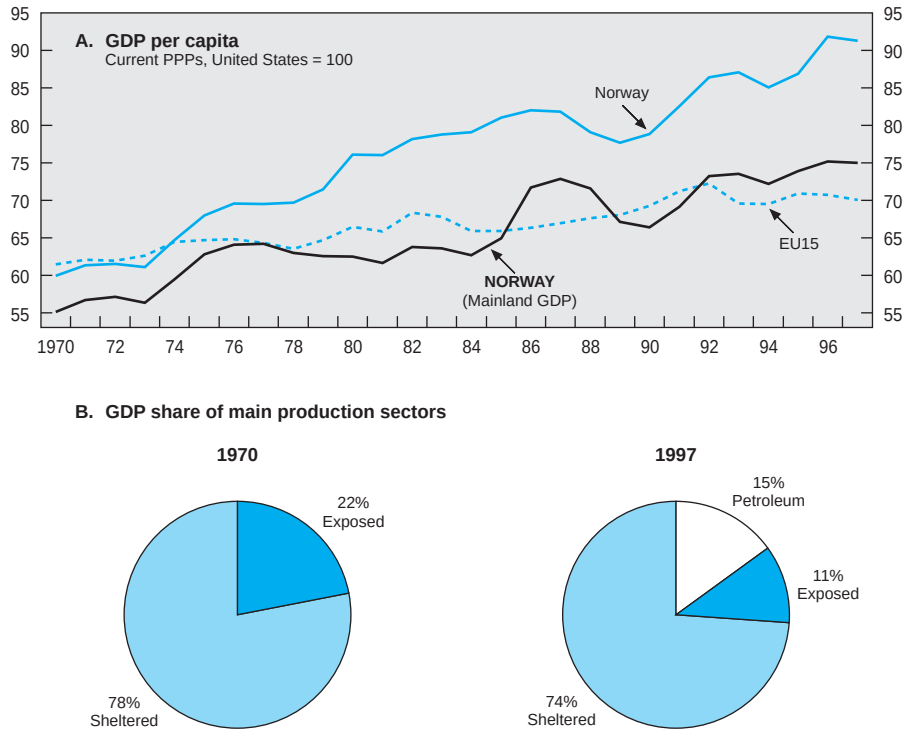
still offer useful service in the years ahead provided, however, that the flexible, medium-term orientation of exchange rate policy is maintained, allowing controlled exchange rate movements around a medium-term target to cushion the "asymmetric shocks" to which the Norwegian economy is prone. This would reduce the risk of being trapped again in a pro-cyclical stance of monetary policy. A pre-condition for this strategy to be successful is that the social partners deliver moderate wage deals in the years ahead, as this would ease financial markets' concerns over competitiveness and inflation pressure. Such efforts would usefully complement structural reform on a broad front to raise overall economic efficiency, the focus of the next chapter.

II. Implementing structural reform: a review of progress

Norway's large stores of oil and gas distinguish it from other OECD countries. Since Norway began exploiting its petroleum resources in the mid 1970s, overall per capita GDP has grown faster than the EU average and has approached that of the United States (Figure 12, Panel A). Income per head on the mainland (excluding the offshore petroleum sector) has, however, grown broadly in tandem with that of the EU and remained well below US levels. While the increased oil dependence has led to greater volatility in the mainland economy, the net long-run impact of the offshore sector on the level of mainland activity is less clear. Indeed, as highlighted in the 1995 *Economic Survey* (OECD, 1995), "demand pull" from income, investment and intermediate demand of the offshore sector on the mainland economy has been partly offset by lagging growth in "traditional" export-oriented and import-competing activities, whose share in GDP declined from over a fifth in 1970 to a tenth at present (Figure 12, Panel B).[7]

Although many attribute Norway's economic success in achieving practically full employment to prudent economic policies, such an assessment understates the role of petroleum wealth. Oil wealth allowed the expansion of public employment, while avoiding the ratcheting-up of public debt that has been typical for most European OECD countries, and permitted a substantial reduction in the average working week.[8] While the orientation of economic policies since 1993 has been to contain the expansionary effects of the oil bonanza – through the co-ordination of fiscal, monetary and incomes policies – it has, nevertheless, been associated with the maintenance of significant inefficiencies. The policy framework has resulted in a continued narrowing of the wage distribution (unlike trends in most other OECD countries) and the maintenance of a large number of persons on public support in view of the high employment and low unemployment rate. Furthermore, ample direct and indirect transfers have been provided to agricultural activities in sparsely populated regions while the large public sector has thwarted the release of labour from government funded to market-based activities.

Figure 12. **Structural indicators of the Norwegian economy**



Source: OECD, *National Accounts* and OECD Secretariat.

As highlighted in the 1998 *Economic Survey* (OECD, 1998*b*), the Norwegian economy would benefit from wide-ranging structural initiatives – by making producers in the mainland economy more exposed to competition and by improving the ability of firms to absorb economic or financial shocks more smoothly. While it is true that structural change in product and financial markets – mainly due to EEA and WTO membership – has resulted in increased contestability of these markets, a large public sector and the high tax burden create constraints on the mainland business sector. Against this backdrop this chapter reviews the structural initiatives in labour, product and financial markets since the previous *Survey*.

The labour market

Overview

Building on the conclusions of the OECD *Jobs Strategy* (OECD, 1997*a*), the 1998 *Survey* emphasised that the incentives for people on public income support to seek and accept work should be enhanced, in order to relieve the labour supply constraint on non-inflationary economic growth. A review of unemployment and related benefits was singled out as an important first step. At the same time it was considered that the scope for firm level wage bargaining should be enhanced to improve the allocation of labour, in particular by aligning the wages of young and low-skilled workers to their work experience and skill level; that employment practices should be liberalised by easing employment protection legislation; and that labour market training be better targeted with a view to reducing mismatches in an increasingly tight labour market. A synopsis of these and other recommendations and their subsequent implementation is presented in Table 8.

As discussed in Chapter I, the exceptionally strong employment growth during the past upswing was accommodated by an amazingly flexible response of the labour supply. The entry of women, students and young graduates into the labour force exceeded all expectations. While the education system had served as a buffer to unemployment for young persons during the recession of the early 1990s, young cohorts have contributed significantly to labour supply subsequently. While a recent survey (Cappelen and Svendsen, 1998) suggests that some scope for further increases in the labour force participation of women and a potential rise in average hours worked (through a shift from part-time to full-time work) exists, labour supply constraints have taken hold, with labour shortages spreading to ever more sectors. Workers from neighbouring Nordic countries have contributed to ease pressures in specific sectors, including the health care, engineering, construction and restaurant sectors. Foreign workers have, however, begun to return to their home countries as the economic situation has improved there.

A key feature of the flexible labour market response thus has been the entry of "outsiders" into the labour force. Indeed, as Figure 13 (Panel B) shows, the biggest shift out of non-employment has concerned home workers and students, which is reflected in rising employment and falling unemployment rates across all age and skill categories. The trend rise in the number of persons on disability and early retirement pensions has, however, continued despite the robust employment growth, indicating that the government's goal of increasing their labour market participation – by restricting access to such programmes – has not been realised. The main culprits are probably the weak financial incentives

Table 8. **Implementing the OECD Jobs Strategy – an overview**

Jobs Study proposal	Action taken since the previous *Survey*	Assessment/recommendation
A. Reform unemployment and related benefits		
a) Consider shorter effective benefit duration	No action since early 1997, when maximum duration was limited to three years, with the possibility of extension if participating in ALMPs	Unemployment benefit reform is slowly going in the right direction. Need to introduce longer waiting period for unemployed benefits and to reform part-time unemployed benefits
b) Limit disincentives arising from benefit and tax system for low-income workers by reducing withdrawal rates and/or by reducing benefit replacement ratios as benefit period increases	Special cash benefit introduced for parents of children two years old who do not use subsidised child care facilities approved by parliament in summer 1998. Access to early retirement eased to include those of 62 years of age in spring 1998	Reform is going in the wrong direction and is serving to reduce the effective labour supply at a critical time
c) Tighten eligibility rules for unemployment benefits	Job search requirements better enforced by the PES	
d) Decrease unemployment benefit entitlements for part-time unemployed persons	In practice henceforth the same maximum benefit duration applies to part-time and full-time unemployed persons	
e) Introduce a waiting period for unemployment benefits	No action	
f) Further limit the inflow of workers into disability pension scheme for reasons which are not strictly medical	Official commission to be set up to examine recent increases in sickness leave and inflows into disability	Need to set more restrictive medical criteria for use of disability pension scheme as inflow into it is rising
B. Increase employment flexibility		
Consider more liberal employment protection legislation for *e.g.* fixed-term contracts	Commission suggested more liberal rules for hiring. Parliament proposal expected in spring 1999	A further easing of job protection legislation is needed
C. Review active labour market programmes		
a) Improve targeting for labour market programmes	Use of these programmes more focused on skill enhancement and as a test on availability for work	Closely monitor labour market developments by occupation and skill level and target training programmes on emerging mismatches

Table 8. **Implementing the OECD Jobs Strategy – an overview** *(cont.)*

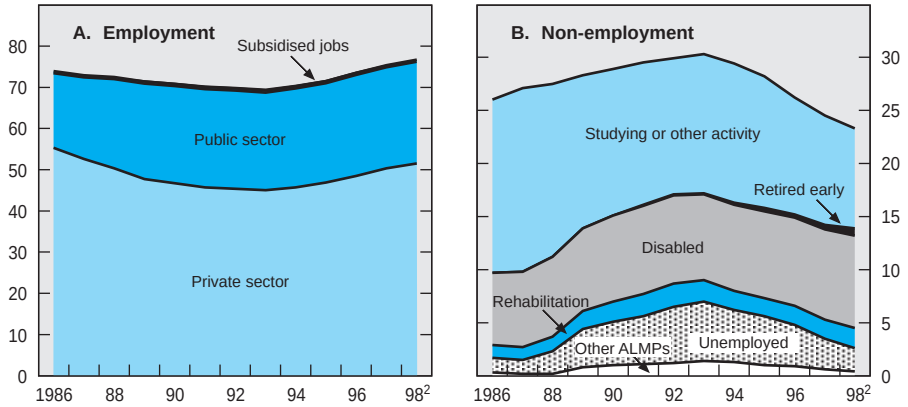
Jobs Study proposal	Action taken since the previous *Survey*	Assessment/recommendation
b) Evaluate ALMPs to better identify those which raise human capital of participants in a cost-effective way	Evaluation in progress	Studies show that ALMPs are effective in raising job finding probabilities
c) Consider liberalising job placement services, currently monopolised by the PES	Official commission presented a proposal suggesting the PES monopoly on placement be abolished. Parliament proposal envisaged in spring 1999	Open up job brokerage to private services
D. Increase wage and labour cost flexibility		
Wages for youth and low skilled workers should be aligned more closely with their experience and skill level	Official commission suggested a significant extension of the sectors in which temporary work agencies are allowed to operate, with a view to enhancing the employability of young workers	Review existing wage formation arrangements in order to enhance employability of young and low-skilled workers other than those enrolled in apprenticeships
E. Enhance product market competition		
a) Remove regulations that restrict competition in sectors dominated by state enterprises	General compliance with EEA and WTO agreements	Review opportunities for the privatisation of state-owned enterprises, especially the two leading commercial banks
b) Reduce direct support for agriculture	The Producer Subsidy Equivalent of public support to the agriculture sector increased from 71 to 75 per cent of the output value	Reform going in wrong direction. Significantly reduce the very high support for agricultural sector
F. Enhance technology diffusion		
a) Improve research links between public and private sectors	ENT and FORNY programmes to promote commercialisation of new products and research results	Review opportunities for further promotion of technology diffusion
b) Evaluate and rationalise various technology diffusion programmes	Support for existing programmes for technology diffusion (most importantly TEFT, FORNY, REGIN, NT and SME-competence programme) is projected to be around NKr 100 million in 1998	

Table 8. **Implementing the OECD Jobs Strategy – an overview** *(cont.)*

Jobs Study proposal	Action taken since the previous *Survey*	Assessment/recommendation
c) Establish R&D co-operation networks for SMEs	New measure to provide incentives for SMEs with less than 100 employees to hire people with higher education and to establish contacts with universities and colleges	
G. Promote entrepreneurship		
Encourage formation of SME clusters and networks for more effective competition with larger firms and globally	NETWORK programme, managed by SND amounting to NKr 40 million in 1997 expired in 1998. No prolongation envisaged based on advice from private consultants	Proceed with efforts to strengthen SME sector
H. Improve labour force skills and competencies		
Promote industry-education co-operation for apprenticeship training and bolster career planning in secondary schools and afterwards	Reform 94 offers all vocational students that have completed basic education two years of in-work training in co-operation with private employers. Government proposal to allow adults who have not completed primary or secondary education to do so, and to establish a system of certifying non-formal learning on the job (Competence Reform)	Ensure that education and training meet evolving needs of labour market. Apprenticeship programme functioning well but some mismatches

Source: OECD Secretariat.

Figure 13. **Composition of the working age population**
Population aged 16-74,[1] per cent



1. Excluding persons registered as retired (of standard retirement age).
2. Estimate based on average for first three quarters.
Source: Ministry of Labour and Government Administration, and OECD Secretariat.

for benefit recipients to work, along with the still easy entry conditions, especially for the widely used disability scheme.

Progress has been slow overall

In 1997 the benefit duration in the unemployment insurance scheme was effectively restricted by abolishing the possibility of beneficiaries requalifying for a new benefit period through participation in active labour market measures. The maximum duration of the benefit was limited to 1½ to 3 years depending on work history, with the possibility of extension of the benefit at roughly the same level (after tax) made contingent on participation in a labour market programme. While this change should be considered as an improvement, as the contingency of participation in a labour market programme to some extent acts as a work-availability test, practically no action was taken concerning the *Jobs Strategy* recommendation to introduce a longer waiting period before receiving unemployment benefits and to reform unemployment benefits for part-time workers.[9]

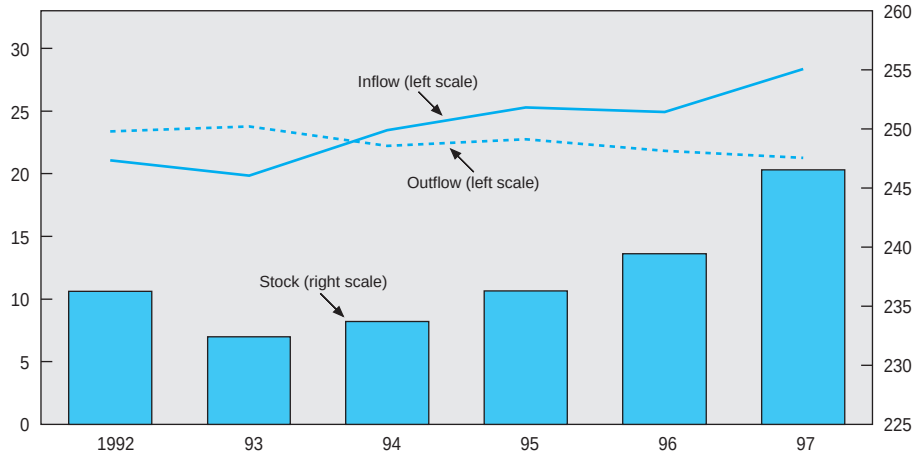
In Norway, as in some other countries, the disability pension system has been used to provide income support to workers who withdraw from the labour market. As a result, participation in this scheme rose from an average of 160 000

(6.2 per cent of the working-age population) in 1980 to 240 000 (8.5 per cent of the working-age population) by 1991. In 1991, the disability pension scheme was reformed in an effort to limit the use of the system to the purpose it was designed for by requiring a clear medical justification for benefit claims. While the reforms initially resulted in a declining number of disability pensioners, their number began to rise again in later years, reaching almost 250 000, or 8.5 per cent of the working-age population, in 1997 (Figure 14).

The majority of the disability pensioners tend to be women, have a low level of labour market skills, are of prime age or older, and have entered the scheme with mostly muscular and skeletal illnesses. In many cases, the relatively generous disability pension scheme is used as a bridge to the retirement pension. The authorities have stepped up efforts to integrate more of these people into the workforce. These efforts focus on preventing people on sickness benefits from dropping out by providing an adapted medical treatment. In addition, rather than automatically being granted a disability pension, claimants are required to participate in vocational rehabilitation training. Participants in this training are being given the same benefit as in the regular disability pension scheme, to ensure they have no disincentive to participate. In the 1998 *Economic Survey* it was recommended that, in addition to such efforts, it would be advisable to also

Figure 14. **Disability: inflow, outflow and stock**

Thousand persons



Source: Ministry of Health and Social Affairs.

undertake a thorough medical re-evaluation to ensure that participation in this scheme follows its original intention.

Norway introduced an extensive system of apprenticeship training as part of the 1994 education reform, with the aim of providing all students in vocational secondary education – almost half of the relevant age group – with an apprenticeship. The scheme involves two years of preparatory school work, followed by a two year apprenticeship in industry. Although the design of the system made it attractive for employers to take on apprentices, there has been an excess demand for training places from the outset, exacerbated by a mismatch between demand and supply. An increase in the number of training places over time has eased the situation somewhat, but mismatches persisted while it proved difficult for the authorities to influence the composition of training places. Mechanisms to achieve a better allocation of apprentices across employers are needed, but the authorities prefer gaining further experience before embarking on new reforms. To complement the 1994 education reform, a government White Paper presented to parliament in May 1998 proposed to enable adults who have not completed upper-secondary education to take up a study leave and to certify "non-formal" learning in the work place. This "Competence Reform", which was adopted by parliament in January 1999, is part of a tripartite effort to raise competencies of all individuals, both inside and outside the labour force, in line with recommendations by the OECD (1996). While such investment in human competences may prove valuable in the long run, there is some reason for concern that it could reduce labour supply in the short run.

The Public Employment Service (PES) in Norway has a national monopoly on job placements.[10] The monopoly has been increasingly challenged by private job placement firms – including headhunter firms and temporary work agencies – currently operating without a clear legal status. Moreover, a recent convention of the International Labour Organisation advocates liberalisation of this activity. An official commission formed in March 1998 to review the system with a view to opening it up to competition has delivered a legislative proposal, suggesting *inter alia* to abolish the PES monopoly on job placements. The commission has also been charged with reviewing elements of existing employment protection legislation and the relatively stringent job hiring restrictions. The commission proposed to liberalise the rules on hiring out of staff to other employers and, based on these recommendations, the government is expected to present a proposal to the parliament in the first half of 1999.

Some measures went in the opposite direction

Several measures introduced or planned since 1997 have actually gone in the opposite direction of the policy strategy advocated in the 1998 *Economic Survey*, as they tended to reduce the labour supply at a time when the labour

market was already overheating. These measures included a lower entry age into early retirement and new cash benefits to families who do not use public day care centres. Due to participation in these measures, around 30 000 people (almost 1¼ per cent of the labour force) are officially expected to have withdrawn from active labour market participation by the end of 1999. While these policies may be justified by their longer-run objectives, such as ensuring an equitable distribution of welfare spending, their timing is unfortunate.

In the 1997 tripartite wage negotiation it was agreed to reduce the legal entry age for early retirement in two steps, from 64 to 63 years in 1997 and to 62 in 1998, as part of a longer-term project to extend the early retirement scheme (AFP scheme) on a step-by-step basis. Participation in the early retirement scheme has almost doubled since, from below 10 000 in 1997 to around 18 000 by the end of 1998, notwithstanding the reduction in the average benefit level from above 100 per cent of prior income to 70 per cent (before tax; the after-tax replacement rate normally reaches 80 per cent) effective from the start of 1998.[11] The benefit level is not actuarially adjusted as the same rate is given irrespective of age and contribution years – which is one of the factors explaining the popularity of the programme. In addition to the regular early retirement scheme, a new "part-time" early retirement pension scheme offers a very high replacement rate, currently exceeding 100 per cent, giving a strong financial incentive for participation.[12] Moreover, in 1998 the (flat-rate) minimum pension, affecting workers who have built up low earnings-related entitlements more than proportionally was raised. This will further weaken the link between lifetime earnings and pension entitlements. All these measures combine to provide strong incentives for early retirement and Norway may, therefore, rapidly converge towards the low average effective retirement ages found in most European Union countries – especially if Norway enters a period of slow economic growth.[13] A motivation for the extension of the early retirement scheme is to allow older workers in sectors with demanding working conditions (heavy industries and education) to withdraw from the market. However, employees in the banking sector, who are generally in good health, also make heavy use of the early retirement scheme.[14]

As part of the programme of the new government, which took office in late 1997, a Family Cash Benefit scheme was approved by Parliament in June 1998. This Cash Benefit, which is offered to all families with two and three year old children, amounts to NKr 3 000 per month per child. The grant, which will be automatically paid to all families with children who do not use the publicly funded child care system, is intended to make it financially feasible for parents to care for their small children at home after the maternity leave has expired. As the benefit is approximately one-third of an average full-time salary, it is expected to represent a strong financial incentive for married mothers working part-time to give up their job. By 1999, some 10 000 to 15 000 women are expected to receive

the benefit. Although the scheme aims to relieve congestion in public day care centres, it was not accompanied by measures to stimulate the use of private day care such as increased tax deductibility of enrolment fees, possibly to avoid competition with the public system.

Product markets

The previous *Survey* highlighted the extensive state involvement in the Norwegian economy – including state ownership of manufacturing firms and commercial banks, substantial public employment, sizeable regional and sectoral subsidies, as well as the onerous regulation of some sectors in the pursuit of regional policy objectives. Of particular concern were the significant distortions associated with high direct support and trade protection offered to agriculture. While the power supply and distribution sectors have been forerunners in Europe in terms of applying free market principles and efficiency, state monopolies have remained prevalent in *e.g.* communication, transportation and postal services. At the same time, state involvement in the banking sector has remained significant. While rules governing competition have been reformed and have moved in the right direction, the establishment of a separate appellate body for competition regulation and arbitration has not yet been achieved. The sections below review recent progress in these areas.

Public support remains extensive

Regional and sectoral support remains a considerable element of government intervention in the economy. In particular, extensive and growing support to agriculture and regulations controlling the fisheries sector are based on the desire to protect such traditional activities from the forces reshaping the global economy. A plethora of subsidies and government regulations ensure the provision of jobs, health, education, transportation, postal and banking services to citizens in outlying and scarcely populated regions.[15] Budgetary support includes various forms of tax expenditure, benefiting primarily the energy-intensive and shipping industries, agriculture and the self-employed, which has led to the gradual erosion of some of the virtues of the comprehensive tax reform in 1992 (see Box 3). For instance the neutrality and fairness of the system have suffered from modifications of the "split model" and the introduction of the special tax regime for shipping companies. As shown in Table 9, over two-thirds of subsidies (excluding tax expenditure), NKr 10 billion or 1 per cent of GDP, are directed to the agricultural sector. Including border protection measures, public support for agriculture is estimated to amount to around 2 per cent of GDP. While this substantial support reflects the predominance of regional goals in agricultural policies, there is scope for achieving these goals in more cost-effective ways.

Box 3. The 1992 tax reform

Following the base-broadening and rate-flattening tax reform movements of the late 1980s in the OECD area, four Nordic countries – Norway, Sweden, Denmark and Finland – adopted considerable changes in their personal income tax systems, involving a separation of all income into either capital or labour income. Norway introduced this reform in 1992.

In the Norwegian dual income tax system, capital income is taxed at a low, flat rate, while labour and pension income are subject to a higher, progressive rate. All these forms of income are first taxed jointly at the corporate income tax rate of 28 per cent, while labour and pension income are, subsequently, taxed at a rate of 13.5 per cent above a certain threshold (NKr 269 100 per year in 1999). In addition, there is a social security contribution on labour, self-employed and pension income of 7.8, 10.7 and 3 per cent, respectively. Noteworthy is that all earnings-related costs and all allowances are deductible only against the income that is taxed at the rate of 28 per cent. Compared to the other Nordic countries, notably Sweden and Denmark, Norway has adopted a fairly pure dual income tax system, which avoids double taxation of dividends and capital gains. In essence, the dual income tax system rests on two presumptions. Firstly, the free capital movement necessitates low, flat tax rates on capital income and secondly, the immobility of labour and the unequal distribution of human capital make it possible to levy progressive tax rates on labour. In an open economy like Norway with a wide social safety net and high income transfers through the budget, and hence high taxes, capital income taxed as heavily as labour income would most likely reduce capital formation in the economy and increase capital outflows leaving labour less productively employed.

The equalisation of tax rates across sources of capital income coupled with the broadening of the tax base and more effective taxation of capital income have prevented tax avoidance, distributed the tax burden more evenly than before and even allowed reductions in income tax rates without causing a loss in total tax revenue. In addition, the reform has largely eliminated the different taxation of capital income across sectors, assets and types of finance, and by doing so, restored tax neutrality: economic factors rather than tax motives determine the behaviour of entrepreneurs. Indeed, by *i)* imposing the same rates for corporate and personal capital income; *ii)* allowing the full imputation of dividend income, thereby eliminating the double taxation of dividends; and *iii)* allowing investors to reduce their capital gains by the amount of retained earnings that is attributable to their shares to eliminate double taxation of retained earnings, the Norwegian tax system is among the most neutral in the OECD area (Gordon and Tchilinguirian, 1998). The reform has thus improved the efficient allocation of resources and reduced disincentives to invest in human capital, even though it may have increased disincentives to invest in fixed capital due to the fact that capital costs on average are higher. Before the reform, the taxation system was relatively neutral towards investment because the impact of taxes was mitigated by a generous range of tax allowances which helped companies to shield their investments from tax. The abolition of these allowances may have increased the impact of taxation on investments. However, the burden can be considered tolerable as capital income is taxed uniformly at a moderate rate (Cnossen, 1997).

(continued on next page)

(continued)

A main shortcoming of the dual income tax system in Norway is the absence of withholding taxes on interest and royalty income which provides an incentive for tax avoidance and reduces neutrality. Changes in the split model, which is used to divide assessed income of the self-employed into capital and labour components, each taxed in different ways, has made it relatively easy to declare a large part of income as capital income which is taxed at 28 per cent rather than at 52.2 per cent. Furthermore, the new special tax credit regime allowing shipping companies to pay taxes on profits only when taken out as dividends, violates the principle of tax neutrality and may, therefore, erode the acceptance of the system.

Table 9. **Support programmes**[1]

1997

	Subsidy amount		Authorities responsible
	Million NKr	Per cent of GDP	
Total	**13 034**	**1.20**	
Industry	2 599	0.24	Ministry of Trade and Industry through Eksportfinans ASA
Sectoral[2]	1 009	0.09	Ministry of Finance and Customs, Ministry of Cultural Affairs
Regional	868	0.08	Ministry of Local Government and Regional Development, SND
Exports[3]	103	0.01	Ministry of Trade and Industry through the Norwegian Trade Council, Eksportfinans ASA and Guarantee Institute for Export Credits
SMEs	619	0.06	Ministry of Trade and Industry through SND and public advisory institutions
R&D	827	0.08	Norwegian Ministry of Trade and Industry through the Research Council and SND
Agriculture	10 069	0.93	Ministry of Agriculture
Price support	2 599	0.24	
Decoupled support	7 469	0.69	
Fishing[4]	157	0.01	Ministry of Fisheries, SND
Forestry[5]	209	0.02	Ministry of Agriculture
Environment[6]	231	0.02	Ministry of Environment, partly the SND and the Norwegian Pollution Control Authority, Ministry of Oil and Energy through the Norwegian Water Resources and Energy Administration

1. See Annex Table A3 for a detailed listing.
2. Includes NKr 79.8 million interest subsidies and NKr 147 million from a 1996 tax concession.
3. Includes NKr 47 million interest subsidy and NKr 405.6 paid-out guarantees minus fees received for export credit guarantees.
4. Includes NKr 8.1 million interest rate subsidy.
5. Includes a NKr 15 million tax concession which could be increased up to NKr 20 million.
6. Includes NKr 31 million of loan guarantees for the treatment of hazardous waste.

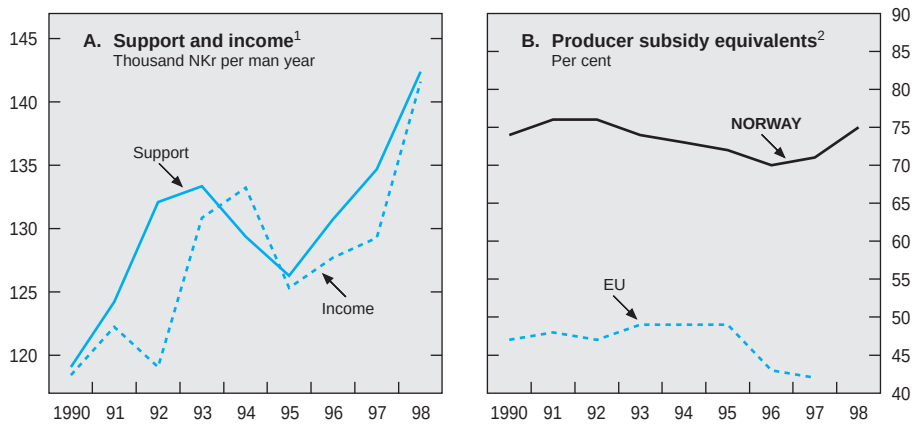
Source: WTO Report on Subsidies from Norway, for the year 1997.

Support to agriculture

Due to its northerly location, Norway is a "low yield-high cost" producer of agricultural products. With some of the cleanest air, water and soil in the world (see Chapter III), agricultural policy could have allowed the development of what seems to be the country's comparative advantage in the production and export of high quality agricultural goods. However, the authorities have created and maintained a system of support and regulation with the objective of producing a wide range of traditional agricultural products, including a surfeit of meat and milk. This policy has resulted in one of the highest overall agricultural support levels in the world, second only to Switzerland. In fact, average agricultural support per farmer in Norway has exceeded the average reported earned income in this sector almost every year since 1990 (Figure 15).[16] Although farming is formally market based, Norwegian farmers could be considered to be quasi-public employees.

The system of agricultural support is comprehensive and complex. It includes price support, subsidies per animal, support per acreage for landscape management purposes, production support for market crops and transport subsidies (see Annex II, Table A3). This reflects the predominance of several policy objectives, including a high degree of self-sufficiency in food production, ensuring an economic base for rural economies, preserving rural landscapes and agro-

Figure 15. **Agricultural support**



1. Data for 1998 is estimated. Income covers remuneration to labour and own capital.
2. Total value of transfers as a percentage of the total value of production (valued at domestic producer prices), adjusted to include direct payments and to exclude levies. Data for 1996 is estimated and later years are projections.

Source: Budget Committee for Land Use and OECD (1998), *Agricultural policies in OECD countries.*

biological diversity, reaching high sanitary standards and offsetting the high costs stemming from local climate conditions. Overall support to the agriculture sector has declined slightly in recent years, but the new measures adopted in 1998 are estimated to raise the total Producer Subsidy Equivalent (PSE) by some 5 per cent and the percentage PSE by around 1 percentage point. The percentage PSE is almost twice the level in the EU. Price support accounts for roughly one third of total agricultural support. The amount is subject to annual negotiations between the government and farmers' organisations which result in so-called Agricultural Agreements. The latest such agreement implies a growing scope for the system of administered prices. In fact, the price support level was not only raised somewhat, but more restrictive regulations were also adopted. For example, further restrictions have been introduced in the milk-quota trading system, making it less flexible in the spatial dimension.[17] Moreover, as target prices in the meat sector were not easily reached in the past two years due to excess supply, quantitative restrictions were introduced for beef imports.[18] Other measures include increased tax expenditure, by abolishing the investment tax for the agricultural sector, included in the 1999 draft National Budget. In addition, a special pension scheme has been created for farmers, bringing their early retirement age to 62 years. An objective of this measure is to make way for younger farmers and thereby raise the skill level in the sector. The cost is estimated to amount to NKr 100 million per year.

It is clear that a radical overhaul of the system of agricultural support, long overdue, faces formidable political difficulties. From an economic point of view, a system of regulation and central planning frequently results in a sub-optimal allocation of resources. A high support level over the long-term does not represent good policy, due to the heavy financial burden it places on consumers and taxpayers, and it undermines the incentives for farmers to innovate and produce high quality consumer goods and search for new markets. It should be acknowledged that agricultural policies in Norway, in fact, aim to sustain traditional labour intensive farming activities, which would otherwise be discontinued. However, even if the regional policy goals of agricultural policy are accepted, more cost-effective ways to achieve these goals should be sought, *e.g.* by further moving away from price support (including quotas and tariffs) to support based on land size or head count, as this would widen the scope for Norway to benefit from international comparative advantages in food production.[19] While the share of non-price support in total support is already high by international standards, this should be taken further, the more so since non-price support contributes to sustainable farming and low-polluting practices (see Chapter III).

Support to manufacturing industries

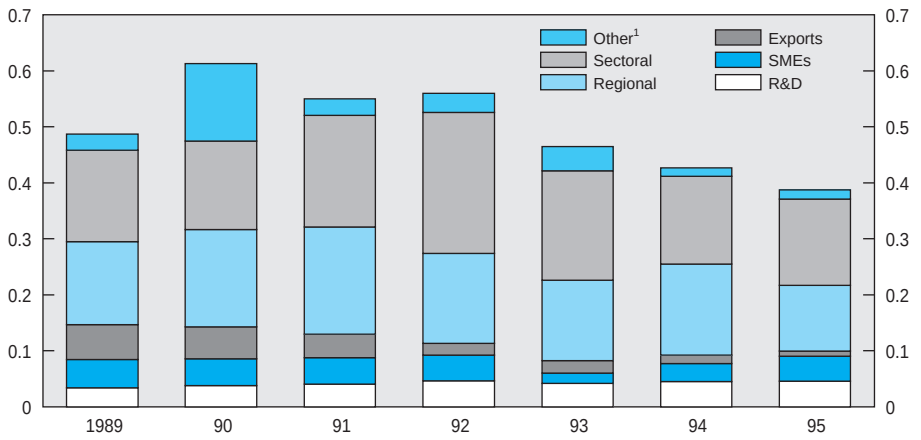
Although support to the manufacturing sector remained around NKr 3½ billion in the first half of the 1990s, it declined in real terms over the

period. Moreover, measured as a share of manufacturing output, support to the manufacturing sector fell from its 1992 peak of 5 per cent to 3.2 per cent by 1995 when it was, nevertheless, almost three times as large as the OECD average (OECD, 1998*c*). Sectoral aid has shown a sharper trend decline, even though this has been hampered by the continuation of shipbuilding subsidies following the non-ratification of the OECD Shipbuilding Agreement – accounting for 85 per cent of sectoral aid in 1995 (Figure 16).[20] Due to Norway's membership in the EEA agreement, however, it will comply with EU regulations and phase out support to shipbuilding by the year 2001. Industrial support policy has increasingly focused on "horizontal" objectives, such as boosting research and development and small and medium sized enterprises (SMEs), whose share in industrial support had risen to over 25 per cent by 1995.

One rationale for the shift towards such horizontal state aid is to correct market failures affecting in particular SMEs in the areas of:

– *technology diffusion.* An important factor in technology diffusion is firm spending on research and development (R&D). In Norway, as in most other small economies, business sector R&D spending, at around 1.4 per cent of production, remains well below the OECD area average of 1.8 per cent. This reflects Norway's industrial structure, which is

Figure 16. **Industrial support**
Per cent of GDP



1. Environment, energy and crisis aid.
Source: OECD (1998*c*).

characterised by a relatively large share of SMEs that typically have a low propensity for such investments. In recent years, the authorities have introduced many initiatives to enhance technology diffusion, notably to improve the interaction between science and industry, to raise competencies of SMEs and to increase co-operation among SMEs for R&D. Since its founding in 1993, the Research Council has focused on making delivery of support measures more efficient.[21] The financial contribution of the Research Council to industry-related R&D, including technology-diffusion programmes, was around NKr 600 million in 1997, more than half the total state support in this area which amounts to around NKr 1 billion per year. In 1997, a new measure was adopted to foster technology diffusion at the regional level through the establishment of regional clusters, with the aim of mobilising co-operation among regional institutes, colleges and companies;

– *the provision of risk capital to start-ups in remote areas.* Four industrial schemes were reorganised into one entity, the Norwegian Industrial and Regional Development Fund (SND), in 1993. The SND provides loans, grants, guarantees and equity capital – up to 50 per cent of external finance – to SMEs mainly in rural and remote areas in order to improve the quality of their production, management skills and competitiveness. SND has outstanding loans currently amounting to around NKr 10 billion, or 1 per cent of GDP. The main SND programmes are: *i)* FRAM, to improve management skills; *ii)* ENT, to assist commercialisation of new product ideas and concepts; *iii)* FORNY, to assist commercialisation of research results; *iv)* BIT, to increase competitiveness and profitability in SMEs through IT-solutions at the branch level; *v)* NT, to contribute to the development of technically advanced products or production methods; and *vi)* a regional restructuring programme, to stimulate new economic activity in regions with a weak economic basis. In 1997, furthermore, a new measure was introduced which aims to enhance SME competencies, by increasing the incentives of enterprises with less than 100 employees to hire people with higher education, and by establishing a system of contact persons for such firms in universities and colleges.[22]

The support in the area of R&D may be justified by the spillover from R&D activities into substantial positive externalities for firms and the business environment in general. However, there are questions about what role governments should play in influencing technical progress. While it is commonly accepted that government can stimulate technical progress indirectly through the provision of a stable macroeconomic environment, non-distortive and low taxation, and good regulatory frameworks, there is less consensus about direct measures. In particular, when a government decides on a mix of financial support

measures – aiming to improve the capability and incentives of firms to innovate and to accelerate the diffusion of new technologies – it is assumed that government action enhances welfare overall. This may not always be a realistic assumption. First, direct financial support implies that the government is capable of identifying the market failure necessitating intervention. Moreover, even if it could do so, it is not clear that the benefits would exceed the potential costs of government failure in solving the market failure. In particular, it is doubtful whether the government can design the right mix of financial support, due to its lack of knowledge of the associated synergies and externalities. At the same time, it is hard to control dead-weight losses, where firms would have gone ahead with investment in technological development in any case, in the absence of direct government support. These problems suggest that it might be better for the government to focus on creating a stable, unencumbered operating environment for firms, one that is conducive to long-term investment, including R&D.

More generally, while government support in Norway is geared to regional and social goals that are widely considered important, there are concerns about the cost-effectiveness of this support. Government support in Norway serves to sustain dispersed economic activities, which might possibly otherwise be discontinued. This approach is faced with a major challenge in the decades ahead, given the need for the government to accommodate an expected fall in oil and gas revenues and a subsequent increase in welfare expenditure due to the ageing of the population. Continuing this support will then require a greater transfer from income earners in urban areas to workers in rural areas. The alternative is to allow faster structural change to occur, which may result in urban regions and activities having to absorb these workers more quickly. The high level of subsidies to agriculture and other activities in the remote and sparsely populated regions of the country should thus be reviewed in terms of their overall welfare implications. The very high and growing support to the agricultural sector represents an inefficient way of pursuing regional objectives and clearly needs to be reassessed with a view of reducing the burden of costly over-production on consumers and taxpayers. On the other hand, as in most OECD countries, industrial policy has become gradually more geared towards the subsidisation of "intangible" investment – *e.g.* investment in R&D, education and training, acquisition of entrepreneurial and organisational skills, marketing and advertising, and software used in firms – in order to boost productivity gains.

State ownership continues to be substantial

In past decades, state-ownership policy in Norway has been guided by the principle that state ownership should continue for enterprises operating in the areas of transport and communication infrastructure provision, and petro-

leum and hydro energy production and distribution. In value terms, around three-quarters of the asset value of state-owned enterprises relate to the oil and gas sector, while the remainder relates to mainland business activities. Many state-owned enterprises are now managed as normal joint-stock companies and, in principle, there is no presumption that they would be bailed out by the government in case of bankruptcy. Dividend pay-outs to the state are based on a target rate of return on capital which varies between companies. In principle, there is no interference from government in the operations of state-owned enterprises.

Despite privatisations during the 1980s, state involvement in commercial activities remains very high. There are over 200 (fully or partly) state-owned companies in Norway, including many small ones, which in aggregate represent around 20 per cent of business sector employment and over 25 per cent of the capitalisation on the Oslo stock exchange. The total value of state-owned companies, including those not listed on the stock exchange, is estimated at over 50 per cent of GDP, representing more than half of the productive assets in the country. In 1996, the state fully owned 35 non-financial enterprises with an assessed value of NKr 165 billion, eight banks or lending institutions worth NKr 26 billion, and owned more than half of 26 firms with a market value of almost NKr 17 billion. In addition to the large state involvement in the energy sector, the state owns partly or fully an opera, theatres, film production companies, airlines, banks, wine and spirit outlets, and many manufacturing firms. Around 15 ministries are active in managing the state's diverse business interests. The Ministry of Trade and Industry is responsible for the state's 51 per cent stake in Norsk Hydro with annual revenues of NKr 96 billion (1997) and a market value of almost NKr 62 billion.[23] Meanwhile, the Ministry of Energy manages the state's interests in the petroleum sector – notably Statoil and the State's Direct Financial Interest (see Chapter III) and some power supply companies. The Ministry of Transport and Communication oversees the state's former telecommunications monopoly, Telenor, which is scheduled to be partly privatised. It, moreover, manages the National Rail and Civil Aviation Administration.[24] In spring 1998, the Ministry of Trade and Industry issued a White Paper concerning the government's policy on state ownership (Ministry of Trade and Industry, 1998). While no impending privatisations are signalled in the White Paper, the government intends to pursue a "pragmatic" approach to privatisation: if there is no compelling reason to warrant state ownership, including concerns over headquarters moving out of the country, such enterprises should be privatised when market conditions are favourable. Companies that utilise vital national resources, such as Statoil and Norsk Hydro, may not be (fully) privatised. The heavy state involvement in industries should be reconsidered, however, with its implications for efficiency being carefully analysed and opportunities for privatisation of commercial assets being explored.

Deregulation progresses steadily

Deregulation has progressed rapidly in recent years. In line with the EEA agreement, the state monopoly of Telenor in the telecommunications industry was abolished on 1 January 1998, permitting market access for both domestic and foreign competitors. As a result, entry of private firms has taken place in the industry. In local telephony, however, access to the local grid remains a problem because Telenor is in a monopoly position and controls access pricing and other conditions. The Competition Authority has challenged the separate telecom regulator PTE, which resides under the Ministry of Transport and Communication, to treat all companies alike. An outstanding problem concerns the four-number prefix customers need to dial in order to get access to the services of other providers than Telenor. According to the Competition Authority this poses a competition problem – even though as from 1 June 1999 customers of private providers will no longer be obliged to dial such a prefix. There are also concerns about the fixing of interconnection prices and technical matters over which Telenor still has monopoly power.[25] This issue is currently before both the Competition Authority and the EFTA Surveillance Authority.

The restructuring of the electricity sector pursuant to the Energy Act in 1991 has become a role model for electricity sector reform in a number of other countries and has involved the creation of the world's first international electricity spot and forward exchange market with Sweden at the start of 1996. The state energy company *Statkraft* was split into separate generation and transmission functions. The grid owner operates an electricity pool exchange with its Swedish counterpart. With an integrated Norwegian-Swedish market, there are now plans to include Finland and Denmark into a Nordic wholesale market.[26] At the retail level, the local distribution companies' right to monopoly supply has been removed and consumers at all levels can choose the supplier. Households are able to change suppliers every week, with prices posted for the coming week – without any transaction costs as a change of meter is no longer required.[27] Moreover, the Competition Authority has created an information system to educate consumers to use the new competitive environment effectively. Power generation remains dominated by public ownership, however, and there are concerns about inadequate incentives for private sector investment in new capacity and sluggish progress in rationalising 200 municipally-owned hydro-power plants, which account for 80 per cent of electricity supply to the grid. There are, in particular, potential problems related to companies in remote areas seeking to lock in local customers through price discrimination. Recently a case of price discrimination was filed whereby a municipal company gave those customers it defined as their "own" a lower price than those it considered as "foreign". As the board of the municipal company is largely a political body, the Competition Authority could not intervene and had to ask the relevant ministry to intervene in the matter.

However, it was not willing to do so, considering this a temporary problem. Concerning the regulation of local monopolies, an "efficiency factor" prevents efficiency gains from reducing employment, especially in publicly-controlled utility companies. In this regard, the 1997 IEA *Review of Energy Policies in Norway* (IEA, 1997) recommended that electricity operations be separated from other municipal or public operations. Such a change would reduce local government interference and allow the market mechanism to fully function in this sector.

With the new Gardermoen airport near Oslo, the problem of equal landing rights to carriers from other EEA countries, related to capacity constraints at the old airport, has been solved. However, the Ministry of Transport and Communication grants exclusive landing rights to domestic carriers on small regional airports. Since the Competition Authority allowed a merger between SAS and Wideroe Flyveselskap, a smaller regional aviation company, there are only two major "domestic" airline companies, SAS and Brathens, in Norway. This approval, however, required assurances of non-discrimination against the remaining purely domestic airline, Brathens, concerning transfer of passengers coming from these regional airports. Based on a bid tender deal, which ends in 2001, the carriers SAS, Wideroe Flyveselskap and Helikoper Service receive a government subsidy of about NKr 700 million in exchange for the regional aviation services.

Competition policy needs a stronger framework

With the opening of Norwegian markets in recent years, their contestability has improved – with the exception of public services, banking, postal services, railways, and alcoholic beverage distribution – and remaining competition problems relate mostly to the application of technical standards in some industries and the enforcement of competition rulings. As in most other OECD countries, the "prohibition" principle has been adopted as the basis of Norwegian competition law. Since EEA membership, the jurisdiction of EU competition law extends to cross-border trade and merger control issues, with the Competition Authority acting domestically. The functioning of the Competition Authority may be hampered by the fact that half of its staff of 140 persons are in the regional offices, which formerly dealt with price regulation, and as such may need training to deal more effectively with competition violation issues. The Competition Authority is fairly independent, with a mandate to investigate infringements and initiate corrective action, including the assessment of penalties, but in matters where the Competition Authority may intervene against terms of business mergers which are restricting competition, the decision of the Authority may, according to the Competition Act, be appealed to the Ministry of Labour and Government Administration. The Minister has occasionally overruled the decisions of the Competition Authority for reasons other than enforcing competition, and if this occurs

too often, its credibility may erode. There is, however, a growing understanding about the need for adequate competition in the Norwegian economy and its role in improving economic efficiency. In fact, the Minister of Labour and Government Administration has initiated a review of the appeal process, examining the possibility of an independent appellate body. He is expected to report to parliament by the end of 1999.

The Competition Act applies to all companies, including state owned-enterprises, and no cases of discrimination in favour of such companies have been documented. There are, however, unclear demarcation lines between individual ministries and the enterprises they regulate on the one hand, and the appellate body in the Ministry of Labour and Government Administration on the other. On these issues, therefore, Norway has yet to establish a clear separation of powers between the executive and judiciary branches of government. A conflict also exists between the competition authority and regulating ministries, in that the technical standards imposed by the regulator may have important effects on competition, often adversely affecting private firms entering a "liberalised" sector containing a former state monopolist. Even the state-owned company, Telenor, recently objected to the regulatory burden associated with the technical details imposed by its supervising ministry, the Ministry of Transport and Communication. Other sector-specific agencies regulating national monopolies are NDE in the electrical sector; PTE in the post and telecommunications sector; and *Kredittilsynet* in the financial sector. There is a need to negotiate the division of labour in the cases of overlapping (and sometimes conflicting) competence.[28] The Competition Authority has reached such agreements with NDE, PTE and Kredittilsynet. Australia and New Zealand are often quoted as examples of competition enforcement and regulatory systems that seem to deal better with the problem of overlapping competencies, by having only one official body for competition policy and enforcement.

An outstanding issue concerns the enforcement of competition policy and dispute settlement in the international setting. In the Nordic market for electricity, some enforcement issues and challenges are hard to meet with the present set-up. With international mergers, for example, between Swedish and Norwegian utility companies, there is a growing need for increased co-operation on the exchange of information, even allowing one country's Competition Authority to waive the right to intervene while the Competition Authority in the other country fully assumes this responsibility. Under the present Competition Act there is a limited ability to take into consideration the welfare implications of cross-border competition under the EEA, as the EFTA Surveillance Authority is not able to address all problems.

Financial markets

The previous *Economic Survey* noted that the Norwegian banking industry has benefited from substantial improvements in its operating environment – including a new deposit guarantee scheme, changes in reserve requirements and more effective bank supervision – amid increased foreign competition. However, the rationale for the government's continued direct involvement in the banking industry – a remnant from the banking crisis of the early 1990s – was questioned. Moreover, the ownership regulations concerning both commercial and savings banks were found to hamper the participation of the Norwegian banking industry in the ongoing internationalisation of financial services. It was concluded that there remains scope for rationalising the financial industry by reaping economies of scale through further integration into the Nordic financial market. Developments since the previous *Survey* have gone in the direction of further concentration in the banking industry, while concerns have remained over the government ownership of the *Postbanken*, the still considerable government stake in the two largest commercial banks and the role of the "traditional" state banks that provide "soft loans" to households and businesses. Furthermore, with credit markets weakening due to the expected downturn, the reforms to financial market supervision, undertaken in the wake of the banking crisis at the turn of the decade, may be thoroughly tested.

State ownership of banks continues amid increased foreign competition

As a result of the EEA agreement, the operating environment of financial firms has become more competitive in recent years. With the opening up of the Norwegian financial industry following the EEA agreement, competition from foreign banks and non-bank financial firms in the domestic lending market has gradually intensified, and is reflected in the consolidation of branches and overall job losses in this decade. At the same time, while recovering from the banking crisis, net interest income remains low, as reflected in low profits before tax, although since the mid-1990s Norwegian banks have been performing better than those in many other European countries (Table 10). The rising activity of foreign banks and non-bank financial firms in the Norwegian market has been reflected in a growing number of bank branches rather than subsidiaries. By 1998, the number of foreign banks with establishments in Norway had increased to eight, from five in 1996, with two banks, however, providing only a limited amount of services. All eight establishments are foreign bank branches operating in Norway under home country rules.[29] In 1997, foreign banks accounted for over 2 per cent of household borrowing, around 7 per cent of corporate borrowing, over one third of lending abroad. Foreign banks have mainly been active in the lending market for large corporate borrowers; they have found it more difficult to penetrate the SME

lending market due to a lack of appropriate infrastructure and informational disadvantages linked to their less well established local presence.

Several mergers of financial institutions initiated in the wake of the early 1990s banking crisis have served to reduce operating costs through the elimination of overlapping branch networks (Table 10). This process has stalled in recent years, due in part to concerns over increasing concentration in the Norwegian financial industry, but a major new merger was announced in 1998 – between Norway's second-largest bank *Kreditkassen* (*Den norske Bank* is the largest bank), the smaller *Fokus Bank* and Postbanken. Competitive bids from, subsequently, *Svenska Handelsbanken* and *Den Danske Bank* for Fokus Bank – which was fully privatised in 1994 – however, upset this plan.

While there are no formal restrictions on mergers and acquisitions in Norway regarding financial institutions, takeovers require approval by the Norwegian authorities and strict general ownership rules apply – except for the government.[30] This indicates that the Norwegian authorities view the Norwegian banking market as a domestic market, despite the significant integration of financial institutions within the EEA area. One motivation for this policy is that the authorities want to keep the headquarters in the country. With regard to the legislation shielding savings banks from takeovers and mergers, the issue is

Table 10. **Banking resources in selected OECD countries**

Per 10 000 inhabitants

	Norway	Finland	Sweden[1]	France	Germany	Italy	United Kingdom[1]
Branches							
1985	4.82	5.99	1.72	..	6.37	2.04	2.40
1990	4.23	5.66	1.57	4.60	6.26	2.59	2.26
1995	3.58	3.16	2.53	4.58	5.39	3.64	1.81
1996	3.71	2.75	2.47	4.51	5.32	3.73	2.05
Employees							
1985	72	75	27	..	89	55	60
1990	74	92	29	78	105	58	71
1995	63	60	44	70	89	59	65
1996	66	53	44	69	87	57	71
Profits before tax[2]							
1985	0.55	0.39	0.34	..	0.72	0.97	1.09
1990	–0.66	0.40	0.22	0.36	0.48	1.00	0.70
1995	1.36	–0.37	1.33	0.15	0.57	0.36	1.17
1996	1.23	0.41	1.30	0.18	0.53	0.50	1.15

1. Commercial banks only.
2. As a per cent of the average balance sheet total.

Source: OECD (1998), *Bank Profitability.*

whether savings banks should be permitted to transform voluntarily into commercial banks, which would allow them to participate in mergers and takeovers. Such changes, as proposed by an official commission reviewing the Banking Law and currently considered by the government, would create scope for further integration of the commercial and regional savings bank sectors. The rules of bank ownership are, however, expected to change little.

Financial market supervision may face new challenges

As the authorities consider supervision to work well, there have been no new reforms in this area. Given that the Norwegian financial sector is likely to undergo further significant structural change, successful execution of this function remains especially important. The financial market regulator Kredittilsynet focuses on sound operating practices of financial institutions, including banks, finance companies, credit institutions, insurance companies, investment firms, real estate brokers, debt collection agencies and auditors, but excludes the four specialised state banks and the public pension system.[31] Kredittilsynet is of the view that the existing set of regulations is well balanced around maintaining competition, while prudential rules prevent unsound banking practices. While Kredittilsynet supervises all professional market operators on the stock exchange, the stock exchange itself remains self-supervised. An official commission reporting to the Ministry of Finance has suggested that the stock exchange should be supervised by an external agency, notably Kredittilsynet.

As the economic boom unfolded until mid-1998, the rapid growth of domestic credit started to raise concerns by the supervisor, as it risked undermining capital adequacy by eroding the "core capital" of commercial and savings banks, although the latter still have a relatively high ratio. Some banks had been growing very fast indeed, despite a relatively low ratio of core capital, and there were indications suggesting that loans had become riskier. As the capital adequacy and related rules for banks are decided by the Ministry of Finance, the government proposed two measures that Kredittilsynet should adopt: a higher risk weighting for mortgage loans – from 60 to 80 per cent – and a tightening of conditions for subordinated debt. After these measures came into effect last summer, credit growth has significantly slowed down due to tighter lending conditions (see Chapter I).

Some uncertainty exists over how Norwegian banks will adapt to the new EU payments settlement system, TARGET, in order to enable them to clear euro payments. The banks would prefer a link to TARGET *via* the Bank of Norway, but this appears not to be feasible in the short run. As a provisional solution most commercial and savings banks are considering setting up a domestic clearing and settlement system. The Oslo Stock Exchange is also keen on integrating into a euro-area clearing system once it is established, to be able to offer a cheap entry

to and exit from the euro area for portfolio diversification purposes. This change would also make government borrowing abroad less costly, and thus diminish the need for reliance on the small Norwegian bond market.[32]

Concerns over the role of the "traditional" state banks remain

Activities of the traditional state banks are mainly geared towards serving regional and social policy objectives. Activities of the state banks may lead to unfair competition *vis-à-vis* commercial banks and the question has arisen as to whether private banks would be better in providing risk capital. The traditional state banks (SND, the Housing Bank, the State Educational Loan Fund and Kommunalbanken), mostly serve students, housing, SMEs and the municipalities. Their weight in overall lending has decreased in importance since financial market liberalisation. Social considerations dominate lending of the state banks and reforms have strengthened this focus. Since 1997, SND's primary loan activity has focused on high-risk financing and SMEs in remote areas, and as such complements the activities of the commercial banks. Regional seed capital funds initiated and co-financed by the Ministry of Trade and Industry and which are privately held and operated with some involvement of SND, also target SMEs in rural areas.[33] The State Housing Bank currently issues around 15 per cent of the total housing loans in Norway. Its activity is limited to new housing in the low-priced segment – up to an upper limit on the size of houses and standards. Its share in new housing is around 50 per cent.

The perceived market failure, *e.g.* the lack of financial risk capital outside the main urban areas, may well be better dealt with through a full privatisation and further liberalisation. Such reform would better enable the Norwegian financial industry to fully share in the economies of scale associated with the ongoing integration of the Nordic financial market. If subsidised credit is considered a useful instrument in achieving rural development goals, its delivery should ensure a level playing field, while also enhancing the efficiency of the domestic capital market.

III. Sustainable economic growth: natural resources and the environment

Since 1989, sustainable development has become a cornerstone of policy-making in Norway and it has recently been made a top priority area for the work of the OECD (Box 4). Sustainable development touches on a wide range of policy issues, including how to raise human and physical capital and how to manage natural resources. OECD country surveys have always covered a wide range of policy issues concerning potential output growth, the sustainability of pension systems or of the macroeconomic policy setting. While these issues are important in ensuring sustained economic growth, this chapter looks to a broader set of issues concerning natural resource management, including non-renewable and renewable resources as well as the environment, which are also necessary for ensuring sustainable development.

The management of natural resources involves fairly extensive public policy decision making. The key economic policy objectives concerning non-renewable resources in Norway, of which crude oil and natural gas are the most prevalent, are two-pronged. First, economic policy aims at maximising the wealth associated with the non-renewable resource, by choosing an extraction path and achieving cost-effectiveness. Other things equal, conditions of great uncertainty over future market developments tend, rationally, to encourage relatively rapid extraction. Second, economic policy is geared towards achieving a fair distribution of wealth across present and future generations, *i.e.* by establishing a mix between present and future consumption which maximises long-run economic well-being. This has entailed setting up a vehicle through which the resource wealth can be transmitted to future generations. The policy objectives associated with renewable resources, of which fisheries are the most important, are not fundamentally different. However, the ecological dynamics involved and the international sharing of stocks form additional conditioning factors. To date, economic policies in Norway have by and large achieved these objectives, but there remains scope for further improvement.

The environment is largely a public good which is available to all, despite its exhaustible character. Hence economic policies are needed to ensure that an

Box 4. The OECD's work on sustainable development

In a paper for the 1998 meeting of Ministers from OECD countries, the Secretary-General outlined a three-year work programme on this topic, covering four key aspects: follow-up of the Kyoto agreement; analysis of how prices could better reflect the social costs of environmental damage, including through subsidy reduction; the role of better exploitation of knowledge, technology and innovation for resource productivity; and improved measurement of performance. For instance, in the field of climate policy, one important objective for the OECD is to assess developments and facilitate discussion on good practice as Member countries are developing policies to achieve Kyoto targets.

As part of this programme, a series of in-depth special chapters in the OECD *Economic Surveys* are planned. They will focus on how countries go about achieving their environmental and resource management objectives with a view to strengthening sustainable economic growth in the long run. An important aspect in this context is to aim at a comprehensive policy approach, for instance, a better integration of sectoral policies. An analytical framework for these chapters will be established by spring 1999. It will subsequently provide a benchmark for policy analysis in the following *Surveys*. The experience gained from the reviews of individual countries will then be presented in a report to OECD Ministers in 2001. In this process, the present chapter on Norway is a pilot case, with the Norwegian experience influencing the work on the general framework. This choice has been motivated by Norway's strong focus on sustainable development. Indeed, already in its 1989 *Long-Term Programme* the Norwegian government adopted sustainable development as an overriding goal of economic policy.

optimal production level and structure is achieved, taking into consideration the (properly valued) effects of economic activities on the environment (*i.e.* ensuring externalities are properly recognised). Environmental targets set by national choice or international agreements should reflect these objectives, and economic policies should subsequently contribute to attaining these targets in a cost-effective way. There is no simple rule for ensuring cost-effectiveness, but it is, to a certain extent, a function of the choice of environmental policy instruments. Direct regulation, which relies on emission, process and product standards, is sometimes appropriate – for example in the case of heavily toxic substances. In other instances the use of economic instruments, such as taxes, subsidies and tradeable emission rights may be preferable as they leave more freedom for economic actors to seek the cheapest method of abatement and contribute to continuing innovation. Several economic instruments (taxes, auctioned quotas) raise revenue for the government in a non-distortionary way and thus may have an impact on overall economic performance by enabling a lower tax burden on human and fixed capital. Conversely, economic policies in other domains than the environment can have significant consequences for the environment, agricultural and

transportation policies being prominent examples. Given the complex interactions, the Norwegian authorities have set up an extensive framework for co-ordination of economic and environmental policies, which aim at a sustainable development of environmental resources and at reducing policy conflicts.

The discussion of non-renewable resources in the first section of this chapter draws on earlier OECD work (the 1995 *Economic Survey* and the IEA's 1997 *Norway Review)*.[34] The environmental section that follows also builds on previous OECD work (1993 *Environmental Performance Review* for Norway).[35] Compared to the IEA and the OECD *Environmental Performance Reviews*, which provide more detail on energy provision and environmental performance *per se*, this chapter focuses on sustainable development in its broadest economic sense – *e.g.* including, for example, considerations of inter-generational equity and cost-effectiveness – and are thus complementary. The chapter concludes with an overall assessment in the light of the challenges that lie ahead.

The policy issues – an overview

What is sustainable development?

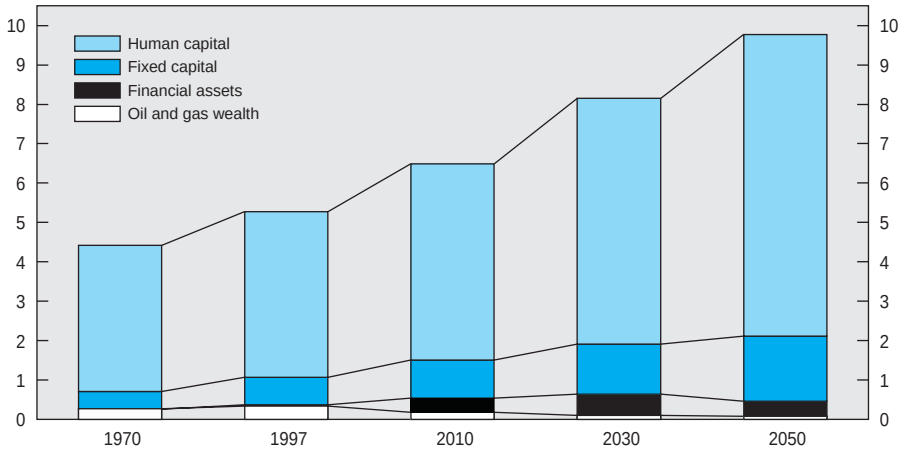
The 1987 Brundtland Report (UN, 1987) defined sustainable development as "development that meets the needs of the present without compromising the ability of future generations to meet their own needs". Concerns over equity between and within generations are central to this definition in particular as the world's resources including global natural resources are so unequally distributed, but the focus to date has mostly been on inter-generational equity. Hence a policy for sustainable development is typically defined as one that leaves future generations with the opportunity to attain similar or higher levels of well-being than the present one. In an economic sense, "opportunities" left to future generations depend on the total stock of wealth they inherit, including natural (resource and environmental) capital as well as man-made physical and human capital. "Social capital", *i.e.* the whole system of social values and institutions, could also importantly influence sustainable development, but is less amenable to economic analysis and beyond the scope of this chapter. Nonetheless, institutional aspects bearing on the setting of resource and environmental objectives and policies are covered.

A key issue is the degree of substitutability of the various forms of capital. To the extent they are substitutes, sustainability can be achieved by offsetting a decline in natural capital by a proportionate increase in man-made capital. This is the "weak" concept of sustainability. The "strong" concept requires that some level of the stock of natural capital be preserved under any circumstance. In reality some forms of natural capital are critical for development and can be

replaced with man-made capital to only a limited extent, while others are more fully substitutable. Most non-renewable resources are of the former category, as are renewable and environmental resources that are prone to very slow regeneration processes (fishery, forestry, biodiversity, ozone layer, etc.). As long as science is unable to provide reliable rules for sustainable depletion paths or viable alternatives for these forms of natural capital, their depletion could present a high cost for future generations, because of irreversibility.

Natural capital other than the non-renewable resources are virtually impossible to value, but skimming the large amount of indicators below suggests that national assets of this sort may have decreased until the early 1980s and are likely to have risen again since. Market valuations of non-renewable resources and man-made capital are to some extent available. According to official estimates, the decline in oil and gas stocks in recent decades has been outstripped by far by the simultaneous substantial increase in human, financial and fixed capital (Figure 17). Of these, human capital has shown a significant increase, rising from NKr 3.7 million per capita in 1970 (1997 prices) to NKr 4.2 million in 1997, while on current projections it could soar to NKr 7.7 million per capita

Figure 17. **National wealth**
Million NKr per capita, 1997 prices



Source: Ministry of Finance, Long-term Programme 1998-2001.

(1997 prices) by 2050. As discussed in the 1997 OECD *Economic Survey* of Norway, educational qualifications of the population in Norway have risen strongly and are now among the highest in the world. The fact that Norway's mainland potential output growth has been relatively subdued by OECD standards in recent decades mirrors the comparatively low rate of return on human capital in the extensive public sector, which may be associated with a lack of market scrutiny (OECD, 1997*b*). Nevertheless, overall per capita GDP, including oil and gas output, is close to the top in the OECD.

How to raise wealth accumulation?

Sustainable development as defined above constitutes a major goal for policy, but does not necessarily imply that the welfare of both current and future generations is maximised within the constraints imposed by sustainability requirements. Cost-effective policies aim to correct market failures, by properly pricing the use of natural capital while prompting cost minimisation by the individual economic actors. Cost-effective policies, moreover, ensure that the extraction of the proceeds from the use of natural capital (the "rent"), and their distribution among present and future generations does not conflict with the objective of optimal allocation. Cost-effectiveness is, indeed, vital to achieve sustainability, as it will allow faster wealth accumulation. In this respect, the OECD has always cast its net widely. Major examples concern the horizontal work on ageing, labour and product markets. Concerning Norway, work with a clear connection to sustainable development in recent years concerned special chapters on the environment, the oil and gas sector, pensions and human capital formation. In addition, the major aim of the structural surveillance chapters is to analyse structural conditions and to encourage policies that boost potential output growth.

Policies interact

Policy goals can conflict. There is, for instance, a strong emphasis on regional policy in Norway. As part of regional policy, a high degree of agricultural subsidisation not only boosts agricultural output, but also emissions to the soil and air, while the rent of the fishing industry is not taxed because of the desire to generate income for fishermen. At a minimum the cost of such policies should be made transparent, so as to be better able to evaluate trade-offs. In many cases, there should also be possibilities to design better policies which will reduce distortions, while keeping to the initial goals. Policy goals can also be mutually reinforcing. Taxing the natural resource rent or environmental externalities provides government revenues which are not distorting and can be used to reduce other taxes, which could influence potential output.

Resource management

Norway is endowed with major non-renewable and renewable resources. These comprise the important crude oil and natural gas resources on the Norwegian continental shelf, a very large hydro capacity, substantial forest resources and significant fish stocks in the Atlantic Ocean and the North Sea.[36] The government is strongly engaged in the management of these resources and, in the case of oil and gas, draws significant rents from it. Yet, it is difficult to design strategies for the sustainable development of these resources:

- since the government is the sole owner of the oil and gas stock, it could, in theory, aim to maximise the associated wealth by optimising the pace of extraction in view of the expected development of oil and gas prices over time. In practice, however, the uncertainty over future price developments has been overwhelming and, moreover, the size of the resources has been systematically underestimated. The pace of production of oil and gas fields has, therefore, largely been left to market forces, with the government acting as a regulator;
- since the early 1990s, the government has transformed part of the rent from the development of the oil and gas resources into foreign financial capital in order to improve the options open to future generations. In contrast, in the 1970s and 1980s, priority was given to investing the rent domestically in physical, in particular infrastructure, and human capital. All along, moreover, the oil and gas rent was also used to expand the public sector and subsidise economic activities – including most notably fisheries and agriculture;
- in the case of fisheries, property rights are subject to international agreements on shared stocks and quotas. The imperfect design of these agreements has implied that overfishing problems have not been resolved. The implementation of quotas in Norwegian waters is embedded in an extensive regulatory framework, which is mainly geared towards achieving sustainable harvesting – with fish stocks recovering since the 1980s – and regional policy objectives.

Non-renewable resources – oil and gas

Government policies allow market forces to determine oil and gas extraction rates

Norway is endowed with a large oil and gas wealth (Annex I), but at current extraction rates and technology, oil reserves could be exhausted in the next 20 years, while gas reserves are likely to last much longer (more than 80 years). An optimal management of the petroleum wealth involves decisions on the speed of oil and gas extraction, as well as when and how the rent should be distributed. These decisions are, in principle, separable. The optimal speed of

extraction is determined by comparing the return on "keeping oil and gas in the ground" to the return from extracting it and investing the rent. An income maximising producer will increase extraction if the returns from investing the net proceeds from additional oil and gas sales exceed the appreciation of the oil reserves in the ground and *vice versa*. The introduction of uncertainties modifies the optimal extraction rate. For example, the risk of sustained price declines of fossil fuels due to environmental policies, provides an incentive to increase the speed of extraction. On the other hand, political economy arguments – governments may be short-sighted in the use of the oil wealth – would argue for a slower speed of extraction.

In practice, the Norwegian government has chosen not to steer the extraction rate according to these text book principles. When production on the shelf started in the 1970s, the dominant view was that reinvesting the oil and gas rent in financial assets was more risky than preserving the petroleum stock itself and hence would not be a viable option. Calls to slow down the issuance of licences in the 1970s also stemmed from inter-generational considerations, while currently such calls are also based on the climate change issue. The point of view of the early "oil conservation adepts" has been superseded by subsequent developments: a faster extraction rate than actually realised combined with a policy of reinvesting the proceeds in securities would have yielded higher returns. After the oil price collapse in 1986, the idea of transforming the oil and gas stock into financial capital emerged. The world-wide bull run on stock markets since the 1980s raised hopes that the conversion of the oil and gas stock into financial capital would be very beneficial (see below).

Oil and gas resources are state-owned, regulation being based on the Petroleum Act. Licenses awarded by the state give companies the right to explore, produce and sell the resources for a certain period of time. Afterwards, ownership reverts to the state, unless the license is extended.[37] When issuing acreage licences for exploration, the authorities invite companies who meet high standards of security, solvency and reliability to participate in a consortium. Following new discoveries, production licenses are granted if exploitation is profitable. Licences are not auctioned off, unlike in some other oil and gas producing countries in the OECD area, such as the United States (Gulf of Mexico), Canada and Australia. Companies may apply in groups in the Barents Sea and in the North Sea, while they are obliged to apply individually in the Norwegian Sea. However, the authorities usually grant a joint licence to a group of companies designated to form a consortium also in the Norwegian Sea.

In recent years, Norway's approach to developing oil and gas fields has evolved in a number of areas, providing a clearer distinction between government regulatory functions and those of commercial operators, including the state-owned ones. In particular, the government has reduced its direct involvement in

the set-up of consortia for field development. Some of the changes are related to the need for compliance with competition rules of the EEA, and some are aimed at providing incentives for exploration and development in frontier regions.[38] The joint applications for blocks in the Barents Sea in 1997 and in the North Sea in 1999, rather than the government designating the consortia (so-called "arranged marriages") is a reflection of the evolution of Norway's policy in this regard. Moreover, from the 1996 15th licensing round, the state-owned operator Statoil is no longer awarded an automatic share of every license (see Annex I).[39] The price of crude oil is determined in the global market, and the government does not directly control exports of oil (apart from adhering to oil sanctions decided by the United Nations). There are no quotas, tariffs, product or crude restrictions.

The authorities have argued that the strong state involvement in gas sales and contract allocation can be justified by the fact that there is a need for co-ordination of the development of gas fields and gas pipelines due to economies of scale, in order to make gas development economically viable and to ensure optimal resource management. Recently governments in a number of EU countries have stated that they are opening up their local energy markets to increased competition, and the current structure of the EU gas market may change significantly as a result – *i.e.* become more "atomistic". This could have a substantial impact on Norway's market for future gas sales. It is, thus, important for Norway to keep under review the existing structure for gas transport and sales so that it is consistent with good resource management and changing market conditions. Such a review could provide an assessment of the pros and cons of different ways of managing the gas transportation grid and negotiating gas contracts.

The government increasingly transforms the oil and gas wealth into financial capital

The stock of oil and gas in the continental shelf represents a large source of wealth. In managing this wealth, the Norwegian authorities have aimed to: *i)* channel a substantial part of the revenues originating from the production of oil and gas through the budget to ensure that the whole society benefits from it at a given point in time; and *ii)* preserve an equitable share of these revenues for future generations through the build-up of public assets.[40] In line with these objectives, the government has set up an extensive welfare system while ensuring a comfortable net asset position. In this respect, Norway is a unique case in the OECD area. However, as has been highlighted in previous *Economic Surveys*, vigilance is needed as the projected fall in oil production in the first half of the next century will coincide with an expected increase in public welfare expenditure as the population ages.

Significant revenues accrue to the government due to the transfers of net income from the State's Direct Financial Interest – the SDFI – and the special tax regime applying to oil companies, including the fully state-owned company

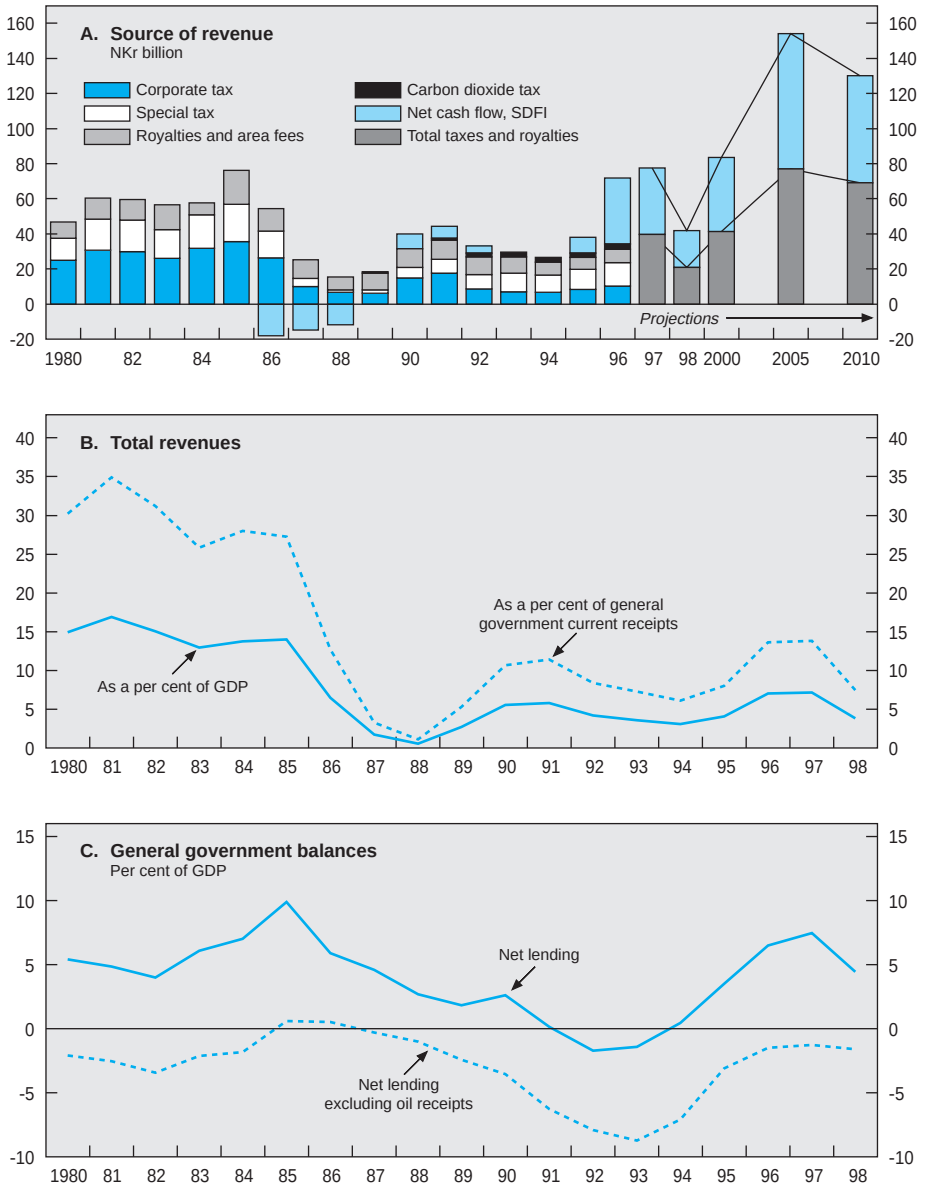
Statoil (see Annex I). In the past, government revenues stemming from the oil and gas sector have shown large swings (Figure 18). In recent years, for instance, they soared from 3 per cent of GDP in 1994 to 8 per cent in 1997, before plummeting to an estimated 4 per cent in 1998 in the wake of the recent oil price collapse.

Producers of petroleum earn a rent as profits will exceed average profits in other industries because they are given access to a scarce resource.[41] In principle, all the oil and gas rent should accrue to the government. The government can achieve this *via* the auctioning of licenses, taxation or by developing the resources on its own behalf. In practice, the Norwegian approach aims at extracting the rent *via* taxation and by the use of the SFDI. Estimates by Statistics Norway, however, indicate that about 80 per cent of the cumulative rent accrued to the government between 1980 and 1995. An alternative would be to combine these instruments with auctioning of exploration rights. This would also partly transfer the risk concerning the flow of oil revenues away from the state to the producers and reduce taxation non-neutralities associated with the need to maintain two different tax regimes: offshore and onshore (see Annex I). However, it is also very important for the companies to have the right balance between risk and reward. Higher price risk for companies associated with auctioning could imply a lower level of exploration, while the government's tax system spreads the risk over a large number of wells.

The view that not only present but also future generations should benefit from the oil and gas activities has led to the estimation of the remaining "petroleum wealth in the ground", both of the nation as a whole and the part which accrues to the government in the form of future petroleum proceeds – roughly 80 per cent of the total. Estimates of the petroleum wealth have tended to be volatile as they are very sensitive to oil price developments: in the 1980s and 1990s, they have varied between half and 2½ times GDP.[42] Until the mid-1990s such wealth estimates were used to set rules for the amount of spending out of government petroleum revenues that would not affect the wealth position of future generations. For example, the revised national Budget of 1992 stipulated that the non-petroleum deficit (the fiscal balance excluding petroleum revenues) should not exceed the implicit return on the assessed value of the remaining petroleum wealth, estimated at the time to be in the range of 5 to 7 per cent of mainland GDP.

Such rules of thumb have received less attention in recent years as the non-oil fiscal deficit virtually disappeared. Instead, budgets routinely include "generational accounts" as a way to assess the inter-generational distribution of government policies, including petroleum proceeds.[43] These accounts suggest that, on current demographic projections, social transfers and government services such as health, pensions and education, major inter-generational imbalances are unlikely to emerge. Such estimates are, however, very sensitive to the

Figure 18. **Net petroleum revenues of the government**



Source: Ministry of Petroleum and Energy and OECD Secretariat.

underlying assumptions with regard to the oil price and the projected social expenditure entitlements.

The preservation of petroleum wealth for future generations calls for a vehicle for transmission of the wealth to future generations. While the government had accumulated significant financial assets prior to the adverse oil-price shock in 1986, it was reluctant to create a fund, but finally established the Government Petroleum Fund in 1990. It initially remained empty due to the recession, but the favourable fiscal outcomes later on prompted the government to start transferring assets into the Fund in 1996. By design, the annual allocations to the Fund correspond to the fiscal surplus of the central government, and have been of the order of 5 per cent of GDP per year. As a result, by mid-1998 the capital in the Fund amounted to a market value of NKr 136 billion, or 14 per cent of GDP and is officially projected to grow to NKr 600 billion (close to 46 per cent of GDP) by the year 2002.[44] The Fund, as currently managed by the central bank in co-operation with external fund managers, invests exclusively in foreign currency denominated assets in order to: *i)* offset the impact of the currency inflow associated with oil exports on the exchange rate; and *ii)* avoid the risk of a combined fall in oil prices and the market value of domestic assets, as many domestic asset values are highly correlated with the oil price.

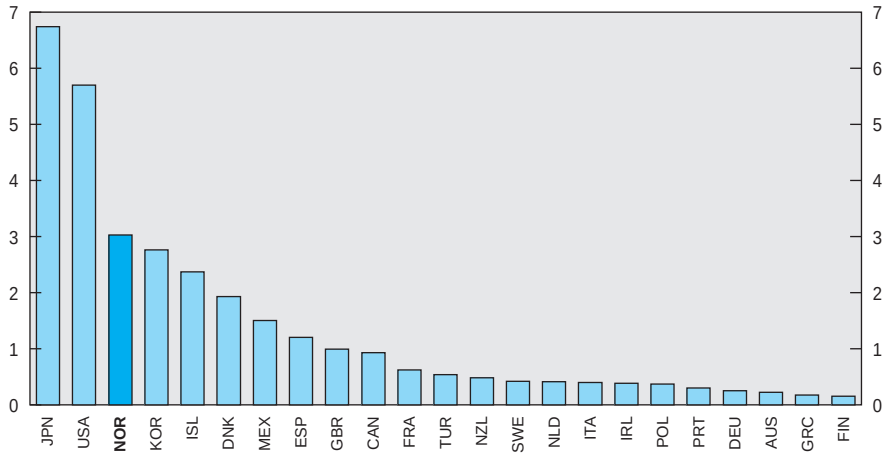
Renewable resources – fisheries and forestry

Fisheries

Norway is the tenth largest fishing nation in the world and the third largest in the OECD, after the United States and Japan[45] (Figure 19). Norway is, moreover, the world's leading producer of farmed Atlantic salmon (serving 50 per cent of the world market). The bulk of Norwegian fish is exported: Norway is the second largest seafood exporter in the world, following Thailand. The fishing industry, including the fish-farming industry, is Norway's second-biggest export industry – but it represents only one-tenth of oil and gas exports. Other key features of the Norwegian fishery sector are the limited processing activities, which are small scale and decentralised, the remoteness from markets (and closeness to resources); and its seasonal character. Fishery activities involve 50 000 workers (2¼ per cent of total employment), including fishermen, fish farmers, workers in processing plants and marketing agents. The fishing industry underpins the coastal area's economic activity, including private and public services. The total catch has grown considerably during the 1990s, from a trough of 1.6 million tonnes in 1990 to 2.8 million tonnes in 1997 even though the number of fishermen has continued its long-run decline.

Fish catches in Norway are constrained either by nationally fixed targets or by the internationally-agreed Total Allowable Catches (TACs) to prevent overfishing – *i.e.* the collapse of harvestable stocks.[46] These TACs are based on

Figure 19. **Fish catches in OECD countries**
Per cent of total world catch, 1996[1]



1. Marine catches of fish, crustacea and molluscs.
Source: FAO Fisheries Department, 1998.

scientific research of the size and age composition of each stock as well as changes in migratory patterns, although the estimates have not been very robust to date. International agreements have increasingly been used to settle disputes concerning the distribution of TACs for such species as the Atlantic herring, which migrates seasonally and unpredictably between the Norwegian, Faeroese and Icelandic 200 mile exclusive economic zones as well as the "Ocean Loop" outside the 200 mile zones.[47] Apart from the international Atlantic Herring Agreement, Norway has bilateral agreements with Russia and the EU on the sharing of fish stocks. For herring, for example, the second most important species for Norwegian fisheries, the country received 57 per cent of the Atlantic herring quota in 1997 while the rest was distributed among the EU countries, Russia, Iceland and the Faroe Islands. The use of TACs to limit fishing in the North Atlantic region remains the dominant management tool. They do not, however, avoid incidental catches of unwanted fish of other species and young fish, or of illegal "black-fish" landings. Overfishing in the North Sea remains a serious problem, as several species have almost been driven to extinction and the ecosystem is severely damaged, as reflected in increasing fishing mortalities. The tendency towards overfishing in this region is heightened by the overcapacity in the European fishing fleets. Indeed, both in the EU and in Norway, over-capacity is considered

to be the single most urgent issue hindering the transition towards sustainable fisheries.[48] Overall, existing fishery control policies and measures are either inadequate or not sufficiently enforced.[49] Despite failures, the overall resource and economic situation of the Norwegian fishing industry has improved. Some fish stocks have been rebuilt, for instance, Atlantic herring from practical extinction in the late 1960s to the biggest North Atlantic fish stock in 1997, while government financial transfers to the fishing industry peaked in 1981 (at 25 per cent of the export value) and has been reduced to practically zero in 1997.

The right to fish in the Norwegian 200 mile zone by Norwegian fishermen, is based on a quota system to allocate the TACs, managed by the Ministry of Fisheries.[50] The Norwegian quota system allows a fleet of traditional small fishing boats to operate alongside a fleet of larger and capital intensive ships (trawlers and purse seine vessels), with small vessels receiving the largest quotas in proportion to their capacity. In addition, vessels registered in the remote northern counties enjoy a favoured treatment. The quotas require renewal from the ministry each year and are not transferable at the sale of fishing boats and ships. Since 1996, however, owners who agree to withdraw a vessel from the fleet in perpetuity have been allowed to allocate part of their quota to the remaining fleet for a period of 13 years in order to achieve scale economies. This has induced owners of bigger vessels to buy smaller ones in order to combine two licenses for one ship, and has sparked a sharp price increase of ships with a license. While contributing to greater cost effectiveness, this system implies that the increased rent is reaped exclusively by the fleet owners. Not surprisingly, the current system of allocating the annual fish quotas for free among owners on the basis of historic rights is debated. A system of auctioned and freely tradeable quotas would allow a further re-allocation of production to the most efficient suppliers and would transfer the rent to the public coffers. These objectives could also be achieved by other means, for instance, tradeable licences and fees or resource taxes. The rent is currently estimated to be of the order of NKr 1.5 billion, if fisheries were managed in an efficient way. Given the inefficiencies generated by the current set-up, the actual rent is probably much smaller.

Forestry

Forests cover 37 per cent of the Norwegian land area, with 80 per cent privately owned. After suffering from excessive harvesting, the volume of timber has doubled since 1900. Since 1920, the aim of policy has been to increase the forestry resources and to improve their condition. More recently, policy has focused on environmental aspects, in particular biodiversity. Employment considerations in rural areas also play an important role. The measures to achieve these objectives include legislation, taxation, financial support and training. The authorities justify subsidisation by the very long growing period for trees in Norway (70 years on average). Subsidies have decreased in nominal terms since

1990 (from NKr 350 million to NKr 250 million). Over-harvesting is not an issue and there were only marginal changes in forest area over the last decade.

Environmental management

Norway probably has the most extensive framework for environmental policy of all countries in the world:

- it was among the first countries to establish a Ministry of Environment, in 1972. The ministry is responsible for identifying environmental problems, assessing and reporting on environmental trends and proposing cross-sectoral measures and national goals. It is assisted by several environmental management bodies under its authority, including the important State Pollution Control Authority which, pursuant to the Pollution Control Act of 1981, rules on emission permits for industrial activities and monitors compliance, either directly or on the basis of annual environmental reports of companies that have set up their own emission control and monitoring procedures. In clear cases of violation, companies are prosecuted and fined and profits may be confiscated;
- co-ordination on environmental matters is supported by various interministerial committees, that aim to: *i)* establish a common knowledge base on the environmental effects and their valuation in a wide range of policy areas, and *ii)* ensure that this knowledge feeds through into official policy proposals.[51] Such co-ordination is important so that, as far as possible, policy objectives of other ministries do not conflict with environmental goals. In particular, government support to economic activities in environmentally vulnerable remote areas, including fishery, agriculture, mining and heavy industries, have been susceptible to such conflicting goals;
- in the late 1980s, the Ministry of Environment, in co-operation with the Ministry of Finance, established a framework for an annual document annexed to the National Budget examining the "Environmental Profile of the State Budget", identifying all expenditure items that are wholly or partly motivated by environmental policy objectives. For this purpose, the spending ministries have been asked to classify outlays by "result areas", *i.e.* according to the environmental policy goals they should help achieve. There are no attempts to calculate a "green Net Domestic Product" (GDP corrected for depreciation of natural capital). However, environmental expenditure was estimated in the OECD *Environmental Performance Review* 1993, showing outlays of 1.2 per cent of GDP.

The main body of environmental legislation was established in the 1970s and early 1980s. Most importantly, the Pollution Control Act of 1981 and subsequent amendments instituted integrated pollution control for all stationary sources (industry, agriculture, municipalities and the continental shelf), endorsing four main principles: *i)* avoid irreversible developments and decisions (precautionary principle); *ii)* give priority to prevention over restoration; *iii)* assess the environmental properties of products at all stages, including their production, use and disposal; *iv)* seek cost-effective solutions. The Act contains provisions on implementation, enforcement, inspection, penalties for non-compliance, civil fines and compensation for environmental damage. Polluters must meet regulations, respect limits set by emission permits or pay taxes on certain emissions. Moreover, the Constitution was amended in 1992 to include considerations relating to the right to a clean environment and natural resource management, and several new Acts were legislated.

Since the early 1990s, an ongoing evaluation effort of environmental policies has been undertaken. The government has mandated several high level committees to investigate ways to improve the policy framework. In 1992, the *Environmental Tax Commission* published its report "Towards more cost-effective environmental policies in the 1990s: principles and proposals for better pricing of the environment" (Ministry of Finance, 1992). The report called for a differentiation of several environmental taxes to better take account of polluting substances contained in fuels and other products, in order to enhance the cost-effectiveness of these tax measures. This was followed up by the *Environmental Policy Instruments Committee* (Ministry of Environment, 1995*a*), which recommended to improve the cost-effectiveness of a wide range of environmental policies. The *Green Tax Commission* established in 1994 released in 1996 its report "Policies for a better environment and high employment" (Green Tax Commission, 1996). It suggested ways to capture better the "rent" from the use of natural capital. It recommended, *inter alia*, to impose a rent tax on hydro power stations, which was actually implemented in 1998, and to examine the possibility of auctioning emission and catch quotas. It also proposed changes to a number of existing environmental taxes, most prominently to extend the CO_2 tax to all sources exempted to date, and to introduce new environmental taxes, *inter alia* on waste. Being a large net importer of air-borne pollution and a small open economy, Norway has a clear interest in international co-operation and co-ordination of environmental policies and the country has been very active in this regard. Indeed, the implementation of, and compliance with international agreements figures very high on the political agenda.

The extensive institutional framework for environmental policy has helped to integrate policies and to build a consensus about the need to make these amenable to sustainability in the long run. However, there remains considerable scope for improvement, which may not be easy to achieve since policies

pursuing sustainability in the long run can conflict with vested interests in the short run.

Global environmental problems

Climate change

Norway has been one of the main proponents of putting climate change on the international agenda and has actively participated in the conclusion of the Kyoto Protocol for reducing greenhouse gas emissions (see Box 5). Prior to the

Box 5. **The Kyoto Protocol**

As a follow up to the 1992 Convention on Climate Change at the Earth Summit in Rio de Janeiro, it was decided to hold a Conference in Kyoto in December 1997 to agree on legally binding quantitative targets. The result was a protocol which involved the following major provisions:

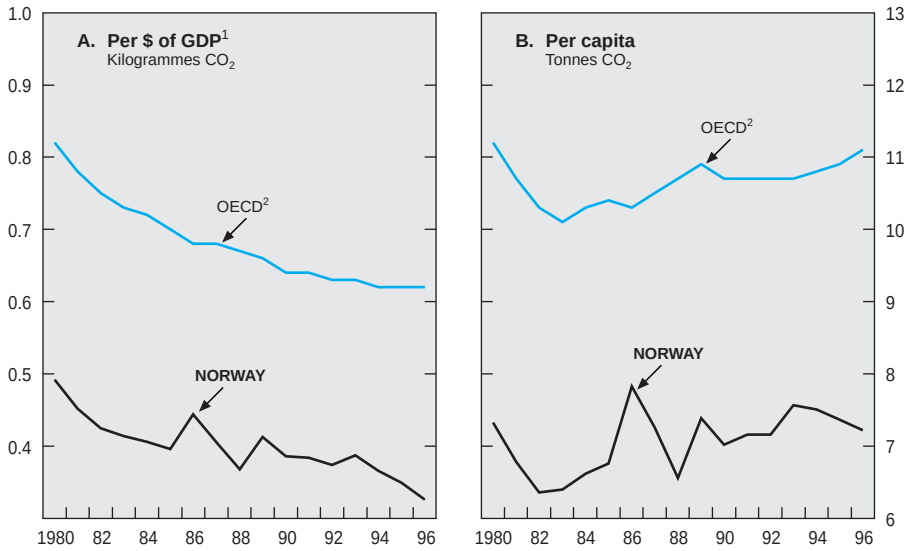
- Annex I group countries (OECD countries except Mexico, Korea and Turkey, plus Russia, Belarus and the countries of central and eastern Europe) will cut their greenhouse gas emissions by at least 5 per cent relative to the 1990 level in the 2008 to 2012 period. The reduction commitments are differentiated by country, but the protocol mentions that they can be met individually or jointly. The reduction targets range from an increase in emissions of 10 per cent in Iceland to reductions of 8 per cent in the European Union. Norway is one of the few countries that has been granted an increase in greenhouse gas emissions (1 per cent relative to 1990 outcomes), with the rationale that Norway's relatively clean system of power generation (hydro power), leaves limited scope for further reductions in CO_2 emissions at reasonable cost. The Protocol covers a basket of six greenhouse gases: carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O) and three synthetic fluorinated compounds (SF_6, HFCs and PFCs);
- the Protocol allows for emission trading and joint implementation among Annex I countries. Emission reductions can be "banked" in the sense that countries that more than meet their commitments in the "first commitment period" (2008-12) can use the surplus reductions for future commitment periods (to be defined). In addition, the Protocol contains a provision whereby abatement investments in a non-Annex I country financed by an Annex I country could count against the target of the latter (Clean Development Mechanism) but rules for such a mechanism are yet to be defined. Similarly, to some extent changes in a country's absorption of CO_2 emissions due to a change in its land use and forest surface, would count against its abatement requirement set by the Protocol;
- the Protocol will enter into force 90 days after 55 Parties accounting for 55 per cent of total CO_2 emissions of Annex I countries in 1990 have ratified it. Future meetings will define rules and guidelines for emission trading; ways to verify compliance with agreed commitments; and other specific rules for implementation of the Protocol.

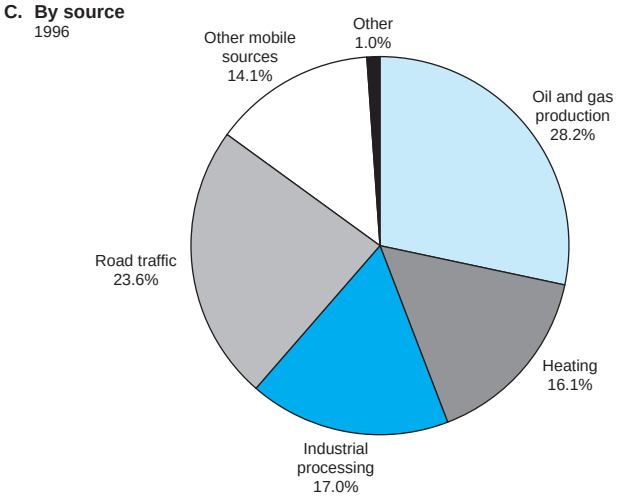
Protocol, moreover, Norway was one of the five countries to implement a carbon tax to curb CO_2 emissions which are the main source of climate change. Norway's active stance on climate change policies may seem at odds with its small contribution to global greenhouse gas emissions and its interests as a major oil and gas exporter (even though its hydropower industry would benefit from increased demand for clean energy sources). However, irreversible change in the earth's climate is an important preoccupation of the Norwegian people, even though the country is not particularly vulnerable to a sea level rise.

As in most countries, carbon dioxide is by far the most important greenhouse gas emitted in Norway, accounting for 70 per cent of total emissions in 1996. Norway is a median OECD country when it comes to CO_2 emission per capita, and in the lower range with regard to emission per unit of GDP – in part reflecting the mainland economy's reliance on hydro power (Figure 20, Panels A and B). The extraction and transportation of oil and gas on the continental shelf generate large greenhouse gas emissions (Figure 20, Panel C). Hence, the decline in CO_2 emission per unit of GDP achieved since the early 1980s is remarkable in view of the soaring oil and gas production, and reflects strong improvements in energy efficiency. Carbon dioxide emissions have, nevertheless, continued to rise in absolute terms since 1990. As concerns other greenhouse gases, Norway has significantly reduced the emissions of perfluoridised carbons (PFCs) and sulphur hexafluorides (SF_6) in the aluminium and magnesium industries to a considerable extent due to the implementation of "no-regret" measures which were financially beneficial as well (Figure 21). By contrast, emissions of HFCs have been increasing since 1990 due to the banning of the ozone-depleting gases (CFCs and HCFCs). Nitrous oxides and methane emissions have been roughly stable in the same period.

Under the Kyoto Protocol, Norway has been permitted a 1 per cent increase in greenhouse gas emissions in the period 2008 to 2012 relative to the 1990 outcome, while the national baseline projection points to an increase of 23 per cent. Thus, a reduction of 18 per cent from the baseline will be required. The rise in the baseline scenario is mainly due to increasing carbon dioxide emissions (Figure 21). Oil and gas production is largely responsible for the bulk of the projected increase. The rest of the expected increase in CO_2 emissions stems from mainland economic growth, changes in production structure and the possible introduction of gas-fired power generation.[52] The emissions of the other greenhouse gases are expected to remain broadly stable in aggregate, but this masks diverging tendencies among them. In particular, methane emissions from waste deposited in landfills and animal manure are expected to fall whereas emissions of HFCs will continue to increase as ozone-depleting gases are phased out.[53]

Figure 20. **Carbon dioxide emissions**

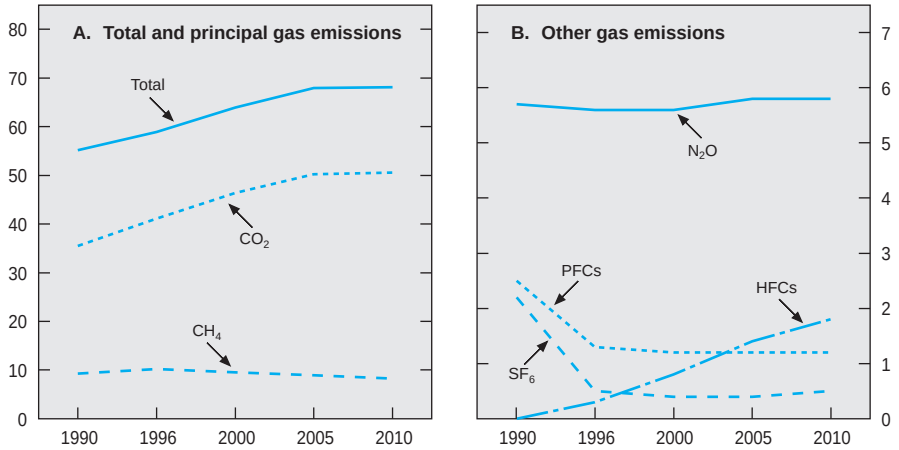




1. Using 1991 prices and purchasing power parities.
2. Average excluding eastern Germany, Czech Republic, Hungary and Poland.

Source: IEA (1998), *CO_2 Emissions from Fuel Combustion*; OECD Environmental Indicators 1998; Statistics Norway and OECD Secretariat.

Figure 21. **Projections for greenhouse gas emissions**
Million tons CO_2-equivalents



Source: Ministry of Finance and State Pollution Control Authority.

The key climate policy instrument used to date is the *carbon dioxide* tax introduced in 1991. The tax rate differs across fossil fuel category and geographic location of the activity (mainland and offshore), see Table 11. Process emission from several export-oriented mainland manufacturing industries, such as the ferro-alloy, aluminium, fertiliser, petro- and other chemicals industries have, moreover, been exempted. Exemptions are also granted to the fishing fleet, aviation, coastal shipping of goods and international shipping.[54] As a result, only about 60 per cent of CO_2 emissions are subject to the tax, and only about 20 per cent of emissions from manufacturing. The exempted manufacturing sectors are all heavy CO_2 emitters and most of them are strongly exposed to international competition. Similar arrangements exist in the other countries with a CO_2 tax, and are motivated by concerns over international competitiveness of individual firms. In Norway such concerns are heightened by the aim to maintain industrial activities in remote areas – CO_2-intensive industries being particularly important there. As noted by the Green Tax Commission, it is difficult to avoid a loss in competitiveness in individual sectors, but this could be offset by overall tax reductions in a revenue-neutral tax reform package.[55] It also pointed out that structural change is the inevitable consequence of the introduction of a CO_2 tax, as its basic aim is to change industrial structures and consumption patterns. An introduction of a

Table 11. **CO_2 tax structure and coverage**

1 January 1998

	Tax rate[1]	Tax per ton of CO_2 emission NKr	*Memorandum:* As a per cent of total CO_2 emissions in 1995
Mainland			
Petrol	0.890	384	13.8
Mineral oil			
Heavy oil	0.445	143	23.8
Light oil	0.445	168	
Pulp, paper and herring meal	0.222	n.a.	
Air transport	..	..	11.7
Shipping of goods[2]	..	..	
Supply fleet in the North Sea	..	..	
Shipping fleet[2]	..	..	
Coke and coal			
Coal used for energy purposes	0.445	183	0.4
Coke used for energy purposes	0.445	382	
Production of cement and leca	..	..	1.3
Coke and coal used for non-energy purposes	..	..	13.3
Gas	..	..	7.5
Offshore			
Oil	0.890	336	18.9
Gas	0.890	382	

1. NKr/l for petrol and mineral oil; NKr/kg for coke and coal; NKr/scm oe for offshore oil and gas.
2. Coastal fishing and coastal goods transport are fully compensated for the CO_2 tax paid on fuel oil consumption.

Source: Ministry of Finance, St prp nr 54 (1997-98), *Grønne skatter,* Oslo.

relatively high and broad-based carbon tax could prove costly if similar policies are not adopted by a significant number of other countries, *e.g.* in the framework of internationally binding commitments. This could imply adjustment in the medium term which may prove undesirable in the long term, if other countries introduce similar measures later on. The industry argues that, in addition, the closing and relocation of CO_2-intensive production to other countries resulting from such a unilateral broad-based CO_2 tax would be ineffective in terms of reducing global emissions. Norwegian companies prefer "voluntary agreements" with the government to reduce greenhouse gas emissions over a CO_2 tax. One exempted industry (aluminium) has recently concluded such an agreement.[56] As with all such agreements, the government lacks a legal basis to enforce them.

Taxes on fossil fuel consumption typically take the form of excise duties and VAT, and represent up to 70 per cent of the market price (Figure 22). Although these taxes do not primarily aim at reducing emissions, they provide the same type of incentive as the CO_2 tax, even though they do not take into account the emission content of the fuel. Taxes are lowest for fuel consumption in industry and highest for car fuels, mostly affecting households. Whereas other tax elements on mineral oil products have been eliminated in steps after the introduction of the CO_2 tax, the taxation of petrol has increased. Norway, therefore, has a relatively high level of petrol tax compared to the North American and Pacific OECD countries although it is in line with the European average (Table 12).[57] By contrast, the purchase tax on cars is high compared to other European countries and since 1996 has been differentiated according to car weight, engine output and engine volume.

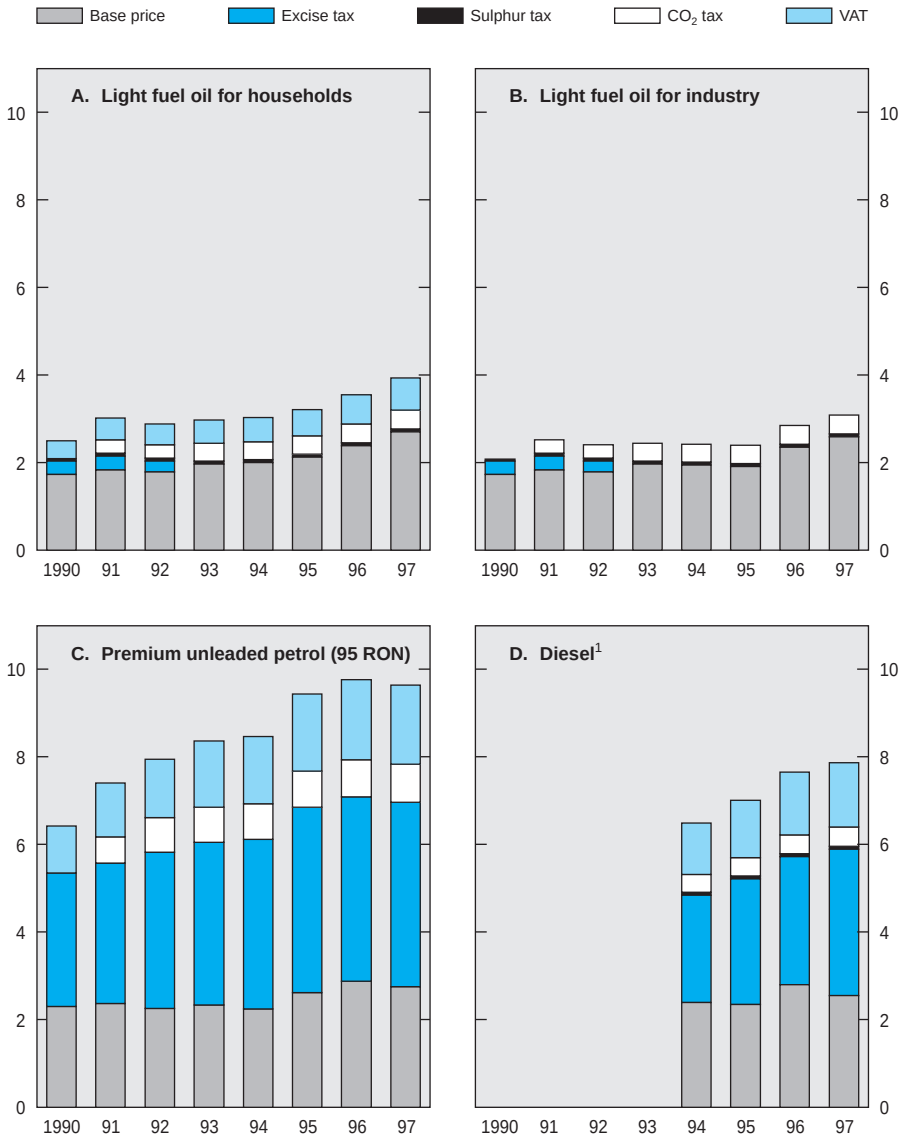
The size of the impact of CO_2 and related taxes on the emission of greenhouse gases is uncertain. A study by Statistics Norway suggests that the CO_2 tax may have reduced CO_2 emissions from households, transport and stationary sources by 3 to 4 per cent in the period 1991-93. There is, moreover, evidence suggesting that the CO_2 tax has prompted lower CO_2 emissions on the Norwegian continental shelf by encouraging the use of energy-efficient gas turbines and the replacement of pilot flames for flaring by electronic ignition systems.

Even if the CO_2 tax may have contained growth in CO_2 emissions, studies by the Environmental Tax Commission in 1992 and the Green Tax Commission in 1996 show that the structure of the carbon tax should be adapted to raise cost-effectiveness. The exemptions and the weak link between the rates of taxation and the carbon content of products imply that the incentives for reducing carbon emission are weakest in industries where marginal abatement costs are lowest. The overall cost of meeting a given CO_2 reduction target, therefore, tends to be too high – in any case higher than if the tax were levied in proportion to carbon emissions and applied across the board.[58] The majority in the Green Tax Commission thus advocated a rapid change to a carbon tax system without exemptions. The weak incentives for emission reduction in the exempted sectors is also conflicting with goals of stimulating environmentally-friendly technological change and may lead Norwegian households and businesses to move towards more polluting economic activities. Although they are probably of less importance compared to environmental policies *per se*, some transport subsidy schemes provided to agriculture, the fisheries and petroleum products used in remote areas as part of regional policy were also identified by the Green Tax Commission as environmentally harmful.

Policies focusing on curbing emissions of *methane* have relied essentially on changing waste treatment, as waste disposal in landfills has been the main source of rising emissions in recent years. Since 1994, landfills generating large

Figure 22. **Prices and taxes for fuel oil and petrol**

NKr per litre



1. Automotive diesel for non commercial use.

Source: IEA (1998), *Energy Prices and Taxes* and OECD Secretariat.

Table 12. **Petrol prices and taxes in international comparison**

US$ per litre,[1] 1996

	Diesel fuel			Unleaded premium		
	Price excluding tax	Tax	Price including tax	Price excluding tax	Tax	Price including tax
Norway	**0.26**	**0.32**	**0.58**	**0.22**	**0.58**	**0.80**
Austria	0.22	0.24	0.46	0.23	0.46	0.70
Belgium	0.22	0.26	0.48	0.21	0.56	0.77
Denmark	0.20	0.15	0.35	0.20	0.46	0.66
Finland	0.21	0.24	0.45	0.20	0.60	0.80
France	0.18	0.32	0.50	0.17	0.67	0.84
Germany	0.18	0.26	0.44	0.18	0.49	0.67
Greece	0.20	0.27	0.48	0.23	0.50	0.73
Ireland	0.46	0.34	0.80	0.28	0.53	0.81
Italy	0.25	0.41	0.66	0.26	0.72	0.98
Luxembourg	0.19	0.22	0.41	0.20	0.37	0.57
Netherlands	0.31	0.29	0.60	0.23	0.60	0.83
Portugal	0.29	0.41	0.69	0.31	0.76	1.07
Spain	0.25	0.31	0.56	0.26	0.54	0.80
Sweden	0.25	0.23	0.48	0.19	0.52	0.71
United Kingdom	0.20	0.47	0.66	0.18	0.58	0.76
Australia	0.20	0.08	0.28	0.22	0.27	0.50
Canada	0.23	0.16	0.39	0.22	0.21	0.42
Czech Republic	0.45	0.48	0.93	0.54	0.85	1.39
Hungary	0.35	0.63	0.98	0.38	0.72	1.10
Japan	0.14	0.17	0.31	0.24	0.29	0.53
Mexico	0.35	0.05	0.40	0.47	0.07	0.54
New Zealand	0.24	0.00	0.25	0.27	0.25	0.52
Poland	0.35	0.21	0.55	0.32	0.47	0.79
Switzerland	0.11	0.31	0.42	0.15	0.33	0.48
Turkey	0.36	0.57	0.93	0.44	0.85	1.29
United States	0.18	0.10	0.29	0.24	0.09	0.33

1. 1991 prices and PPPs.
Source: IEA-OECD.

amounts of combustible gas have been equipped with systems to extract gas for heating purposes or flaring, a technology which has proved successful in reducing emissions. Limiting landfilled waste through greater emphasis on separation, recycling, composting and incineration of waste will also contribute to reducing methane emissions. By contrast, methane emissions from animals in agriculture have not been tackled to date, and the shift from production-related subsidies to subsidies per animal are unlikely to lead to an improvement. Nor have the

authorities established policies to reduce emissions of the other greenhouse gases, apart from a voluntary agreement with the aluminium industry to reduce emissions of PFCs, even though the abatement costs per CO_2 equivalent of these other gases may be lower than for CO_2. The authorities have, however, for many years had a close dialogue with industries including the possibility to reduce these emissions.

In April 1998, the government proposed a reform of environmental taxes which was only partly endorsed by parliament. The proposal comprised the following measures:

- to introduce a CO_2 tax of NKr 100 per ton of CO_2 for the previously exempted sectors in line with the recommendations of the Green Tax Commission. The processing industries (aluminium, ferro-alloy, carbides, etc.), however, would be granted a flat rate compensation per ton of output for carbon emissions unrelated to combustion.[59] Fisheries and air transport were also to be compensated. The compensation schemes aim at avoiding a fall in profitability, but do not undo the incentives for CO_2 emission abatement. The compensation was proposed to be gradually phased out after the entry in force of the Kyoto Protocol, and to be abolished by 2010;[60]
- to introduce a tax of NKr 300 per ton of waste delivered to landfills or combustion plants in order to reduce methane emissions;[61]
- to exempt investment in renewable energy sources (biofuels, windmills and heat pumps) from the 7 per cent investment tax, and to remove the exemption of the automobile diesel tax on diesel oil used in buses, with a compensation scheme for bus services to avoid negative effects on public transport.

The proposed extension of the CO_2 tax to the exempted mainland industries met strong opposition in parliament, which decided to limit the extension of the tax to air traffic (with a compensation), cargo shipping in coastal waters and shipping activities on the continental shelf. Parliament also requested the government to appoint a special Committee to prepare a national system of tradeable emission quotas, and proposed a CO_2 emission reduction target of 30 per cent for the currently exempted mainland industry for the 1990-2010 period (corresponding to 12 per cent for the economy as a whole). This Committee, which should report by the end of 1999, is mandated to examine several options, as a minimum including the introduction of tradeable emission quotas in mainland processing industries, currently exempted from CO_2 tax.[62] The domestic quota system shall be linked to the Kyoto-mechanisms. In addition, a choice needs to be made between handing out emission quotas for free – which amounts to a subsidy from other parts of society – or to capture part or all of the "resource rent" by selling the quota, either at a pre-set rate or through auctioning or tendering.[63] The

parliament endorsed the government proposals on diesel, waste and tax exemptions for renewable energy sources. Since the reform is intended to be revenue neutral overall, the net proceeds should slightly reduce non-environmental income taxes.[64]

Even if Norway could achieve the target of reducing CO_2 emissions by 30 per cent in mainland industry by 2010, this would not suffice to comply with the Kyoto target. In fact, to achieve it through a broad-based CO_2 tax would impose a significant burden on the economy, as this would require an estimated tax rate in the range of NKr 250 to NKr 350 per ton of CO_2, a level which is currently attained only in the offshore oil and gas sector and for car petrol.[65] Such a tax rate would dwarf the theoretical world-wide quota price of NKr 125 per ton of CO_2,[66] reflecting the much higher abatement costs in Norway. On the other hand, a CO_2 tax would also have ancillary benefits by leading to reductions in other pollutants. Norway would greatly gain from buying CO_2 emission quotas abroad if that were possible. Indeed, the Ministry of Finance estimates, in a partial exercise that does not include the response of other signatories, that attaining the Kyoto targets by using emission trading and other "flexibility mechanisms" (investing in clean-up projects abroad to obtain additional emission rights) would reduce the annual costs of compliance with Kyoto to a third (from 0.6 to 0.2 per cent of GDP), compared to the most cost-effective domestic solution (Table 13).

If all signatories to the Kyoto agreement were to faithfully implement reduction targets, this could have a considerable effect on the oil and gas price. The impact is highly uncertain, however. It would, for instance, strongly depend on whether countries implement cost-effective policies or not. In a partial exercise, excluding Norway's response to climate change, contained in the Long-term

Table 13. **Macroeconomic effects in 2010 of meeting the Kyoto Protocol obligations**

In million tons CO_2 equivalents as compared to the reference scenario

	Scenario 1 Cost-effective using flexible mechanisms	Scenario 2 Cost-effective domestic solution
Total greenhouse gas emissions	68.1	68.1
Reduction in non-CO_2 greenhouse gases	3.7	4.2
Reduction in CO_2	1.5	8.1
Emission reduction abroad through different flexibility mechanisms	7.1	0.0
Total yearly costs[1] (1997 billion NKr)	2.0	6.0

1. Change in real net disposable income in Norway compared to the reference scenario as a result of: abatement and adaption costs in reducing CO_2 emissions, abatement costs in reducing emissions of other greenhouse gases and costs following the use of the flexibility mechanisms.

Source: Ministry of Finance.

Programme 1998-2001, it was assumed that crude oil prices could fall by 15 to 20 per cent in 2010, while gas prices would not be influenced. Such an oil price fall would reduce real national income by NKr 15 to 20 billion. Adding the two partial exercises suggests that the overall income loss could be more than 2 per cent of GDP, thus imposing a heavy burden on the economy.

Ozone depletion

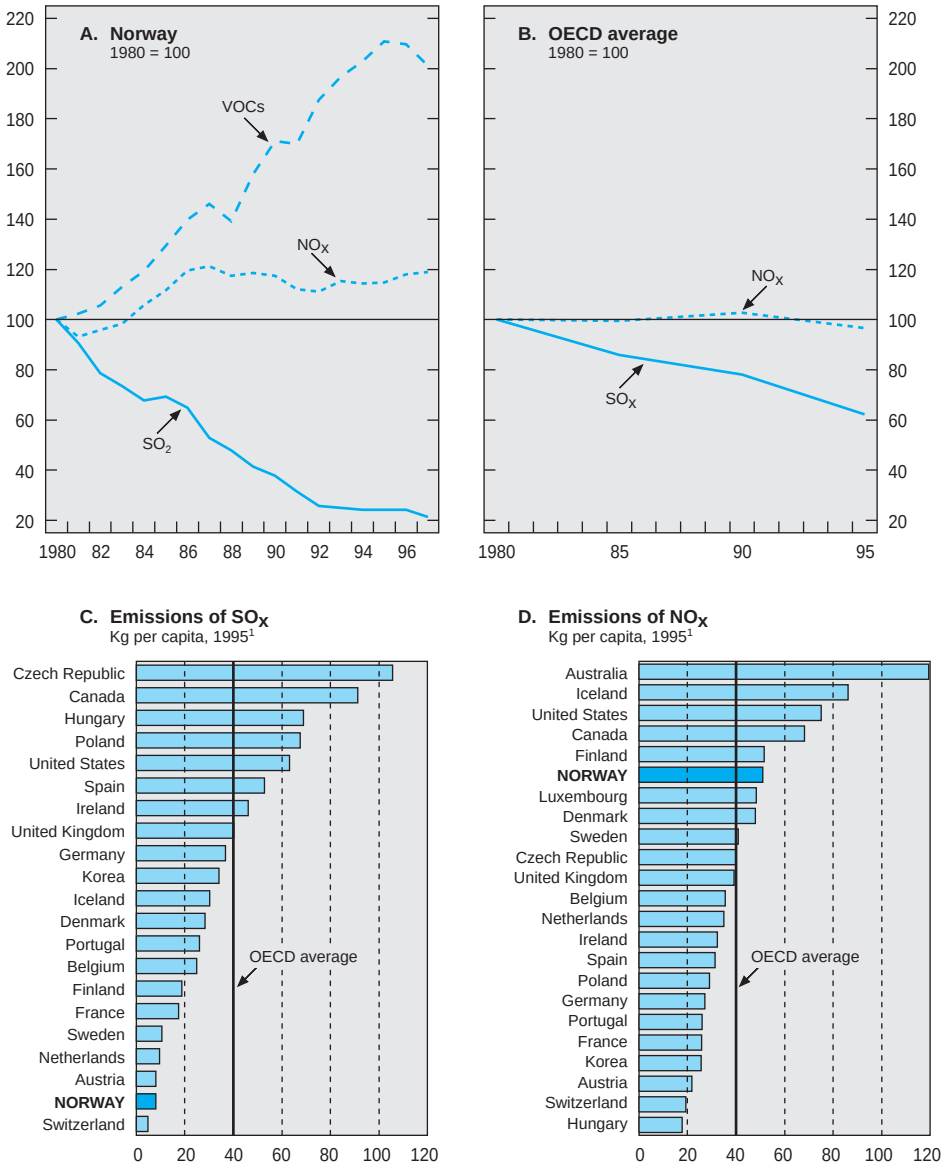
Subsequent to the Montreal Protocol signed in 1987 and revised in 1995, Norway has reduced and gradually eliminated consumption of all substances susceptible to depleting the stratospheric ozone layer protecting the earth from UV radiation. The most important ozone-depleting substances are chlorofluorocarbons (CFCs) and halons which are used mainly in cooling systems, dry cleaning and fire-extinguishers.[67] National regulations in Norway imply a faster phasing out for several substances relative to the Protocol's timetable, in line with EU regulation.[68] As a result, the imports of newly-produced halons and CFCs have been entirely eliminated, while recycled substances are still accepted.[69] The tight timetable required strict regulation and reduced the scope for the use of economic instruments such as a tax.[70] However, a tax to meet the reduction targets for ozone-depleting substances with a longer phasing out period – *e.g.* extending to 2015 for HCFCs – is being considered.

Transborder pollution problems

The Norwegian territory is affected by a number of transboundary environmental problems, most prominently:

- *acidification*, especially in the southern and western parts of the country and close to the Russian border, caused by atmospheric emissions of sulphur dioxide and nitrogen oxides which, through "acid rain", lead to deposition of sulphur and nitrogen in the soil and water surfaces. The damage in Norway is relatively important due to its lime-poor rocks and thin soils which have little capacity to neutralise acid deposition – *i.e.* low "critical loads". This combination of factors has caused death of aquatic life in surface waters and weakened the vegetation resistance capacity against drought, cold and frost, of Norwegian forests.[71] The bulk of the sulphur and nitrogen deposition, 95 and 86 per cent, respectively in 1994, stems from long-range air transport from the United Kingdom, Central Europe and Russia. Domestic emissions, in turn, play a relatively minor role, mainly because there is no SO_2 emission from coal and oil-fuelled power generation, as reflected in a low SO_2 emission intensity by international standards (Figure 23);
- high concentrations of *tropospheric ozone*. Ozone is formed in the lower layers of the atmosphere when NO_x and Volatile Organic Compounds

Figure 23. **Emissions to air**



1. Or latest year available.

Source: Statistics Norway, OECD Environmental Indicators 1997 and OECD Secretariat.

(VOC) react when exposed to strong sunlight. A too high concentration of ozone in the air can cause respiratory problems and damage to vegetation and the ecosystem at large. Contrary to acidification, pollution by ozone in the lower atmosphere is less pronounced in Norway than in the rest of Europe, due to the specific climate conditions in the region.[72] Ozone concentration in Norway is largely due to long range transport of ozone from other European countries and varies widely from year to year, depending on emissions abroad and meteorological conditions. The highest ozone concentrations are found in southern coastal areas of Norway;

– *eutrophication* in the North Sea area. The release of nitrogen and phosphorus lead to nutrient enrichment which promotes the growth of plants and algae; the algae reduce light penetration and, when they die, consume oxygen in the water, thereby damaging marine life. Both Norwegian discharges and long-range transport of nutrients by ocean streams contribute to marine eutrophication but, unlike the cases of acidification and ozone, Norwegian sources are predominant in the areas most affected – including municipal and industrial waste water, fertilisers and animal manure.[73] Norway's use of fertilisers is rather intensive compared to other OECD countries,[74] owing partly to its highly protectionist agricultural regime. A recent shift in agricultural subsidies from price support to income support related to acreage size has reduced the incentive for intensive cultivation somewhat. The Norwegian Producer Subsidy Equivalent has remained one of the highest in the OECD (see Chapter II), however, and is likely to be a major influence on production and environmental pressures. In addition, it is largely offsetting the effect of the fertiliser tax.

Transboundary pollution problems are dealt with within the framework of international treaties (Table 14). Norway is co-operating with other European countries, the United States and Canada under the framework of the UN-ECE Convention on Long Range Transboundary Air Pollution (CLRTAP). The Sofia and Oslo Protocols deal with the acidification problem, and the Geneva Protocol with tropospheric ozone. With the 1987 North Sea declaration, North Sea countries addressed the problems of eutrophication and pollution with toxic substances in the North Sea. Apart from these international agreements, Norway has co-operated bilaterally with Russia, regarding air pollution produced by a Russian nickel smelter located near the Norwegian border, in the arctic Finmark region.

Once targets are agreed among the countries concerned, the design of policies to reduce the emissions is rather complex, since the gases and substances contributing to transboundary pollution problems have different local pollution effects. Hence, policy instruments should be differentiated according to the damage they cause in the different parts of the country. The best way to

Table 14. **Norwegian policy targets for transborder environmental problems**

Type of emissions/inputs	International targets	National targets
Sulphur	Oslo Protocol (1994) – 76 per cent reduction by 2000 with 1980 as a base year	
NO_x	Sofia Protocol (1988) – Stabilisation by 1994 with 1987 as a base year	30 per cent reduction by 1998 with 1986 as a base year
VOC	Geneva Protocol (1991) – 30 per cent reduction by 1999 with 1988 as a base year	
Nitrogen and phosphorus	North Sea Declaration (1987) – 50 per cent reduction with 1985 as a base year in the nine North Sea countries	

Source: OECD Secretariat.

proceed would be to solve local problems as a priority, and use national instruments if necessary to respect the international commitment. Norwegian policies in this regard, indeed, apply this principle to a certain extent.

Acidification

Prompted by its exposure to acid rain from foreign sources, Norway has been a major driving force behind the protocols on acidification. The early protocols (Helsinki 1985 and Sofia 1988) set uniform targets for participating countries for reducing SO_2 and NO_x emissions, and hence were not geared towards a cost-effective solution. In order to raise cost-effectiveness, Norway supported the initiative for the more recent Oslo (sulphur) Protocol, which established targets per country according to critical loads and abatement costs, and participated in the scientific underpinning of the protocol.[75] For Norway, this implied a reduction target of sulphur emissions of 76 per cent by 2000 relative to the 1980 level. In the same spirit, Norway is strongly involved in the preparation of the future NO_x Protocol, which is expected to be concluded in 1999. This protocol is to be multi-target, coherent and science-based, covering acidification, eutrophication and tropospheric ozone, and including emissions of NO_x, SO_2, NH_3 and VOCs. Helped by the various protocols, European emissions of SO_2 have declined steadily since the early 1980s while a less rapid fall in nitrogen emissions has been observed since 1990.

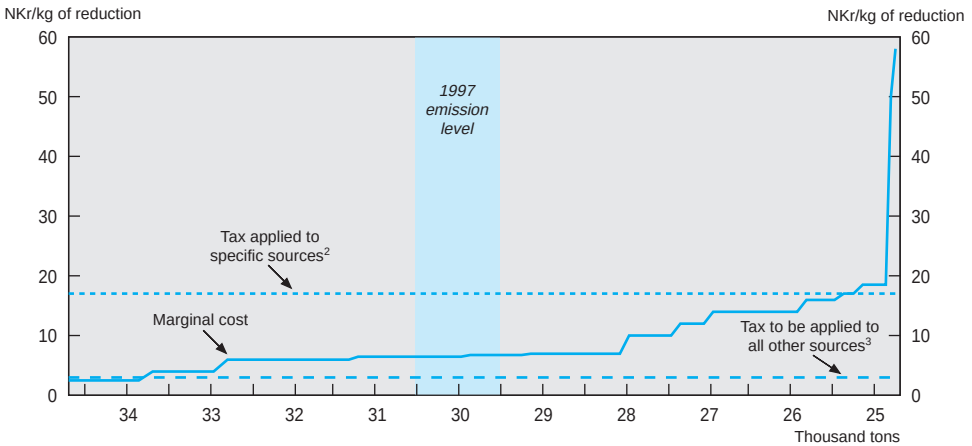
The Norwegian authorities have concentrated major efforts on reducing SO_2 emissions from large industrial sources. In 1996, 60 per cent of the SO_2 emis-

sions originated from these sources, particularly heavy metal industries, while the remainder was due to stationary combustion and mobile sources, contributing 25 and 13 per cent of total SO_2 emissions, respectively. Pursuant to the Pollution Control Act, discharge permits have been the main instrument used to reduce industrial emissions.[76] Emissions from stationary combustion have been reduced mainly through ceilings for the sulphur content of heating oil, which are differentiated by region to reflect variation in local problems associated with sulphur emissions.[77] To supplement the sulphur content regulation, the tax on sulphur in heating oil, in use since 1975, has been gradually increased up to a level of about NKr 17 per kg of SO_2.[78] The tax covers about 70 per cent of the combustion emissions of SO_2 or around 20 per cent of total emissions. Coke and coal are not subject to the tax, and non-combustion emissions from industrial processes are exempted. Tax exemptions are motivated by the fear of competitiveness losses of heavy energy-consuming industries – the same as with the CO_2 tax.

These measures have allowed Norway to reduce its current SO_2 emissions below the target set by the Oslo Protocol, as SO_2 emissions decreased by 78 per cent between 1980 and 1997, slightly more than the target set for the year 2000 (Figure 23). However, emissions are expected to rise again towards the year 2000, due to a projected increase in fuel oil consumption. Hence, more needs to be done to respect the Oslo Protocol. The creation of a market for SO_2 emission quotas has been considered as an alternative, but would, among other things, imply higher administrative costs than a SO_2 tax and distortions may arise due, for instance, to market power, given the small size of the market. Last spring, following the recommendation of the Green Tax Commission, parliament approved the extension of the SO_2 tax to the sectors and products previously exempted, at a low rate of NKr 3 per kilogramme of SO_2. Moreover, refineries will be taxed directly on the basis of their emissions. For the time being, the level of the tax is, however, too low to have a strong effect on emissions given that the rate is above marginal abatement costs only for small emission reductions (Figure 24).

While SO_2 emissions in Norway have been substantially reduced to date, NO_x emissions have increased by 19 per cent since the early 1980s. Mobile sources generate 75 per cent of the NO_x emissions – more than half of it from shipping and the rest from road traffic – while the oil and gas sector on the continental shelf is the dominant stationary source for NO_x emissions. The quantity of emissions depends on the combination of four main factors: the nitrogen content of the fuel, the combustion technology used, the operation and maintenance of the equipment and, since NO_x can be removed, the purification technology. Hence, there is only a weak link between the amount of emission and the amount of fuel combusted, and therefore a fuel tax is highly inefficient. At the same time, since the emissions originate mostly from mobile sources, which are very difficult to monitor, a tax based on measured emissions would also be costly.

Figure 24. **Marginal cost of reduction in sulphur dioxide emissions**[1]



1. Calculated in 1994-95, starting out from an emission level of 36 000 tons.
2. Primarily covering diesel oil and heating oil.
3. The parliament has decided to implement a tax of 3 NKr/kg SO_2 for sources not covered by the current tax as from 1 January 1999.

Source: Ministry of Environment.

This means that the instruments need to be adapted to the processes underlying the NO_x emissions, and based on an analysis of the costs and benefits associated with the abatement options available for each sector. Moreover, as NO_x emissions contribute to a range of environmental problems – acidification, eutrophication, tropospheric ozone and local pollution – the measures should be tailored to local conditions.

In order to comply with the target set in the Sofia Protocol – the stabilisation of NO_x emissions at their 1987 level by 1994, a variety of instruments have been implemented in Norway:

- as regards *road traffic*, the major focus has been to reduce NO_x emissions through exhaust emission criteria for vehicles, which has resulted in a decline in emissions from gasoline-powered motor vehicles since 1987. Since 1989, all new gasoline-powered cars have had to be fitted with a three-way catalytic converter, but due to the relatively long lifetime of private cars[79] in Norway, only 39 per cent were equipped by 1997. In the years ahead, emissions from road traffic are expected to decrease further as the car park is renewed;
- emissions from *stationary combustion* have been tackled by discharge permits for some industrial plants.

Table 15. **Emission intensity of various forms of transport**

Emissions per person/kilometre, 1993/94

	CO_2 gram	NO_x mg	Particulates mg
Air transport	182	516	..
Train[1]	61	189	20
Cars using petrol[2]	106	606	11
Cars using diesel	86	313	119
Buses[3]	80	950	62
Car ferries	926	14 600	146
Other passenger ships	818	18 060	129

1. Estimates based on electricity produced in gas-powered plants.
2. High emissions of NO_x are due to the fact that only a minor part of the fleet was equipped with catalytic converters at this time.
3. Average of 11.32 passengers.

Source: Institute of Transport Economics, 1997.

Limited measures have been taken regarding shipping, even though emissions associated with shipping have become the dominant source of NO_x emissions: emissions of NO_x per person-kilometre of traditional passenger ships and high speed passenger ships are, respectively, three and 14 times those of private cars and buses (Table 15). In the case of gas flaring on the continental shelf, emission reductions of NO_x have been achieved as a side effect of the introduction of the CO_2 tax. Emissions from the petroleum sector are, nonetheless, expected to rise due to increased activity. The authorities have given priority to reducing NO_x emissions from urban traffic and from combustion for heating as their impact on the local environment is important. On the other hand, the government has preferred to protect the shipping industry, including fishing, for regional policy purposes. Fishing vessels, however, do receive a subsidy targeted on increasing energy efficiency.

Excessive concentration of tropospheric ozone

In combination with NO_x, VOC emissions generate tropospheric ozone. VOC emissions have risen steeply in Norway since the late 1970s with the development of the offshore oil and gas fields. Indeed, per capita emissions of VOCs are currently among the highest in Europe. Offshore VOC emissions, which account for more than half of the total in Norway, result from evaporation during the shipment and transfer of crude oil in the North Sea. To date, government action has focused on support for research and development to reduce these emissions.[80] The main onshore source of VOC emissions is road traffic. These emissions have been contained in the 1990s by the adoption of catalytic convert-

ers, emission standards for trucks and measures to reduce emissions during filling at petrol stations. Despite these measures, the target of reducing VOC emissions by 30 per cent from 1988 to 1999, established in the Geneva Protocol, will be largely missed.[81] Development of new technology which will make it possible to mitigate these emissions has taken longer than expected. However, rapid deployment of such technology is expected to be stimulated through the agreement under preparation between the government and the petroleum industry. The emission of VOCs from loading crude oil accounts for 60 per cent of total emissions.

Eutrophication in the North Sea

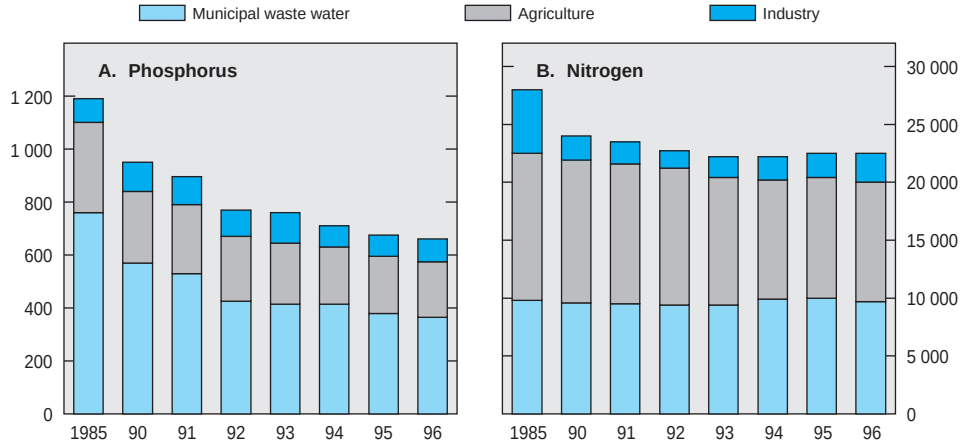
To ease the marine eutrophication problem, the North Sea countries have agreed to a series of far-reaching commitments in the North Sea Declaration to reduce discharges of nutrients.[82] Norway is committed to reduce discharges of nutrients by about 50 per cent compared to 1985 levels in areas of the southern part of the country.[83] A series of measures have been taken:

- since the mid-1980s, the construction and upgrading of sewage water treatment plants is facilitated by government loans and grants to cover the capital costs of the required investments.[84] Consequently, the density and quality of the waste water treatment capacity in the North Sea region is much higher than in the rest of the country. Moreover phosphate detergent has been banned;[85]
- to reduce nutrient runoff in agriculture, regulations for better utilisation of animal manure have been adopted, and support is given to farmers for leaving areas that are particularly vulnerable to erosion under stubble during the winter. The use of fertilisers, moreover, is taxed, at a rate of 19 per cent of the purchase price for nitrogen and 11 per cent for phosphorus. As noted above, the current agricultural subsidisation scheme tends to offset the effect of the fertiliser tax.

Nitrogen discharges to Norwegian North Sea coastal water were reduced by 22 per cent between 1985 and 1995, with the bulk of the reduction originating from industry while discharges from agriculture and municipal waste water remained more or less constant (Figure 25). North Sea discharges of phosphorus, however, have decreased by 44 per cent in the same period, reflecting the emphasis of the policy effort in this regard, which is justified by the fact that phosphorus has been the critical factor of eutrophication in Norwegian rivers and lakes. In agriculture, phosphorus fertilisation has been substantially reduced since the 1980s, whereas nitrogen fertilisation has only marginally decreased – which partly explains why runoff of nitrogen has been reduced less than that of phosphorus. Recent research indicates that a significant reduction of nitrogen

Figure 25. **Water pollution**

Waste input to the North Sea coastal area, in tons[1]



1. The area from Østfold to Vest-Agder. Calculated inputs to the coastal zone outside this area, particularly from agriculture, are uncertain.

Source: Norwegian Institute for Water Research.

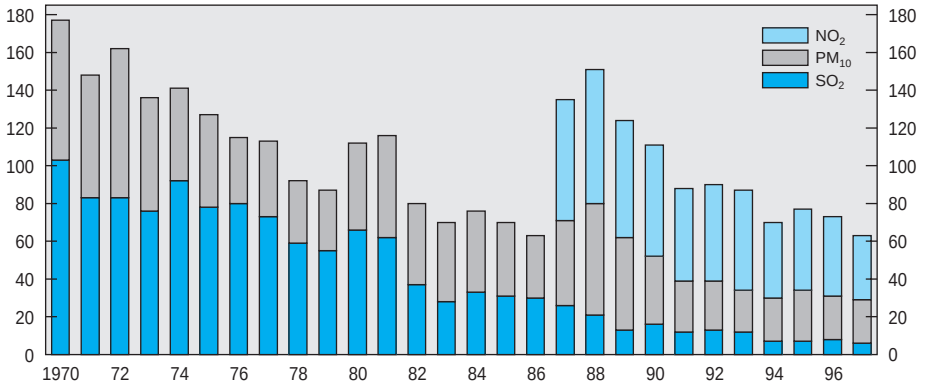
runoff in the agricultural sector could be achieved by combining an increase of the tax on nitrogen fertilisers with specific agronomic measures adapted to local conditions.[86]

Local environmental problems

Norway is privileged by its geographical situation which, together with a low population density, implies a limited burden on the local environment. Local problems, nevertheless, exist:

– *urban air and noise pollution*. Local air and noise pollution concern mainly the urban areas which are concentrated in the southern part of the country. Overall, air quality as measured by the annual average concentration of pollutants in Norwegian cities is relatively good, and the concentration of SO_2 and lead in urban areas has been substantially reduced over the last decades (Figure 26).[87, 88] The Norwegian Pollution Control Authority estimated the annual cost of damage to health resulting from air pollution at NKr 3.8 billion in 1998 for Oslo and Bergen only, and NKr 2.5 billion due to noise for the whole country;

Figure 26. **Air pollution in Oslo**[1]
Midwinter, urban area, µg/m^3



1. The data concerning particles is the measurement of black soot at one location in the period 1970/71-1994/95, and of PM_{10} at another location for the years 1985, 1986 and 1989-97. The two series are not directly comparable. The stations are not directly influenced by road traffic and show background values.

Source: Luftforurensninger i Oslo, Årsrapport 1998.

– in addition, according to the State Pollution Control Authority, about 25 per cent of the population experience noise as a problem at their homes, and 260 000 persons suffer from serious discomfort as a result of road-traffic noise. While road traffic is by far the most important source of local air pollution and noise, there are two emission sources of particles which are specific to Norway: heating systems for buildings – wood-burning stoves in particular – and the use of studded tyres;
– the emission and accumulation of *hazardous substances*, which potentially represent the most acute threat to human health and the environment, and also contributes to transborder pollution, has been significantly reduced. In the past, some fjords were heavily polluted by discharges of toxic substances by local manufacturing industries, which seriously damaged the marine life. However, these "point source" emissions have been significantly reduced through discharge permits, and the most hazardous substances have been banned. Lead emissions are being reduced as leaded petrol is progressively disappearing from the market;
– *waste*, although not an environmental problem *per se*, can give rise to various environmental problems and health damage. As in most other

OECD countries, waste production has been increasing exponentially since the 1970s, reflecting production and consumption trends.[89] Hazardous waste represents slightly more than one per cent of total waste, which needs to be treated separately.[90] Non-hazardous waste, however, may also pollute water, soil and air. Waste deposited in landfills causes leakages into groundwater and the soil, smells, and generates emission of methane and other damaging gases. The incineration of waste, moreover, generates emissions of acidifying substances and toxic gases.

Urban air and noise pollution

The Norwegian government has in 1998 adopted new national goals for the reduction of urban air pollution. New goals were adopted for SO_2, benzene, NO_2 and particulate matter.[91] Regarding *road traffic* which, as noted, is the most important source of air pollution and noise, the Norwegian authorities have deployed a large number of instruments, both national and local. A *first* group of measures aims at reducing emissions per vehicle-kilometre in line with EU regulations. Emission standards have been fixed, which led to the introduction of catalytic converters for the removal of a substantial part of NO_x, VOC and CO_2 emissions from petrol-fuelled cars – but much less so for diesel-driven cars. The standards on the sulphur content of oil, the sulphur tax and the tax on the lead content of petrol have helped to reduce sulphur and lead emissions. Noise criteria for cars have also been adopted. A *second* group of measures aims at limiting the volume of road traffic by increasing its price *vis-à-vis* other means of transport. As noted earlier, taxes on petrol have been raised, but less so on diesel. Toll systems for all inbound vehicles have been established around the city centres of Bergen and Oslo. While these toll rings merely serve to raise funds for investment in the local road infrastructure to reduce congestion,[92] the toll ring in Trondheim is to some extent used for traffic management, as the toll fee is somewhat higher during peak hours and seasonal tickets are not available.[93]

Total support to public transport (subsidies and transfers to buy transport services) amounts to close to NKr 4 billion per year. These subsidies are not primarily directed towards solving local environmental problems, which means that a reallocation of support could have substantial environmental benefits. Support is, among other things, provided to air transport, bus transport in remote areas and high-speed passenger ships, all of which give rise to high emissions per person kilometre (Table 15).

Aside from road traffic, a series of other measures have been taken in recent years to improve the air quality in urban areas. Emission standards for wood-burning stoves have been included in the new building regulations, and they are expected to reduce emissions from this source by 70 per cent in the

long-run. Standards for studded tyres have also been adopted and the local measures to reduce the use of studded tyres is currently discussed. Moreover, parliament passed legislation last spring to entitle local authorities to impose charges on the use of these tyres. An inter-ministerial report concluded that a sharp reduction in the use of studded tyres in the four major urban areas could provide a net benefit of around NKr 500 million per year due to *e.g.* improved health and less corrosion. To address a specific noise pollution problem around the airports of Oslo and Bodø, a charge on aircraft noise has been introduced: the aircraft landing and departure fees are differentiated according to the noise level they create and the noise problem. This charge is earmarked for use within the sector, in contrast with most other environmental taxes in Norway. The opening of a new airport outside Oslo replacing the Fornebu airport has reduced the associated noise problem, but no decisions have yet been taken concerning the aircraft noise charge.

Hazardous substances and waste

The problems regarding air and surface water discharges of hazardous substances by industry have been largely resolved in the past decade. Increasingly, the authorities are focusing on emissions from the use and disposal of hazardous chemical products and substances such as solvents, anti-fouling paint and batteries. This requires the use of instruments which are targeted on the individual substances. Systems of classification, labelling, registration and, if needed, banning of hazardous chemicals have been implemented and appear to function well. The Green Tax Commission, however, saw some scope for greater use of taxes to reduce these product-related emissions, or emissions from individual substances. A refund deposit scheme for nickel/cadmium batteries and a tax on various other products, such as fodder used in fish farming which contains antibiotics, should also be considered.

The overall waste management policy objective, as set out in a 1992 report to parliament, is to ensure that waste problems are solved in a way that minimises environmental and health problems, and also minimises the use of resources. The strategy of the Norwegian authorities to reach this objective is to *i)* minimise waste generation; *ii)* promote the re-use and recycling of material and the extraction of energy from waste; and *iii)* ensure that remaining waste is disposed of in an environmentally sound way. The Norwegian authorities also have specific targets, such as for the recycling of packaging waste which, in line with EU directives, requires that between 50 and 65 per cent of the total packaging waste should be recycled.

Since 1995, the Pollution Control Act requires that municipalities charge fees for waste treatment that fully cover their expenses – an objective that has been largely achieved[94] – and that promote waste reduction and recycling. The

introduction of stricter standards for landfills and incineration in line with EU directives has contributed to increase the costs of waste treatment, and as such gives incentives to recycle. To reduce litter and illegal disposal of hazardous waste and encourage recycling, state-run deposit-refund systems have also been established for lubricating oil and scrapped cars. A tax on beverage containers has been introduced and consists of two elements: *i)* a general tax on all containers differentiated by the degree of recycling potential; and *ii)* a surtax on containers which are not re-useable.[95] The first tax element gives producers an incentive to establish a deposit-refund system. Such systems were introduced for glass (currently 98 per cent return) and plastic beverage containers. The latter tax has turned out to be prohibitive for beverage cans, so they were not introduced. In fact, recycling beverage cans rather than glass would be more friendly to the environment. The government has tried to change the current incentive structure several times, but parliament never passed the legislation.

The Norwegian authorities have also made an active use of regulatory measures, "voluntary agreements" and subsidies to promote waste recycling. If voluntary schemes do not produce satisfactory results, the Ministry of Environment has a mandate to require the municipalities to introduce sorting of waste at source. Industry, which treats 85 per cent of its waste on site or through private waste management companies, may also be required to recycle their waste. "Voluntary agreements" have been signed between various industry branches and the Ministry of Environment regarding paper, cardboard, glass, lead accumulators and car tyres. The government is also providing financial support to municipal recycling and source separation projects.[96] These measures have resulted in a steep increase in the proportion of waste recycled from municipal collection schemes, from 9 per cent in 1992 to 22 per cent in 1996 for households, and from 8 to 18 per cent for industries.

The cost-effectiveness of increased recycling is in doubt. A recent study by Bruvoll (1998), based to a large extent on United States empirical data, for example, suggests that recycling of paper and plastic waste, except for commercial paper waste, is probably less cost-effective than incineration or landfilling. There is some evidence that the costs of collection, transport and production processes associated with recycling, as well as the environmental costs of emissions from transport may be underestimated. Moreover, new technologies have reduced the environmental costs of incineration and landfill. These conclusions may not be valid for every single municipality, as they depend on, for example, the density of population and the distance to the various treatment facilities. The use of purely national targets for recycling are, in any event, inappropriate. Hence the necessity to examine the various options at the local level.

The above-mentioned study also suggested that waste minimisation at the source is the most cost-effective alternative for the categories of waste

examined. This could be achieved by a tax at the source – *i.e.* the products which generate the waste – with the rate varying according to the marginal treatment costs. The introduction of a tax of NKr 300 per ton of waste deposited to landfills or incinerated approved by parliament last spring – as recommended by the Green Tax Commission – constitutes an important step in this direction. The tax level does not cover all environmental costs of waste involved, and is still very crude as there is no differentiation with regard to the treatment cost.[97] However, the introduction of a differentiated tax would be quite complex and costly in practice, while raising the overall tax level would provide a stronger incentive to deposit waste illegally.

Biodiversity

Biodiversity derives its importance from the interaction of mankind with other forms of life, for instance, for recreational reasons, use of genetics by industry or the existence value of species. It is estimated that about 40 000 species of plants and animals exist in Norway. Human-induced extinction of species has continued. About 45 species are known to have become extinct in the last 50 years, and almost 500 are considered to be endangered or vulnerable. Loss of biological diversity is caused by a wide range of factors which influence the environment, such as land use, over-exploitation or pollution. The expansion of towns and the agricultural area has implied serious losses of areas of natural habitat, swamp forests, wetlands and other ecosystems. Policy aims at protecting endangered and vulnerable species, and to restore biodiversity, where possible, for instance, by expanding the area of national parks to 13 per cent of the total Norwegian area by 2008. In addition, ensuring that agriculture, forestry and fisheries are harvesting in a sustainable way is of particular importance. Currently, no cost-benefit analysis is pursued in this area, which would allow prioritisation. Even though uncertainties are large, there is probably scope to establish broad guidelines.

Assessment and challenges ahead

Ensuring sustainability

The evolution of human and fixed capital, of the oil and gas wealth as well as its transformation into financial capital, and of forests, suggests that total national wealth has increased in the past. While trends are much more difficult to judge, natural capital apart from the oil and gas stock and forests appears to have also risen overall since the early 1980s. However, the dwindling of fish stocks has only been partly reversed, keeping biodiversity intact is not easy and local environmental problems persist. While total wealth has probably increased in the past, it could have accumulated faster – or, conversely, consumption could have

been higher – with greater cost-effectiveness of policies in various areas and a better integration of different policies. Finally, the implementation of climate change policies poses a formidable challenge.

Norway is in the very privileged position of possessing a large national wealth in the form of its oil and gas stores and the financial assets ensuing from its development. The move towards channelling a greater part of the rent from the oil and gas sector into foreign financial assets, initiated several years ago, should help ensure the inter-generational balance by providing more freedom of choice for future generations. Accordingly, the generational accounts for Norway, which are routinely updated for every national Budget, suggest that the government's strategy to achieve sound public finance in the long run could succeed. There are some concerns, however, that the government has not fully succeeded in capturing the entire natural resource rent associated with the oil and gas stock. Other countries have chosen to capture some of this rent through auctioning of production acreage rather than only taxing the rent – the approach which is prevalent in Norway. The Norwegian authorities should be encouraged to examine the possibility of auctioning, the more so since a shift from pure taxation to auctioning would reduce the volatility of the annual flow of revenues associated with variations in the oil price.

Norway, being the world's second largest fish exporter, has a clear interest in the application of international agreements to curb overfishing in the Northern Atlantic area. Fish stocks fell considerably between the 1960s and early 1980s, when several species were almost driven to extinction and the marine ecosystem severely damaged. Even though the situation has improved since then, overfishing, in particular in the North Sea, remains a serious problem. The use of Total Allowable Catches to limit fishing remains the dominant international management tool, but there are strong incentives for fishery nations to lobby for large quotas which in aggregate are unsustainable – due mainly to excess capacity of fishing fleets. Moreover, fishery control policies and measures are often either inadequate or not sufficiently enforced. There is little Norway can do on its own, apart from reforming policies that have led to excess capacity in its own fishing fleet.

Over the last two decades, environmental policy in Norway has been fairly successful in reducing the emissions of a wide range of pollutants. In a few areas, however, environmental pressures have continued to rise. Since a substantial part of pollution in Norway is imported from abroad, policy also focuses on negotiating, implementing and complying with international agreements. In this regard, Norway stands out by its valuable efforts to stimulate international co-operation on global and transboundary environment policies. In many domains, this is clearly in the country's best interest, given its vulnerability to *e.g.* acidification and marine pollution originating from neighbouring countries. In other cases,

for instance, climate change agreements, this may conflict with Norway's national interest because it is a major global oil and gas exporter. However, climate change has been an important preoccupation of Norwegian governments, and this preoccupation is shared by the overwhelming majority of the people.

Improvements in Norway's local environment are to some extent interlinked with policies geared towards complying with international agreements addressing cross-frontier pollution – *e.g.* acidification and eutrophication. Local environmental policies are, moreover, in many instances aligned with EU regulations following Norway's membership of the EEA. Pollution from hazardous substances due to industrial activities along the fjords has been significantly reduced, while the most hazardous substances have been banned. Progress in the area of waste reduction and treatment aimed at reducing emissions and leakage into the environment has been marked as well. Nevertheless, from time to time the few urban areas in Norway face serious air and noise pollution associated with car traffic. There are also concerns that the expansion of towns and agricultural activities contributes to further losses in biodiversity. The scope for enhancing the environmental sustainability of local economic activities thus remains significant. Policies in this regard should be guided by extensive and systematic cost-benefit analysis.

Fostering cost-effectiveness

The current policy framework could be improved upon in many respects, which would either raise wealth accumulation and future consumption potential or allow higher current consumption. As concerns oil and gas extraction policies, the elaborate regulatory framework for the licensing of exploration and development acreage has created favourable conditions for participation of foreign operators and participants on the shelf. Foreign companies often welcome a certain degree of state involvement, including the presence of a large state company on national territory. However, there is scope for improvement. In particular, companies bidding in licensing rounds should be allowed to form their own partnerships also in the Norwegian Sea, as the system of "arranged marriages" by the government in which the fully state-owned company Statoil frequently participates does not necessarily yield the best team. In particular, the government should allow a greater variation in participants in each license, possibly based on auctioning of licenses (see above). Concerning licensing in the fishing industry, the existing system of transferring fish quotas between licensees could be changed by delinking quotas from vessels to facilitate re-allocation of quotas to the most efficient parts of the fleet. As a further step, the government could consider the auctioning of the fish quotas to vessel owners in order to increase and extract the resource rent. Such a reform would, however, clash with regional policy objectives.

Climate change policies in Norway will be geared towards meeting the commitments arising from the Kyoto Protocol concluded in 1997. There is a potential for enhancing cost-effectiveness of these policies which resides in a greater use of instruments geared to equalising marginal abatement costs across all economic activities. At present, about 60 per cent of all CO_2 emissions are taxed. Exemptions are still granted to heavy industries, fishing and international shipping.[98] Moreover, rates of taxation vary considerably and are unrelated to the carbon content of products and activities. Ironically, the incentives for reducing carbon emission are generally weakest in industries where the marginal abatement costs are lowest. Moreover, the current tax structure may lead to a bias towards CO_2-intensive activities in the longer run, which conflicts with the objectives of the tax. The new CO_2-taxes with effect from January 1999 have slightly raised cost-effectiveness. However, inefficiencies remain since most of the mainland's energy intensive activities will still be exempted from the CO_2 tax. A market for CO_2 emission quotas is being prepared, but will need to be underpinned by clear rules for trading, banking, verification and compliance, which have not yet been established. Finally, emission permits should be auctioned or tendered, because otherwise the associated scarcity rent would be left in the hands of existing companies. This would penalise new entrants and rob the government of the opportunity to cut distorting taxes. If a unilateral tradeable quota system appears not to be feasible in Norway, a uniform CO_2 tax rate applied to all sectors, at a lower level, should be considered until further international decisions are taken. Compensation schemes for the currently exempted industries, as proposed by the government in April 1998, could provide the necessary time to adapt. Model simulations show that utilisation of the Kyoto-mechanisms would reduce the cost of complying with the Kyoto target significantly – by as much as two thirds compared with an approach based only on domestic measures.

While Norway has succeeded in fulfilling its commitments arising from international agreements on transboundary pollution, there is scope for improved cost-effectiveness also in this area. As concerns acidification, the Norwegian authorities may now need to refocus their efforts towards NO_x abatement, as the scope for SO_2 abatement at reasonable cost has been largely exploited. Increased emphasis on NO_x abatement also opens up opportunities for positive side-effects on tropospheric ozone formation and local air pollution. With regard to NO_x policy, the authorities have so far given priority to reducing emissions from road traffic and combustion for heating as its impact on the local environment is important. To date, measures have been implemented in the areas where the regional and local damage caused by NO_x emissions are the highest. In the future, however, further reductions will probably call for measures in the sectors left unaffected until now, in particular coastal shipping and fishing vessels. Applying an annual tax on all mobile sources based on their NO_x emission intensity could

be a cost-effective way to do this. The introduction of such a tax for vehicles has been considered by the Norwegian authorities, but has so far not been adopted. In any event, an extensive cost-benefit analysis of all sectors is needed. Concerning SO_2 policy, the extension of the sulphur tax to sectors previously exempted represents an important step in the direction of enhanced cost-effectiveness. Current plans for a uniform tax rate should be adopted with priority, especially since the targets of the Oslo Protocol may call for a higher overall level of the tax.

Rebalancing the policy mix may also be useful with regard to eutrophication and waste treatment. Eutrophication policies should now be re-oriented to nitrogen in order to exploit abatement cost differentials that have opened up between nitrogen and phosphorus. An increase in the tax on nitrogen would be a cost-effective way of dealing with this. Concerning waste, the adoption of the state tax on waste is welcome, but the national strategy should be underpinned by a better balance between the various waste treatment options based on region-specific cost-benefit analysis, and the prevention of waste should be sought.

The policies aimed at a reduction of pollution caused by road traffic are subject to debate in Norway. Several issues stand out:

- the current level of car fuel taxes is controversial. The official policy stance is that the tax on car fuels should be used to internalise all external costs of car use and that the region-specific external costs of driving a car should be covered by local taxes and measures tailored to local problems;
- while there is considerable scope for the use of local instruments to restrict car use, their introduction meets resistance. For example, the existing toll-rings could be revamped into road pricing systems – by adopting fees which vary according to the type of car (its polluting properties) and the time of the day or week of the trip. As a first step, the season tickets on the Oslo and Bergen toll rings could be suppressed and time-differentiated fees be introduced. Research by the Institute of Transport Economics indicates that a well designed road pricing system could double the benefits to society as compared to the current situation. Although EU-led test programmes have demonstrated that the technology is available, the operating costs are still considered to be too high and the electronic surveillance system may create privacy problems. Concerns about the distributional impact of road pricing and disagreements on the earmarking of the proceeds are also obstacles to its implementation. Nevertheless, parliament is in favour of the introduction of such variable tariffs in principle. Further initiatives in this regard are planned;
- as elsewhere, the much lower taxation of diesel as compared to petrol is particularly cost-ineffective in environmental terms since the local air

pollution generated by diesel cars (NO_x and particles) is greater than that of petrol driven cars. A study by Statistics Norway estimated that the average social costs of burning one litre of diesel in Oslo is at least 6 to 20 times higher than for a litre of petrol. The low diesel tax – including the CO_2 tax which is also lower for diesel as compared to petrol – has, moreover, induced a shift towards more diesel cars (as in other countries), a tendency which is expected to be exacerbated in the future.

Cost-effectiveness could be enhanced in a number of domains by moving from "command and control" to the use of economic instruments, such as taxes, deposit-refund schemes and tradable quotas. The introduction of taxes, differentiated by the emission properties of the capital equipment should be considered for all mobile sources, with the rate differentiated according to NO_x emission intensity. The taxation of remaining ozone-depleting substances, several hazardous substances and the greenhouse gas HFC, should also be encouraged. Deposit-refund schemes could be envisaged for products containing hazardous elements, such as nickel/cadmium batteries. In addition, the possibility of combining taxes with discharge permits should be further examined.

Policy interactions are important

Environmental and resource management policies in Norway aim to strike a balance between considerations of cost-effectiveness, international competitiveness of individual industries and regional development – with the latter two considerations often overriding environmental and cost-effectiveness concerns. Fishing policies, for instance, have been geared towards maintaining existing settlement patterns by redistributing the resource rents towards smaller vessels registered in remote areas. Environmental policies have been designed so as to limit the impact on the economic performance of industries that are exposed to foreign competition and that are mostly based in remote areas. Hence, there are cases where the scope for cost-effectiveness is not being exploited. It is important that the economic costs and benefits of these choices are carefully evaluated and made transparent to guide future decision making.

There are also interactions between transport policy and environmental pressures. Transport policy aims *inter alia* to provide transport infrastructure. In principle, the user should pay for the use of the infrastructure, and not the tax payer, and congestion fees should be levied to align private and social costs. Concerning road transport, petrol-driven cars probably come close to paying the use of infrastructure and even for externalities (environmental and accidents). This is not the case for diesel-driven cars and even more so for trucks. Raising taxes on diesel would not only make for a better transport policy, but also reduce environmental pressures. At the same time, reductions in support provided to

bus transport in remote areas and high-speed passenger ships could yield substantial environmental benefits.

Agricultural policies have also important environmental ramifications. Agricultural subsidisation is particularly high in international comparison (see Chapter II). While its major aim is to keep farm communities intact, the link between subsidies and production gives incentives to farm intensively, for example, by heavy use of fertilisers. A recent shift in agricultural subsidies from price support to more neutral support has reduced this incentive for intensive cultivation (see Chapter II). A further move in this direction would lead to a considerable further reduction of environmental pressures from intensive farming and enhance the sustainability of agricultural production.

Norway has an extensive institutional framework to foster sustainable development geared towards resolving policy conflicts. In addition, since the early 1990s, an ongoing evaluation effort of the sustainability of policies is underway. This has, for instance, led to the compilation of inter-generational accounts. It has also helped to integrate policies and to build a consensus about the need to foster policies which are sustainable in the long run. While clearly in the interest of society as a whole in the long run, policies pursuing sustainability can clash with vested interests in the short run. Agricultural or transport policies are a case in point. Furthermore, structural policies, financed in part by the oil and gas proceeds, have often served to perpetuate historical patterns of economic activity and settlements that may prove unsustainable once the oil and gas proceeds decline. Hence, even if generational accounts for Norway suggest inter-generational equilibrium of public finances – in itself an important achievement – there is a risk that many forms of public support may turn out to carry a low return in the long-run.

Notes

1. It should be noted that the national accounts underestimate foreign financial assets.
2. Until 31 December 1998, the Bank of Norway used as a reference a composite of the exchange rates in the ECU basket (the "theoretical ECU") rather than the actual ECU market rate. Since 1 January 1999, the euro has replaced the theoretical ECU as the reference for exchange rate policy. The guidelines for monetary policy have remained unchanged, however, with the same trading range applied to the euro/krone rate as to the ECU/krone rate.
3. Additional guidelines in the 1996 national Budget committed the Bank of Norway to purchase foreign currency "at a scale that over time corresponds to the net allocation of capital to the Petroleum Fund" – the fund that was created in 1990 to convert the central government's surplus into foreign financial assets, *inter alia* in order to offset upward pressure on the exchange rate from oil and gas exports.
4. At the same time, the Bank announced that it would abstain from further changes in its use of instruments for the time being.
5. Commercial banks in Norway lack this option due to the thin bond market, which prevents them from refinancing mortgage loans through bond issues.
6. As indirect tax increases envisaged in the 1999 Budget were rejected by the parliament, consumer price inflation in 1999 is now expected to turn out about ½ percentage point lower than projected in the *Economic Outlook*.
7. See Thøgersen (1994).
8. As documented in the latest *Employment Outlook* (OECD, 1998*a*), average hours worked per person in Norway are the second lowest in the OECD area, reflecting a combination of a large incidence of part-time work and a comparatively short standard work week.
9. The lengthening of the waiting period from three calendar days to three working days, while technically an action in line with the recommendations, is expected to have only a negligible impact.
10. In the current situation of labour shortages, the PES has turned its focus on mobilising inactive workers and getting them into work.
11. Over 40 per cent of the early retirement benefit is currently funded by the central government.
12. The replacement rate in the "part-time" early retirement pension scheme is not fixed and differs based on prior income, with a ceiling at 70 per cent of the corresponding prior full-time income.
13. The legal and averge effective retirement ages in Norway are 67 years and 62 to 63 years, respectively.

14. See Thøgersen *et al.* (1998).
15. The large size of the country and small population in remote areas make it difficult to have a good transportation system that is also profitable. As a result, government intervention is seen as being necessary, with subsidised rail services to remote areas costing annually about NKr 300 million. Twice this amount of subsidies are granted to an air carrier to maintain air transport to these areas. Suggestions for reducing the cost of regional public transportation, by use of buses, have met resistance from local politicians.
16. In the 1990s, the number of man-years in agriculture declined by 15 per cent, while direct support per man-year rose 20 per cent.
17. A system of tradeable milk quotas within each of the nine regions (with administered price limits) was introduced in January 1997 to encourage a more efficient utilisation of the existing production capacity. In July 1997, new regulations were introduced to abolish the practice (started in 1936) that milk prices should be equalised for different products (milk, cheese, butter, etc.) and between regions.
18. The government decided to trigger the GSP safeguard mechanism if the annual import of beef from Least Developed Countries (LDCs) exceeds 2 700 tons. LDCs enjoy duty-free access to the Norwegian market. In 1997, imports amounted to 1 925 tons, representing 2 per cent of the market.
19. See, for example, Brustad *et al.* (1995).
20. The high reported support to shipbuilding in 1990 reflects a change in the timing of payments.
21. The Research Council is administered by the Ministry of Education and Research, but is financed by contributions from fifteen ministries. The creation of the Research Council represents the merger of five different research institutes, including those focused on fisheries and agriculture.
22. This programme replaces an older one, which existed for five years and aimed at improving relations between SMEs and research institutions.
23. The Ministry of Trade and Industry is also responsible for the full ownership of Arcus, the alcohol producer and distributor with a nominal share value of over NKr 200 million.
24. Fully state-owned companies included in 1997 Telenor (20 840 employees, NKr 25.5 billion in turnover); Posten Norge BA (30 910 employees, NKr 11 billion in turnover); Postbanken BA (1 664 employees, NKr 68 billion in total resources); the State Railway company (8 714 employees, NKr 6 billion in turnover); and the Civil Aviation Administration (2 084 employees, NKr 3 billion in turnover).
25. The questions are addressed in a government White Paper issued in December 1998.
26. Cables for electrical power may eventually also be laid to the continent.
27. For small retail customers a "profile" of the monthly consumption pattern is created and customers are charged on the basis of this profile.
28. One problem in reforming the regulatory system is the fact that the sector-specific regulators are within ministries and, therefore, have some power to resist change.
29. As foreign bank branches operate on the capital base of the parent bank abroad, they avoid a subsidiary capital requirement associated with operating a bank in Norway, and their capital adequacy is not controlled by the domestic regulator.

30. Assuming the government approves the takeover of a bank, banking ownership regulations restrict the ownership to either less than 10 per cent of the total bank shares, or more than 90 per cent. These regulations do not apply to the government which owns 51 and 52.2 per cent, respectively of the two largest commercial banks.

31. Kredittilsynet, created in 1986, was the first integrated supervision authority in the Nordic countries, ahead of Denmark, in 1988, and Iceland, in 1998.

32. However, there are several obstacles to such a change, including implementation costs. Moreover, Norwegian securitisation laws would need to be harmonised with the recently agreed EU Settlement and Finality Directive, changing the clause on finality of payments settlement, from the current 24 hours to a zero hour (instant) settlement.

33. SND can appoint a member to the board of seed capital funds.

34. OECD (1995) and IEA (1997).

35. OECD (1993).

36. Policies concerning fresh water management are not discussed, because they do not currently raise major sustainability concerns.

37. Parliament has the responsibility for the approval of major developments, while the government is authorised to approve smaller development projects. The overall administrative responsibility for the operations rests with the Ministry of Petroleum and Energy. The Norwegian Petroleum Directorate (NPD), subordinate to the Ministry, has supervisory and administrative responsibility.

38. The new Petroleum Act adopted by Parliament in 1996 entered into force in 1997, and introduced more administrative flexibility in several areas, including acreage management, in order to ensure the attainment of these policy aims.

39. Statoil's regulatory framework was set up in order to manage the public assets on the Norwegian continental shelf. However, the EU license directive prohibits governments giving priority to a national company, and as a result Statoil is now in open competition with other companies on the Norwegian continental shelf. The directive has also created new opportunities for Statoil abroad. Statoil and Norsk Hydro are extending their reach to Russia and other states of the former Soviet Union (in particular the countries surrounding the Caspian Sea where reserves are estimated to be in the range of 50 and 100 billion barrels or two to three times the size of the Norwegian continental shelf). These companies are deemed to no longer need protection within their core Norwegian market.

40. When in the early 1970s the significance of the petroleum resources had become clear, the government stipulated that the profits from the operations on the shelf should "benefit the entire population" and should be used to "create a better society" through increased public spending on social security, culture, education and infrastructure, as well as the development of rural areas. Lower taxes on mainland economic activities, a shorter working week and an increase in development aid were also considered as priority areas, see *Stortingsmelding* nr. 25, (1973-74).

41. The rent is defined as the market price of oil and gas minus the marginal cost of extraction (including capital cost).

42. For example, after the second OPEC shock, the petroleum wealth jumped from NKr 590 billion in 1979 to NKr 2 273 billion in 1981, only to collapse to NKr 694 billion when the oil price plummeted in 1986 (see Thøgersen, 1994).

43. The accounts were set up by Auerbach *et al.* (1993).

44. Based on the assumption that the oil price will recover in real terms by 20 per cent from its 1998 level (see the 1999 Budget proposal released in October 1998).

45. The major fishery nations in the world are China (27 million tonnes per year) and Peru (10 million tonnes per year). Norway catches 3 million tonnes per year.

46. There are a few fish resources considered exclusively Norwegian national stocks of which coastal cod and North-East Arctic saithe are the major ones. For such stocks TACs are determined by the Norwegian authorities.

47. Atlantic herring is a so-called "straddling" stock, *i.e.* various stocks mix on the feeding grounds in the Norwegian and North Sea and it is, therefore, difficult to distinguish between the catches from the Icelandic, Faeroese and Norwegian stocks. At the close of the Third UN Conference on Law of Sea, 1982, straddling fish stock was hardly a problem, as 90 per cent of the world catch was taken within the zones. By the late 1980s a crisis had developed: as stocks in the zones were fully utilised, there was persistent fleet over-capacity and high sea fishing became more profitable. There was growing alarm over the straddling fish stock problem and a UN General Assembly established the UN High Seas Fishery Conference in April 1993. The UN Conference on Straddling Fish Stocks and Highly Migratory Fish Stocks concluded in December 1995 that straddling fish stocks should be managed by regional organisations. If this system succeeds, straddling stocks will be managed as shared stocks and high seas adjacent to 200 mile zones will become high sea in name only.

48. When Norway negotiated the European Economic Area agreement with the EU Commission in 1993, it was agreed to introduce a five year moratorium on investment by firms in this sector. At the discretion of the EU, this agreement may be reopened for negotiation in 1999.

49. European Environment Agency (1998).

50. These national TACs are set as a level consistent with bilateral agreements between Norway, Russia and the EU allowing for fishing in each other's 200 mile zone.

51. Important decisions are taken by the Cabinet, based on proposals by these committees.

52. A consortium of Naturkraft, owned by Statoil, Statkraft and Norsk Hydro is in the process of developing two gas-fired power plants at Karstø and Kollsnes. Norsk Hydro has, however, announced that it could develop such a plant using new technology to eliminate 90 per cent of the CO_2 emissions by re-injecting CO_2 in oil fields.

53. The reduction in emissions of perfluoridised carbons (PFCs) and sulphur hexafluorides (SF_6) (or "the other gases") are expected to continue, albeit at a much slower pace. Nitrous oxide (N_2O) emissions, stemming from fertilisation in agriculture and production of nitric acid, are projected to remain stable.

54. International shipping is not included in the emission commitments.

55. This was proposed by the government but rejected by the parliament.

56. In June 1997, the aluminium industry signed a voluntary agreement with the Ministry of Environment to reduce greenhouse gas emissions per unit of aluminium by 55 per cent in the period 1990-2005, most of which has already been achieved.

57. See OECD (1998*d*), pp. 91.

58. A key feature of greenhouse gases is that the marginal damage of emission is always the same wherever it originates from.

59. In the aluminium industry, for example, greenhouse gas emissions are due mostly to the chemical reactions underpinning the production of aluminium from aluminium oxide.
60. The CO_2 tax extension net of the compensation was estimated to add NKr 1 billion to the NKr 7.7 billion of the total expected CO_2 tax revenues for 1998.
61. Combustion plants utilising methane from waste for energy purposes would pay a lower rate, even though methane emissions will not be reduced.
62. For CO_2 or possibly all greenhouse gases.
63. A combination of several of these possibilities could also be envisaged.
64. With parliament's amendments, the reduction will be less than the 0.1 percentage point decrease in labour tax rates initially calculated by the government.
65. A domestic tax-based solution which would leave the energy intensive sectors unaffected would require even higher tax rates in the non-exempted sectors.
66. Estimate provided in *Storting Prp No. 54*. The study provided a range of between NKr 50 and NKr 200 per ton of CO_2.
67. Other important ozone depleting substances are methyl chloroform, carbon tetrachloride, methyl bromide, hydrobromofluorocarbons (HBFCs) and hydrochlorofluorocarbons (HCFCs).
68. Moreover, additional restrictions on specific uses have been imposed.
69. To enforce the regulations, parliament decided to impose a tax on CFCs if consumption is not reduced. Before the CFC regulations came into force, the industries concerned had organised a reception system for used CFCs. Similar regulations have been used for implementing the phasing out of the other substances with a strong depleting potential.
70. Establishing a market for emission quotas, as in the United States, was also not viable given the small number of potential participants.
71. Critical load is defined as the highest load/deposition in soil or surface water that will not damage the ecosystem. In the mid-1990s, critical loads for the acidification of surface water were exceeded on 37 per cent of the Norwegian territory, and on more than 90 per cent of the southernmost counties area (Aust and Vest-Agder). Between 1960 and 1990, 20 per cent of the fish stocks were lost, and another 20 per cent damaged, in the rivers and lakes of the 12 southern counties. The critical loads for acidification of forest soils are also exceeded in the southern-most parts of the country.
72. The population warning threshold defined by the EU ozone directive has never been reached and the population information threshold is exceeded only a few days a year.
73. There is also nutrient runoff from uncultivated land, partly natural, and partly resulting from nitrogen deposition from long-range transported air pollution.
74. A better measure concerning environmental effects – nitrogen surplus – was developed in the context of the OECD agri-environmental indicators activity. It counts all nitrogen, including animal manure, which passes into the environment. On this indicator, Norway is about average in the OECD, even though the trend is upward.
75. In addition to the stabilisation target of the Sofia Protocol, Norway has, as laid down in the Sofia Declaration, the national goal of reducing NO_x emissions by 30 per cent in 1998 compared to 1986. The Declaration's target is thus more ambitious than that in the Protocol, but not binding.

76. Several plants have been required to install equipment to control emissions, and some of the most polluting plants have been closed.

77. There are regimes for the Oslo region, the twelve most southern counties and the rest of the country.

78. NKr 0.07 per 0.25 per cent of sulphur content per litre of mineral oil.

79. The long lifetime of cars is mainly due to high taxes on car purchases.

80. The decrease in VOC emissions observed in 1997 is partly explained by the installation of a recovery facility for oil vapour at one of the oil landing terminals.

81. In fact, VOC emissions have grown by more than 40 per cent between 1988 and 1997.

82. The North Sea Declaration also includes emission targets for toxic pollutants.

83. The initial timetable for achieving this target was abandoned by the signatories of the Declaration. The Declaration also included a commitment to set up secondary treatment facilities for discharges from municipal sewage treatment plants serving areas of over 5 000 people (in population-equivalent terms).

84. Grants cover between 20 and 35 per cent of the capital costs with the remainder financed through government loans.

85. Indeed, in 1996, treatment plants removed 91 per cent of the phosphorus from waste water in the southern part of the country, against 37 per cent in the rest.

86. A reduction in the nitrogen runoff can be achieved by the introduction of "catch crops" along the coast that absorb the nitrogen in the soil and leave land that is vulnerable to erosion out of crop during the winter (winter "stubbling").

87. Industrial processes were the main source of emissions to air, particularly of SO_2, but their contribution to local air pollution in urban areas has substantially decreased over time. There are, however, emissions of heavy metals and toxic substances in some industrialised areas.

88. Nevertheless, local pollution peaks are sometimes observed in the winter season. Peak values may be very high. The maximum concentrations of the main pollutants (NO_2 and particles), recommended by the Norwegian Pollution Control Authority, are occasionally exceeded for at least 700 000 persons in recent winters.

89. More than 5 million tonnes of waste are generated yearly in Norway – excluding waste from mining, building and construction. 1.3 million tonnes are from private households.

90. Part of hazardous waste is exported, largely to other surrounding Nordic countries in order to reduce hazards from transportation. Other waste exports are small, except for old newspapers, which cannot be treated in Norway because no de-inking facility exists yet.

91. By 2005, the 24-hour mean concentration of sulphur dioxide shall not exceed 90 $\mu g/m^3$; by 2010, the annual mean concentration of benzene shall not exceed 2 $\mu g/m^3$ (urban background value); also by 2010, the hourly mean concentration of nitrogen dioxide shall not exceed 150 $\mu g/m^3$ for more than 8 hours per year; and by 2005, the 24-hour mean concentration of particulate matter (PM_{10}) shall not exceed 50 $\mu g/m^3$ for more than 25 days per year and by 2010 not more than 7 days per year.

92. In Oslo, moreover, 20 per cent of the toll ring proceeds accrue to public transport and the construction of bicycle tracks.

93. Some impact on travel patterns has been observed in Trondheim, with a slight shift in the timing of car trips towards toll-free periods. In Oslo and Bergen, season tickets are available. In Oslo, the toll fees are not differentiated during the day, while in Bergen no tolls are collected during the night.

94. Overall, the income-to-cost ratio of municipal waste collection systems reached 95 per cent in 1995, and it was between 90 and 110 per cent in more than two thirds of the Norwegian municipalities.

95. Packaging of milk and milk products is fully exempted.

96. Pilot projects for waste recycling are also conducted by the SFT in some municipalities.

97. The tax level was set so that the methane emissions would be taxed in line with the average rate of taxation of CO_2.

98. However, international shipping is not covered by the Kyoto agreement.

Glossary of acronyms

AFP	Early retirement scheme
ALMP	Active Labour Market Programme
bcm	Billion cubic metres
CFC	Chlorofluorocarbon
CH_4	Methane
CIS	Community of Independent States
CO_2	Carbon dioxide
EEA	European Economic Area
EFTA	European Free Trade Association
GFU	Gas negotiation committee
GSC	Gas supply committee
HBFC	Hydrobromofluorocarbon
HCFC	Hydrochlorofluorocarbon
HFC	Hydrofluorocarbon
IEA	International Energy Agency
LDC	Least Developed Countries
LO	Central labour union organisation
N_2O	Nitrous dioxide
NAWRU	Non-Accelerating Wage Rate of Unemployment
NDE	State energy authority
NGLs	Natural gas liquids
NH_3	Ammonia
NHO	Confederation of Norwegian Business and Industry
NO_x	Nitrous oxide
NPD	Norwegian Petroleum Directorate
PES	Public Employment Service
PFC	Perfluorocarbon
PPP	Purchasing Power Parity
PSE	Producer Subsidy Equivalent
PTE	Telecommunications regulation authority
R&D	Research and Development
scm	Standard cubic metres
scm oe	Standard cubic metres of oil equivalent
SDFI	State Direct Financial Interest
SF_6	Sulphur hexafluorides
SFT	Norwegian pollution control authority
SME	Small and Medium-sized Enterprises
SND	Industrial and regional development fund
SO_2	Sulphur dioxide
TAC	Total Allowable Catches
UV	Ultra Violet
VOC	Volatile Organic Compound
WTO	World Trade Organisation

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Annex I

Norway's oil and gas wealth

The oil and gas resources

Norway possesses significant oil and gas resources on the continental shelf below the North Sea, the Norwegian Sea (along the Atlantic Coast) and the Barents Sea (close to the polar circle). The oil and gas reserves amounted to 3.0 billion standard cubic meters of oil equivalent (scm oe) in 1997, of which more than half, or 1.8 billion scm oe, consisted of crude oil (including natural gas liquids or NGLs) and the remainder of natural gas (Table A1).[1] Although the Norwegian reserves are large compared to the other oil and gas producing countries around the North Sea (the United Kingdom, the Netherlands and Denmark), they represent only 1.1 per cent of the total oil and gas reserves in the world.[2] By comparison, the CIS and the Eastern European countries control over one-fifth of world oil and gas reserves, and the OPEC three-fifths. Yet, Norway is the second largest oil exporter in the world, after Saudi Arabia, and is also a major gas provider to western Europe – Norwegian gas accounts for 10 per cent of total gas consumption in western Europe, with the remainder largely provided by Russia, Algeria and the Netherlands (Figure A1).[3] This suggests that Norway maintains a very high rate of extraction compared to the other major oil and gas producing regions in the world.

Given the high extraction rate, the production has quickly matured. Most major oil discoveries date from the late 1960s and early 1970s in the North Sea and are close to depletion (Table A2).[4] Hence, extraction activity has moved to newer fields which are smaller, notably in the Norwegian and Barents sea areas.[5] On the other hand, the sharp increase in oil production since the mid-1980s has been coupled with rising estimates of the resources in fields under development, owing to improved exploration and extraction technology – including three-dimensional seismic mapping, the use of flexible drills and floating platforms to develop "satellite" fields in the neighbourhood of existing infrastructures, and the injection of gas or water to maintain pressure. In fact, the enhanced recovery in existing fields, rather than the discovery of new fields, has been the single most important source of growing reserves in the past decade. A main constraint on the use of enhanced recovery technology in the future, however, could be its high energy and hence CO_2 emission intensity.

At current extraction rates and technology, oil reserves span 18 years and gas reserves 85 years, with oil production expected to peak in the early 2000s. Gas production will last for much longer than oil production (Figure A2)[6] and the delivery infrastructure for natural gas from the Norwegian continental shelf to the continent will be expanded accordingly.[7] The prospects for exports of Norwegian gas to Europe are favourable as the delivery is deemed to be secure, with potential new buyers in eastern Europe seeking to diversify their supply from reliance on Russian gas. Natural gas is, moreover, a relatively clean fuel and several countries substitute gas for coal and heavy oil use. Based on commitments

Table A1. **World reserves of oil and gas**

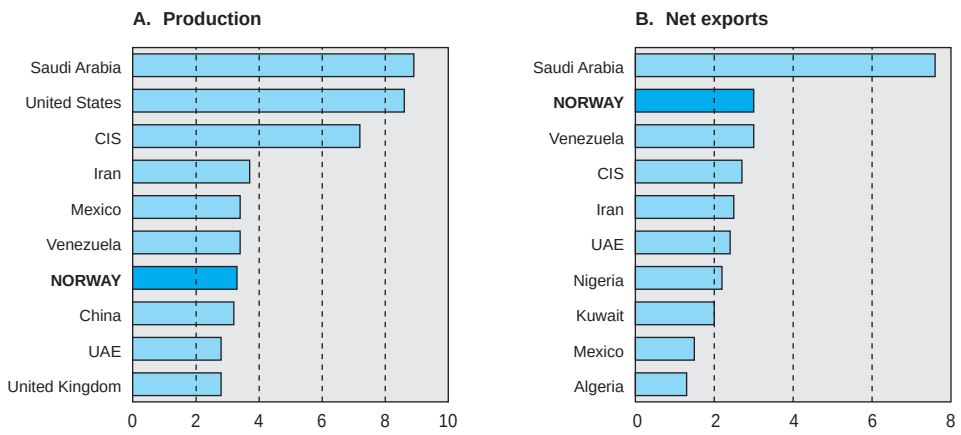
1997

	Oil		Gas		Total	
	Billion scm oe	Per cent	Billion scm oe	Per cent	Billion scm oe	Per cent
Norway	**1.8**	**1.1**	**1.2**	**0.9**	**3.0**	**1.0**
Western Europe	2.9	1.8	4.7	3.4	7.6	2.5
Eastern Europe and CIS	9.4	5.8	56.7	40.5	66.1	21.9
Middle East	107.5	66.4	45.8	32.7	153.3	50.8
North America	4.3	2.7	6.6	4.7	10.9	3.6
Latin America	20.3	12.5	7.8	5.6	28.1	9.3
Africa	10.7	6.6	9.3	6.6	20.0	6.6
Asia and Oceania	6.7	4.1	7.1	5.1	13.8	4.6
OPEC	125.4	77.4	58.1	41.5	183.5	60.8
World	162.0	100.0	140.0	100.0	302.0	100.0

Source: Statistics Norway.

Figure A1. **Production and net exports of crude oil**

Million barrels per day, including NGLs,[1] 1997



1. Natural gas liquids; production level in 1997 was 10.8 million scm compared to 176.8 million scm of crude oil.
Source: Ministry of Petroleum and Energy.

Table A2. **Petroleum resources on the Norwegian continental shelf**

1997

	Total Billion scm oe	Composition, per cent of total			
		Discovered			Undiscovered
		Produced	Reserves	Other[1]	
Oil and gas[2]	**12.8**	**18**	**25**	**29**	**28**
of which:					
North Sea	8.7	25	27	32	16
Norwegian Sea	2.9	2	24	29	45
Barents Sea	1.2	0	16	15	70
Oil[2]	**6.6**	**26**	**29**	**23**	**23**
of which:					
North Sea	5.0	33	27	24	16
Norwegian Sea	1.3	5	40	20	35
Barents Sea	0.3	0	6	10	84
Gas	**6.2**	**9**	**22**	**36**	**33**
of which:					
North Sea	3.6	15	27	42	16
Norwegian Sea	1.7	0	12	35	53
Barents Sea	0.9	0	19	16	65

1. Discovered resources outside fields in operation and expected enhanced recovery.
2. Including NGLs.

Source: Ministry of Petroleum and Energy.

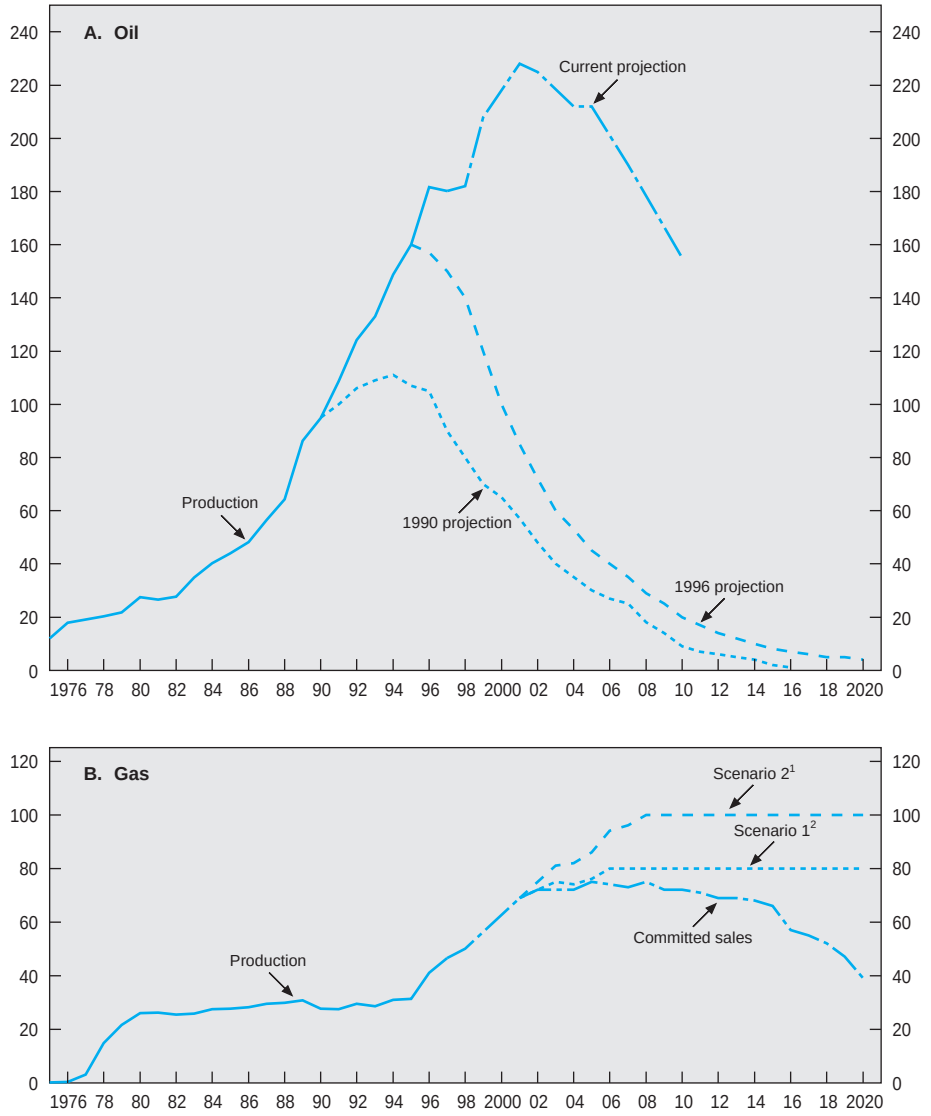
in existing contracts, export volumes should double from current levels by 2005, with Italy becoming a major new customer. By then, Norway is expected to supply an estimated 17 per cent of western Europe's gas consumption.

The supply structure of oil and gas

Some 20 international oil companies are engaged in exploration and development activities on the Norwegian continental shelf. Norway's approach includes, however, the strong presence of government in the exploration and development of fields and majority ownership in two companies: Statoil (fully state owned) and Norsk Hydro (51 per cent state interest). The major official justification for the strong state involvement is that it provides the government with an adequate level of experience and know-how in setting oil and gas policy. Field participation by the Norwegian state takes place through the so-called State's Direct Financial Interest (SDFI), which is operated by Statoil on the state's behalf. Until the 14th licensing round, which took place in 1993, Statoil and SDFI were allocated a stake of at least 50 per cent on every consortium. As a result, Statoil and SDFI were by far the biggest producers on the continental shelf.

Direct regulation of oil extraction by the government is possible in theory but, once investments are made, it is very costly to interrupt production, as oil and gas reserves may be lost forever. In rare cases, for instance, in 1987, production was capped following the

Figure A2. **Oil and gas production on the Norwegian continental shelf**
Million scm oe



1. Depletion by 2065.
2. Depletion by 2075.

Source: Ministry of Petroleum and Energy and International Energy Agency.

price drop in 1986. However, the arrangement broke down in 1989 and was formally ended in 1990. Moreover, while the authorities do have the possibility to postpone the issuing of permits for investments on fields in operation, this instrument is used only occasionally to regulate demand pressure on the mainland economy. The pipelines on the shelf are owned by licensees of which Statoil and SDFI have in most cases the majority share – as with extraction until the 15th Licensing round in 1995.

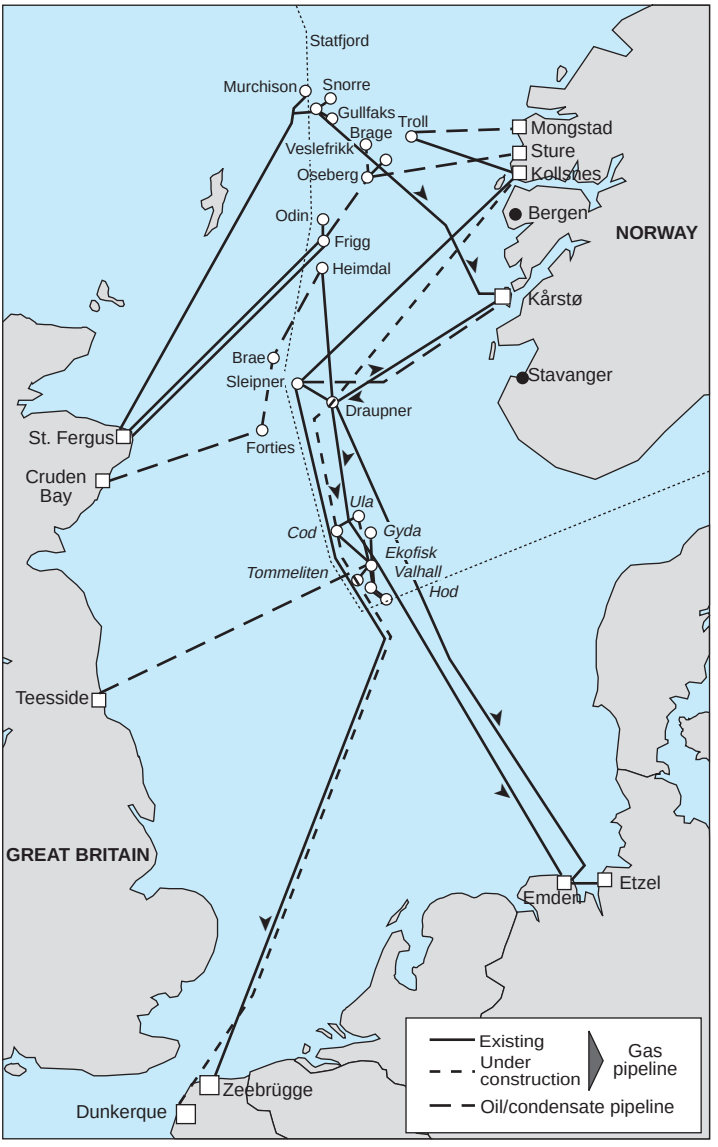
While the production and sale of oil is largely determined on the spot market, gas exports are marketed by the Gas Negotiation Committee (GFU), which since 1980 has acted as the sales agent for Norwegian gas.[8] GFU members are the Norwegian companies Statoil (chair), Norsk Hydro and Saga Petroleum, who hold licenses for three-quarters of Norway's total gas reserves. The GFU is responsible for preparing and conducting all negotiations, including price and volumes, for the sale of Norwegian gas. Prices are set as a weighted average of competing fuels in the destination market plus a mark-up provision for the cost of land transportation, storage, delivery and other services. The two largest customers are the Ruhrgas consortium in Germany (42 per cent of deliveries) and Gaz de France (24 per cent).[9] It is noteworthy that gas contracts are not earmarked for a specific field, which has facilitated the set-up of the integrated gas transportation grid on the shelf (Figure A3). The integrated network has generated important economies of scale and has reduced the need for gas flaring on oil fields, as unused gas can be delivered through the pipeline grid to other operators for enhanced oil recovery projects. It also created the possibility of mixing gas from various fields in order to ensure a uniform quality, both over time and across sales contracts. For this purpose, the Gas Supply Committee (GSC) was established in 1993. The GSC includes representatives from the twelve largest licensees on the shelf. Its task is to ensure that gas – which is produced in a variety of fields with different characteristics and requires a substantial amount of treatment, processing and transportation – can be delivered to the customers in Europe according to standardised quality specifications. The GSC formally acts as an advisor to the Ministry of Petroleum and Energy with the Ministry selecting delivery fields and transportation mechanisms for each contract.

Norway plays a valuable and increasingly important role as a secure provider of natural gas in Europe. However, Norway will be faced with new challenges, with the opening up of energy markets in the EU and the likelihood of significant change in the current structure of the EU gas market. While the current set-up of co-ordinated gas sales and field allocation has generated important economies of scale, and raised the energy efficiency of oil and gas extraction and transportation, Norway will need to take careful stock of its arrangements in this regard so as to reap the potential benefits of market liberalisation. In particular, the Norwegian authorities should assess the cost and benefits of alternative ways of managing the gas transportation grid and negotiating gas contracts.

The taxation of oil and gas

The significant revenues the government draws from the activities on the continental shelf come from the special tax regime applying to oil companies, including the fully state-owned company Statoil. The present tax regime includes a standard 28 per cent corporate tax charged on profits net of depreciation allowance, a special surtax of 50 per cent on those profits minus an uplift and a royalty of 8 to 16 per cent of gross sales on oil from fields cleared for development before 1986. In addition, an acreage charge is levied as a lump sum per square kilometre licensed and companies are also liable for the CO_2 tax.[10] Of less significance, in quantitative terms, is the annual dividend pay-out of Statoil, which usually amounts to 50 per cent of its after-tax operating profits. However, of increasing importance for public finances are the proceeds from the SDFI's stakes in licences on the

Figure A3. **North sea oil and natural gas transportation systems**



Source: Norwegian Petroleum Directorate.

shelf (Figure 19, Panel A). On 1 January 1985, the government created the State Direct Financial Interest which implied that a significant part of Statoil's profits henceforth accrued directly to the state. After this arrangement was implemented, all oil and gas extracted by Statoil had to be split into a part owned by the company and a part owned by the state, with Statoil formally acting as an operator on the state's behalf. The purpose of this measure was to channel a larger part of the petroleum revenues directly to the government as well as to reduce the company's ability to increase the capital base and become too independent from the state.

The effective tax burden on the continental shelf is not out of line with that observed in other gas and oil production areas in the world. It is high compared to the mainland, reflecting the fact that offshore value added is largely a resource rent. The degree to which the current tax system captures the whole rent has been subject to some debate. A majority in the *Green Tax Commission* (1996) commented on several weaknesses in the system which, over time, could induce operators on the continental shelf to avoid offshore taxes through the establishment of subsidiaries on the mainland. They, therefore, recommended to reassess the tax rules on the shelf regarding the valuation of financial assets and liabilities offshore and onshore. A minority in the Commission, however, noted that, in view of the complexity of the offshore tax regime, a major in-depth review of the regime would be required before any firm conclusions within the context of a green-tax proposal could be formulated. The minority, therefore, could not support the recommendations of the majority.

Due to price and volume volatility, the contribution of the state's oil and gas proceeds to the budget have varied substantially in the past two decades. Petroleum revenues were substantial in the first half of the 1980s, of the order of 15 per cent of GDP and 20 per cent of total general government revenues as they were boosted by the high oil price in the wake of the second oil price shock. In the second half of the decade, the oil proceeds suffered considerably from the combined collapse of the oil price and the United States dollar. With the coming on stream of new production facilities, including through the SDFI, however, the government's petroleum revenues had recovered strongly in recent years. Indeed, the state's petroleum revenues soared from 3 per cent of GDP in 1994 to 8 per cent in 1997, before plummeting to an estimated 4 per cent of GDP in 1998 after the recent oil price collapse. The oil and gas proceeds are expected to pick up in the coming years, but should start a long-term decline from the middle of the next decade due to the fall in oil production.

Notes

1. A standard cubic metre of oil equivalent (scm oe) has the same calorific value as 6.3 barrels of crude oil and 1 000 standard cubic metres of natural gas.
2. Discovered gas reserves are estimated at 2 805 billion cubic metres (bcm), over 40 per cent of the total estimated reserves in Europe. Undiscovered resources are estimated at 2 410 bcm.
3. There is almost no onshore use of natural gas in Norway, except for one methanol production facility and apart from a small local distribution network near the Karstø gas-landing terminal, a mainland transportation and distribution network has not been developed.
4. The Troll field, which is located in the north-western part of Norway's North Sea territory and roughly level with the coasts of Bergen and Stavanger, is the largest oil and gas field that has recently come on stream.
5. The average production costs in Norway are relatively high compared with other oil provinces in the world. The weighted average operating costs are US$4.2 per barrel of oil equivalent (including CO_2 tax), compared with US$4.2, US$3.9 and US$3.6 on the British, Dutch and Danish parts of the production region. Average costs in Norway have been on a declining trend since 1986/87, when major gas fields, in particular the Troll field, were brought on stream, but this trend may end in the coming decades. Moreover, while the reserves located in the main North Sea fields are generally of high quality – light crudes with low sulphur content and a low CO_2 content in the case of natural gas – some newer finds are of not so high quality.
6. White Paper St meld 46 (1997-98) *Olje- og gassvirksomheten.*
7. There are four main pipeline systems carrying gas to export markets: the Frigg pipeline to St. Fergus (Scotland), Statpipe/Norpipe from the North Sea *via* the Karstø terminal and the Ekofisk field to Emden, (Germany), Zeepipe from the Sleipner and Troll fields to Zeebrugge (Belgium) and Europipe to Emden (Germany). A new pipeline to Dunkirk in France became operational in 1998 to serve gas sales to Gaz de France. A third pipeline to Germany will be operational in 1999. An agreement reached in April 1997 with the United Kingdom, moreover, opened up the possibility of interconnecting existing pipelines for transportation of Norwegian gas to customers in third countries.
8. Only gas for own use, or if a better price can be obtained, can be sold outside the GFU, but there is no precedent to date.
9. Gas sale contracts usually provide for the gas to be delivered at the border of the buyer's country. Hence the sellers (licensees) also transport the gas. When a new field is developed this usually implies the construction of a new pipeline and a terminal, and normally a joint venture with the same ownership structure as the field

is formed for this purpose in order to avoid conflicts of interest. Norway has no storage facilities for gas. Variations in demand are matched by variation in production, although there are storage arrangements with Germany and France concerning one field (Troll). Pipeline tariffs are set by the government.

10. The royalty, the acreage charge and the CO_2 tax are deductible from the corporate income tax and the special surtax. Investments are allowed to depreciate in six years, at a book value of 130 per cent of the actual value.

Annex II

Government support programmes

Table A3. **State aid programmes or schemes in operation in 1997 and the amount of subsidies reported**

Category	Programme	Authority responsible	Policy objective	Amount in 1997[1], million NKr
R&D	Industrial R&D programmes and projects	The Research Council	Stimulate industrial R&D	628[2]
	Industrial R&D contracts	SND	Stimulate industrial R&D and building business networks	36.7
	Public R&D contracts	SND	Reinforce the capabilities of Norwegian industry to satisfy the needs of public clients	156.5
	Swedish Norwegian Industrial Fund	Norwegian Ministry of Trade and Industry, Swedish Ministry of Industry	Stimulate SMEs and R&D	5.6 (in Norway)
Regional assistance	Regional grants for the development of business and industry	SND	Regional development, job creation, etc.	641.1
	Scheme for restructuring in regions dependent on a single industry	Ministry of Local Government and Regional Development	Restructuring and job creation in municipalities and regions dependent on a single industry in difficulties	91.7
	The Norwegian Co-operation for Industrial Estates and Development	Ministry of Local Government and Regional Development	Regional assistance and development, support for SMEs	17.0
	National Programme for Regional Development	Ministry of Local Government and Regional Development	Co-operation between R&D centres and SMEs	89.0
	Funds for restructuring the community of Rana	Ministry of Trade and Industry	Regional restructuring in the community of Rana	4.9
	Funds for restructuring the community of Sør-Varanger	Ministry of Trade and Industry	Restructuring of the community after the closing of the iron ore/mine industry	10.5
	Funds for restructuring the Grenland Area	Ministry of Trade and Industry; Ministry of Local Government and Regional Development	Regional restructuring	13.0

Table A3. **State aid programmes or schemes in operation in 1997 and the amount of subsidies reported** *(cont.)*

Category	Programme	Authority responsible	Policy objective	Amount in 1997[1], million NKr
Assistance to SMEs	Development grants	Ministry of Trade and Industry/SND	Assistance to SMEs, improve efficiency of industry	472.1
	The Public Advisory System	Ministry of Trade and Industry	Transfer of competence and know-how to SMEs	147.1
Environmental subsidies	Aid for development and diffusion of cleaner technologies	Ministry of Environment	Promote cleaner technologies	5.1
	Grants for waste reduction, recycling and capacity building for sustainable production and consumption	Ministry of Environment	Information promotion for enterprises	72.6
	Guarantee for loan facility to the company responsible for treatment of hazardous waste	Ministry of Environment/SND	Treatment of hazardous waste	31.0 (loans, interest rate is Nibor + 1 point)
	Grant program for new energy technology	Ministry of Oil and Energy	Support introduction of new energy efficient technology	122.0
Export promotion	Strategy programme and programme for export development in SMEs	Ministry of Trade and Industry	Business development and export promotion	56.7
	Export credit financing scheme	Ministry of Trade and Industry	Matching export credits in other nations according to the OECD Arrangement	47.0 (interest subsidy)
	Export credit guarantees	Ministry of Trade and Industry/Guarantee Institute for Export Credits	Promote exports	0[3]
Assistance to specific industries	Interest rate subsidies for the shipbuilding industry	Ministry of Trade and Industry/Eksportfinans ASA	Support shipbuilding industry	79.8 (interest subsidies)
	Contract and construction loan subsidies to shipbuilding	Ministry of Trade and Industry/Eksportfinans ASA	Support shipbuilding industry	569.7

Table A3. **State aid programmes or schemes in operation in 1997 and the amount of subsidies reported** *(cont.)*

Category	Programme	Authority responsible	Policy objective	Amount in 1997[1], million NKr
	Reduced CO_2 tax on mineral oil, coal and coke	Ministry of Finance and Customs	Avoid effect of CO_2 tax on international competitiveness	156 (1996, tax concession)
	Press subsidies	Ministry of Cultural Affairs	Promote a diversified press	203
Assistance to the fishing sector	Aid to structural measures and rationalisation in the fish processing industry	Ministry of Fisheries	Support transport to facilitate fishing activities in certain regions	35.5
	Support for specific fisheries	Ministry of Fisheries	Conversion to more profitable work methods	0.9
	Interest rate subsidies for fishing vessels for domestic deliveries	Ministry of Fisheries/SND	Modernisation of fishing vessels	8.1 (interest rate subsidy)
	Support for fishing vessels for domestic deliveries	Ministry of Fisheries/SND	Modernisation of fishing vessels	91.8
	Grants for building or purchase of vessels to the county of Finnmark	Ministry of Fisheries/SND	Regional development	0
	Support to long-line baiting facilities	Ministry of Fisheries	Support to the fisheries sector in certain regions	13.8
	Research on fishing	Ministry of Fisheries	Research on fishing	7.0
Assistance to the forestry sector	Tax concession in forestry	Ministry of Agriculture	Promote long range investments	15-20 (tax concession)
	Subsidies for forest management planning	Ministry of Agriculture	Sustainable forestry	36.1
	Subsidies for afforrestation and logging	Ministry of Agriculture	Biodiversity, landscape conservation and recreation	84.3
	Subsidy for construction of forest roads	Ministry of Agriculture	Rational forest management in disadvantaged regions	54.7

Table A3. **State aid programmes or schemes in operation in 1997 and the amount of subsidies reported** *(cont.)*

Category	Programme	Authority responsible	Policy objective	Amount in 1997[1], million NKr
	Subsidy for harvesting of round wood in difficult terrain	Ministry of Agriculture	Regional assistance	7.7
	Subsidy for round wood sale and transport in forestry	Ministry of Agriculture	Promote forestry activity in eastern and northern regions	1.6
Agricultural support	Price support	Ministry of Agriculture	Price support for milk and milk products, beef, veal, other meat, cereals, wool and other products	2 599.3
	Structural income support	Ministry of Agriculture	Income support	1 499.6
	Headage support for animals	Ministry of Agriculture	Income support	1 692.4
	Acreage and cultural landscape scheme	Ministry of Agriculture	Income support, environment	3 354.0
	Production support on market crops	Ministry of Agriculture	Income support	66.5
	Transport support, meat and eggs	Ministry of Agriculture	Income support	82.5
	Other income support	Ministry of Agriculture	Income support	213.1
	Market regulation	Ministry of Agriculture	Income support	96.0
	Agricultural Development Fund	Ministry of Agriculture	Income support	515.3

1. Grants unless otherwise stated.
2. Not included in the data reported to WTO in 1997.
3. Paid out guarantees minus fees received and recoveries. Calculated by the referred (net cost) method the subsidy element was negative (–405.6), the subsidy could therefore be set to zero.

Source: WTO Report on Subsidies from Norway, for the year 1997.

Annex III

Calendar of main economic events

1998

January

The national agreement between employer and employee organisations is up for renegotiation in 1998. This year workers are less interested in social reforms than they are in wage increases, which are precisely what the government wants to keep under control. Government prognoses are based on a 3.5 per cent rise in average incomes.

February

As from 1999, the government's Family Cash Benefit scheme will be extended to cover children between one and three years of age (instead of one and two years). The Labour Party takes a stand against this scheme and supports the day-care centres, making a counter proposal which would give parents of small children ten extra days off work per year instead of the cash benefits at an estimated cost of NKr 4 billion.

With the oil price falling (NKr 100 per barrel) a number of oil companies are considering a pause in development projects on the North Sea. The Ministry of Petroleum and Energy is studying means of cutting back oil-sector investments in 1998 which are estimated to increase by NKr 13 billion.

March

The Bank of Norway raises its overnight lending rate by 25 basis points.

As Norway's contribution towards stabilising oil prices, the government takes steps to reduce oil production by around 3 per cent (around 100 000 barrels of oil per day), effective after Easter for the rest of the year.

The Bank of Norway warns that NKr 15 to 17 billion should be cut from the Revised National Budget in June to avert an economic crash landing sometime after the year 2000.

April

The wage negotiations by sector begin. Pay increases won for workers in the mechanical trades and settlement in the hotel and restaurant industry whet the appetite of public sector workers. Employees in the municipalities demand an increase of 8 to 10 per cent, at an estimated cost of NKr 3 billion.

Leaders of the Confederation of Norwegian Business and Industry (NHO) are shocked at the wage settlements and threaten to overturn the lifelong learning reform.

The Norwegian Farmers' Union submits its demands regarding the 1998 Agricultural Settlement. The full cost of the demands is calculated at NKr 2.45 billion.

The NHO and the central Labour Union organisation (LO) now stand together in wishing further EU alignment by linking the krone to the euro.

The government advocates an increase in the standard tax deduction allowed to all taxpayers and a reduction in payroll taxes. The White Paper presented by the government outlines a restructuring of the tax system towards greener taxes and a programme for following up the Kyoto Protocol. It also proposes an industrial carbon tax of NKr 100 per metric ton. In addition to offsetting tax reductions in other areas, the revenues from this tax will go towards alternative energy sources. A scheme for trading in emission quotas among commercial polluters is also studied with favour by a majority in the parliament.

May

The government presents the revised national Budget to parliament. The oil price fall reduces expected revenues from petroleum activities by NKr 25.9 billion (bringing them to NKr 86.3 billion). The State Budget surplus is expected to squeeze to NKr 46.9 billion compared to a projected NKr 67.7 billion in the approved budget and the entire amount is to be transferred to the Government Petroleum Fund.

The Bank of Norway raises official interest rates by 50 basis points.

Confidence in the Norwegian economy is eroding due, *inter alia*, to the lack of austerity measures in finance policy and expansive wage settlements. The krone is weakening and markets are expecting the central bank to raise interest rates further.

June

The Bank of Norway raises its interest rates for the third time in 1998 by 25 basis points.

The government's most highly prioritised project, the Family Cash Benefit scheme, is adopted by parliament.

The price of Norwegian crude oil drops to under NKr 90 per barrel. The oil revenue estimate of the national Budget looks difficult to achieve due to the low oil price. In addition, the six Gulf countries agree to reduce their oil production by almost 415 000 barrels per day but Norway does not follow suit. The government considers earlier reductions as sufficient contribution.

The Bank of Norway presents the first quarterly accounts for the Petroleum Fund. Its value increased by 6.3 per cent, or NKr 7.2 billion, during the first quarter of 1998.

July

Following pressure on the exchange rate of the krone, the Bank of Norway raises its deposit and overnight lending rates by 50 basis points (0.5 percentage points) to 5.0 per cent and 7.0 per cent respectively (a total of 1.5 points in less than four months) bringing them to their highest level since the end of 1993.

The Norwegian Homeowners' Association fears that housing prices will fall 15 per cent within the next year and a half due to interest rate hikes.

August

The Bank of Norway raises its official interest rates on three occasions by a total of 3 percentage points, leaving the deposit and overnight lending rates at 8.0 and 10.0 per

cent respectively. Immediately after the last interest rate increase, the Bank announced that the monetary instruments now were oriented to return the exchange rate over time to its initial range, and that the Bank, for the time being, would make no further changes to its use of instruments.

The all-share index in Oslo falls sharply.

The labour market tightens further with nearly 21 000 jobs remaining vacant, an increase of almost 30 per cent over the last year. The most acute labour shortages are to be found in health care, education, services and industry.

Oil companies are preparing for a long period of low oil prices and putting their activities into low gear: exploration activities are being severely reduced and budgets trimmed, fewer projects will be approved and planned ones postponed.

September

The Bank of Norway considers that a substantial reduction (NKr 15 billion or 1.5 per cent of GDP) in next year's fiscal budget would be desirable.

October

The 1999 Budget Proposal is presented to parliament. It implements budget cuts of NKr 5 billion and tax increases of NKr 4 billion, giving an estimated 1 per cent growth in real underlying expenditure measured from 1998 account figures.

Plunging oil prices, a weaker dollar and rising interest rates, combined with major widespread decline in the European stock markets, precipitate a deterioration on the Oslo Stock Exchange all-share index of approximately 45 per cent (compared to its peak in May 1998) before bottoming out.

The Bank of Norway starts intervention buying in order to shore up the exchange rate of the krone. As a result the krone recovers and restores most of its value, but the index against the ECU still remains below its initial range.

The Norwegian Association of Real Estate Agents indicates that prices on the housing market are already falling all around the country and, if interest rates do not decline, the prices in popular areas should fall 10 to 20 per cent within next year.

November

The NHO will no longer help finance adult education reforms due to the magnitude of the pay increases resulting from this year's labour settlements. Adult education is considered important by the LO and the Labour Party, but the NHO's withdrawal may topple the entire reform.

December

The parliament reaches a compromise on the 1999 Budget. The Centre Coalition Government and the two supporting opposition parties, the Progress party and the Conservative party, agree after lengthy negotiations. The approved budget does not include any increase in taxes and excises, in contrast to a NKr 4.0 billion increase in the budget proposal. Other changes include reduced transfers to the civil service and increased dividends from state-owned companies.

The final Budget bill for 1998 of 5 December shows the non-oil budget deficit falling by NKr 2.2 billion from the 1998 approved Budget.

The Government Petroleum Fund in 1998 will contain NKr 38 billion less than estimated in the 1998 approved Budget due to the low oil price in the second half of 1998.

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Czech Republic	Denmark	Finland	France	Germany	Greece
Population												
Total	Thousands	1996	18 289	8 060	10 157	29 964	10 316	5 262	5 125	58 380	81 877	10 465
Inhabitants per sq. km	Number	1996	2	96	333	3	131	122	15	106	229	79
Net average annual increase over previous 10 years	%	1996	1.3	0.6	0.3	1.3	0	0.3	0.4	0.5	3	0.5
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	8 344	3 737 (94)	3 675 (95)	13 676	4 918	2 593	2 087	21 951	35 360	3 824 (95)
of which: Agriculture	% of TCE	1996	5.1	7.2 (94)	2.5 (95)	4.1	6.3	4	7.1	4.6	3.3	20.4 (95)
Industry	% of TCE	1996	22.5	33.2 (94)	26.7 (95)	22.8	42	27	27.6	25.9	37.5	23.2 (95)
Services	% of TCE	1996	72.4	59.6 (94)	71.4 (95)	73.1	51.7	69	65.3	69.5	59.1	56.4 (95)
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	398.9	228.7	268.2	579.2	56.2	174.9	125.1	1 536.6	2 353.5	91.2 (95)
Per capita	US$	1996	21 812	28 384	26 409	19 330	5 445	33 230	24 420	26 323	28 738	8 722 (95)
At current prices using current PPPs[3]	Bill. US$	1996	372.7	172.4	222	645.1	..	118	96.7	1 198.6	1 736.1	133.5
Per capita	US$	1996	20 376	21 395	21 856	21 529	..	22 418	18 871	20 533	21 200	12 743
Average annual volume growth over previous 5 years	%	1996	3.9	1.6	1.2	2.2	2	2.2	1.6	1.2	1.4	1.3 (95)
Gross fixed capital formation (GFCF)	% of GDP	1996	20.3	23.8	17.3	17.7	33	16.7	16.1	17.4	20.6	17 (95)
of which: Machinery and equipment	% of GDP	1996	10.2 (95)	8.8 (95)	7.5 (95)	6.6	..	7.9 (95)	6.4 (95)	7.8	7.6	7.7 (95)
Residential construction	% of GDP	1996	4.6 (95)	5.9 (95)	4.6 (95)	5.4	..	3.2 (95)	3.5 (95)	4.4	7.3	3.3 (95)
Average annual volume growth over previous 5 years	%	1996	5.6	2.1	0.3	2.2	9.4	2	–4.1	–1.5	0.2	0.5 (95)
Gross saving ratio[4]	% of GDP	1996	18	21.9	22.2	17.8	..	17.6	19.6	18.7	20	16 (95)
General government												
Current expenditure on goods and services	% of GDP	1996	17	19.8	14.5	18.7	21.5	25.2	21.9	19.4	19.8	20.8 (95)
Current disbursements[5]	% of GDP	1995	35.6	48.6	52.2	45.8	..	59.6	55.9	50.9	46.6	52.1
Current receipts	% of GDP	1995	34.9	47.4	49.9	42.7	..	58.1	52.8	46.9	45.9	45
Net official development assistance	% of GNP	1995	0.36	0.33	0.38	0.38	..	0.96	0.32	0.55	0.31	0.13
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	12 596	12 152	13 793	12 959	..	12 027	10 282	12 506	12 244	9 473
Passenger cars, per 1 000 inhabitants	Number	1994	460	433	416	466	282	312	368	430	488	199
Telephones, per 1 000 inhabitants	Number	1994	496	466	449	576	209	604	551	547	483[8]	478
Television sets, per 1 000 inhabitants	Number	1993	489	479	453	618	476	538	504	412	559	202
Doctors, per 1 000 inhabitants	Number	1995	2.2 (91)	2.7	3.7 (94)	2.2	2.9	2.9 (94)	2.8	2.9	3.4	3.9 (94)
Infant mortality per 1 000 live births	Number	1995	5.7	5.4	7.6 (94)	6.3 (94)	7.7	5.5	4	5.8 (94)	5.3	8.1
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	1.7	5.2	2.7	2.4	..	3.2	3.8	2.6	4.2	11.8
Consumer prices	%	1996	2.4	2.9	2.2	1.4	11.9	1.9	1.5	2	3.1	11.6
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	60 288	57 870	170 223[7]	202 320	21 910	51 030	40 576	288 450	521 263	11 501
As % of GDP	%	1996	15.1	25.3	63.5	34.9	39	29.2	32.4	18.8	22.1	12.9 (95)
Average annual increase over previous 5 years	%	1996	7.5	7.1	7.6	9.7	..	6.2	12.1	6.3	5.4	5.8
Imports of goods, cif*	Mill. US$	1996	61 374	67 376	160 917[7]	170 931	27 721	44 987	30 911	271 348	455 741	27 402
As % of GDP	%	1996	15.4	29.5	60	29.5	49.3	25.7	24.7	17.7	19.4	30.4 (95)
Average annual increase over previous 5 years	%	1996	9.7	5.9	5.9	7.7	..	5.6	7.3	3.9	3.3	6.6
Total official reserves[6]	Mill. SDRs	1996	10 107	15 901	11 789[7]	14 202	8 590	9 834	4 810	18 635	57 844	12 171
As ratio of average monthly imports of goods	Ratio	1996	2	2.8	0.9	1	..	2.6	1.9	0.8	1.5	5.3

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Hungary	Iceland	Ireland	Italy	Japan	Korea	Luxembourg	Mexico	Netherlands	New Zealand
Population												
Total	Thousands	1996	10 193	270	3 621	57 473	125 864	45 545	418	96 582	15 494	3 640
Inhabitants per sq. km	Number	1996	77	3	52	191	333	458	161	48	380	14
Net average annual increase over previous 10 years	%	1996	–0.3	1.1	0.2	0	0.4	1	1.3	2	0.6	1.1
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	3 605	142	1 307	20 036	64 860	20 764	212 (95)	32 385 (95)	6 983	1 688
of which: Agriculture	% of TCE	1996	8.4	9.2	10.7	7	5.5	11.6	2.8 (95)	23.5 (95)	3.9	9.5
Industry	% of TCE	1996	33	23.9	27.2	32.1	33.3	32.5	30.7 (90)	21.7 (95)	22.4	24.6
Services	% of TCE	1996	58.6	66.2	62.3	60.9	61.2	55.9	66.1 (90)	54.8 (95)	73.8	65.9
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	43.7 (95)	7.3	70.7	1 214.2	4 595.2	484.8	17	329.4	396	65.9
Per capita	US$	1996	4 273 (95)	27 076	19 525	21 127	36 509	10 644	40 791	3 411	25 511	18 093
At current prices using current PPPs[3]	Bill. US$	1996	..	6.3	68.8	1 148	2 924.5	618.5	13.5	751.1	324.5	63.6
Per capita	US$	1996	..	23 242	18 988	19 974	23 235	13 580	32 416	7 776	20 905	17 473
Average annual volume growth over previous 5 years	%	1996	–2.4 (95)	1.5	7.1	1	1.5	7.1	4.8	1.7	2.3	3.7
Gross fixed capital formation (GFCF)	% of GDP	1996	19.3 (95)	17.5	17.2	17	29.7	36.8	20.8	18	19.7	20.9
of which: Machinery and equipment	% of GDP	1996	..	6.7	5.5 (95)	8.8	10.1 (95)	13	..	8.8	9.4	10
Residential construction	% of GDP	1996	. .	3.9	4.9 (95)	4.5	5.3 (95)	7.6	..	4.7	5	5.6
Average annual volume growth over previous 5 years	%	1996	–0.9 (95)	–1.4	6	–1.4	1.3	6.9	0.2	–0.7	2.2	9.6
Gross saving ratio[4]	% of GDP	1996	..	15.6	21.7	20.5	31.4	34.2	37.5	22.7	25.7	16
General government												
Current expenditure on goods and services	% of GDP	1996	24.9 (95)	20.8	14.1	16.4	9.7	10.6	13.6	9.7[10]	14	14.4
Current disbursements[5]	% of GDP	1995	..	35.1	39.2 (94)	49.5	28.5	15.1	..	..	51.8	..
Current receipts	% of GDP	1995	..	36	39.3 (94)	44.5	32	25.1	..	..	50	..
Net official development assistance	% of GNP	1995	..	..	0.29	0.15	0.28	0.03	0.36	..	0.81	0.23
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	..	14 244	10 020	12 224	13 912	7 354	17 811	5 045	12 477	10 895
Passenger cars, per 1 000 inhabitants	Number	1994	212	434	264	517	342	115	544	91	383	457
Telephones, per 1 000 inhabitants	Number	1994	170	557	350	429	480	397	564	93	509	470
Television sets, per 1 000 inhabitants	Number	1993	427	335	301	429	618	215	261	150	491	
Doctors, per 1 000 inhabitants	Number	1995	3.4	3.9 (94)	3.4	3.0 (94)	1.7	1.6 (92)	1.8 (94)	1.1	2.2 (93)	1.6
Infant mortality per 1 000 live births	Number	1995	11	6.1	6.3	6.6 (94)	4.3	9	5.3 (94)	17 (94)	5.5	7.2 (94)
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	..	..	3.7	3.5	1.8	..	..	–1.6	2.4	1.5
Consumer prices	%	1996	23.2	2.6	2.2	4.5	0.7	5.3	2.4	19.7	2.5	2
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	15 674	1 891	48 416	250 842	411 067	129 715	..	96 000	203 953	14 316
As % of GDP	%	1996	35.9	26	68.5	20.7	8.9	26.8	..	29.1	51.5	21.7
Average annual increase over previous 5 years	%	1996	8.9	4	14.8	8.2	5.5	12.5	..	17.6	8.9	8.2
Imports of goods, cif*	Mill. US$	1996	18 105	2 032	35 763	206 904	349 149	150 340	..	89 469	184 389	14 682
As % of GDP	%	1996	41.4	27.9	50.6	17	7.6	31	..	27.2	46.6	22.3
Average annual increase over previous 5 years	%	1996	9.6	3.4	11.5	2.5	8	13.9	..	12.4	7.8	11.8
Total official reserves[6]	Mill. SDRs	1996	6 812	316	5 706	31 954	150 663	23 670	..	13 514	18 615	4 140
As ratio of average monthly imports of goods	Ratio	1996	..	1.9	1.9	1.9	5.2	..	..	1.8	1.2	3.4

* At current prices and exchange rates.
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2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Poland	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population											
Total	Thousands	1996	4 370	38 618	9 935	39 270	8 901	7 085	62 695	58 782	265 557
Inhabitants per sq. km	Number	1996	13	123	108	78	20	172	80	240	28
Net average annual increase over previous 10 years	%	1996	0.5	0.3	–0.1	0.2	0.6	0.8	2	0.3	1
Employment											
Total civilian employment (TCE)[2]	Thousands	1996	2 110	14 977	4 475	12 394	3 963	3 803	20 895	26 088	126 708
of which: Agriculture	% of TCE	1996	5.2	22.1	12.2	8.7	2.9	4.5	44.9	2	2.8
Industry	% of TCE	1996	23.4 (95)	31.7	31.4	29.7	26.1	27.7	22	27.4	23.8
Services	% of TCE	1996	71.5 (95)	46.2	56.4	61.6	71	67.4	33.1	71	73.3
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1996	157.8	117.9 (95)	103.6	584.9	251.7	294.3	181.5	1 153.4	7 388.1
Per capita	US$	1996	36 020	3 057 (95)	10 425	14 894	28 283	41 411	2 894	19 621	27 821
At current prices using current PPPs[3]	Bill. US$	1996	106.7	..	130.1	587.2	171.4	180.6	383.3	1 095.5	7 388.1
Per capita	US$	1996	24 364	..	13 100	14 954	19 258	25 402	6 114	18 636	27 821
Average annual volume growth over previous 5 years	%	1996	4.1	2.2 (95)	1.5	1.3	1	0.1	4.4	2.2	2.8
Gross fixed capital formation (GFCF)	% of GDP	1996	20.5	17.1 (95)	24.1	20.1	14.8	20.2	25	15.5	17.6
of which: Machinery and equipment	% of GDP	1996	8.4	..	11.7 (93)	6.1 (95)	7.9	9.3	11.9	7.6	8.3 (95)
Residential construction	% of GDP	1996	2.6 (94)	..	5.2 (93)	4.3 (95)	1.9	11[9]	8.4 (95)	3	4.1 (95)
Average annual volume growth over previous 5 years	%	1996	2.8	5.4 (95)	2.2	–1	–2.6	–0.8	6.9	1.3	6.9
Gross saving ratio[4]	% of GDP	1996	29.9	..	21.6	20.7	16	27.1	22.3	14.6	16.6
General government											
Current expenditure on goods and services	% of GDP	1996	20.5	16.9 (95)	18.5	16.3	26.2	14.3	11.6	21.1	15.6
Current disbursements[5]	% of GDP	1995	45.8	..	42.5 (93)	41.2	63.8	47.7	..	42.3 (94)	34.3
Current receipts	% of GDP	1995	50.9	..	39.8 (93)	37.9	57.5	53.8	..	37.2 (94)	32.1
Net official development assistance	% of GNP	1995	0.87	..	0.27	0.24	0.77	0.34	0.07	0.28	0.1
Indicators of living standards											
Private consumption per capita using current PPPs[3]	US$	1996	11 593	..	8 522	9 339	10 096	15 632	4 130	11 865	18 908
Passenger cars, per 1 000 inhabitants	Number	1994	381	186	357	351	406 (93)	451	47	372	565
Telephones, per 1 000 inhabitants	Number	1994	554	131	350	371	683	597	201	489	602
Television sets, per 1 000 inhabitants	Number	1993	427	298	190	400	470	400	176	435	816
Doctors, per 1 000 inhabitants	Number	1995	2.8	2.3	3	4.1 (93)	3.1	3.1 (94)	1.2	1.6 (94)	2.6 (94)
Infant mortality per 1 000 live births	Number	1995	4	13.6	7.4	6 (94)	4	5	46.8 (94)	6.2 (94)	8 (94)
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1996	3.2	..	..	5.8	4.8	..	..	4.9	2.7
Consumer prices	%	1996	1.9	..	5.6	4.7	2.7	2.2	81.6	2.7	2.9
Foreign trade											
Exports of goods, fob*	Mill. US$	1996	49 576	24 417	24 614	102 067	84 836	79 581	23 301	259 941	625 075
As % of GDP	%	1996	31.4	20.7	23.8	17.5	33.7	27	12.8	22.5	8.5
Average annual increase over previous 5 years	%	1996	7.8	..	8.6	11.2	9	5.3	11.1	7	8.2
Imports of goods, cif*	Mill. US$	1996	35 575	37 185	35 192	121 838	66 825	78 052	43 094	287 033	795 289
As % of GDP	%	1996	22.5	31.5	34	20.8	26.5	26.5	23.7	24.9	10.8
Average annual increase over previous 5 years	%	1996	6.9	..	6.1	5.5	6	3.2	15.1	6.5	10.3
Total official reserves[6]	Mill. SDRs	1996	18 441	12 409	11 070	40 284	13 288	26 727	11 430	27 745	44 536
As ratio of average monthly imports of goods	Ratio	1996	6.2	..	3.8	4	2.4	4.1	3.2	1.2	0.7

* At current prices and exchange rates.
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5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
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